

Discrete and Continuous Calculus: The Essentials

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Preface

If you're anything like me you hate to have to read large tomes devoted to *any* subject, even ones you like. When I see a large volume on a mathematical subject, it is even worse; I find even opening the book depressing. I'm often interested in the essential ideas of a subject and I don't want to wade through an encyclopedic reference. My experience from watching others – as well as myself – is that essential ideas get lost in the endless details of such a reference.

Ironically, you can see this in the way people *forget* subject material they've learned. This is not so strange if you think about it. If you've ever had the experience of leaving a town after living there a while I think you'll see what I mean. You'll forget the names of streets after a bit. But, most likely, you will forget the more insignificant streets and places first. If you were to go back to this town, you could probably pick up the information you lost fairly easily. You could do this even if the town had grown or changed a fair amount. If instead, you were to memorize a list of street names with disconnected snapshots of locations I think you would have a different experience remembering/forgetting them. You are just as likely to forget a major road or landmark as a minor one. Of course, if certain roads were set in bold in the list and you were told that they were more important than the others, this would help you remember them. What helps is to arrange the information in a hierarchy. In this way we prioritize the more important bits of information. The idea of hierarchy occurs in other academic areas as well. If you read a novel there will be major and minor themes in the text. These themes help you remember and connect the novel to other works. For instance, the notion of death and rebirth, or free will versus fate. You may forget much of the details of the novel after you have read it but the themes that the book is dealing with help you remember the plot. While you will forget subject material in this book, I would hope that you will remember the larger themes. If exposed to calculus after reading this text, you will (hopefully) be able to recall the essential nature of the subject.

You may be saying “Well of course the subject should be organized in a hierarchy – every text book is arranged that way.” This is certainly true, but in academic subjects, and mathematics in particular, there is a tradition of creating a set of supporting concepts which build to a major result. This process (or series of such) leads to a crescendo with a powerful/amazing result proclaimed. My observation watching students is that they are disconnected from the material. The feeling one gets with mathematical instruction is that education is done *to you*, rather than one being an active participant in the exploration of the ideas one generates when attempting to solve a problem. The feeling is one of being an audience member of a magic act. The magician does his thing and you are left to wonder how it was done.

Consequently, the material becomes a linear sequence of small sections that, step by step, obscure the bigger picture of the subject. I first saw this process when I was a young boy. My mother would try to teach me how to balance a check book. It was a step by step process yet I found I couldn't remember the steps if I hadn't done them recently. The problem was that I didn't see the larger purpose, it was all a jumble of steps that didn't make sense to me. Later on, I understood the purpose and could remember what to do. I was also more *adaptable* to change. If the procedure needed to be tweaked, I could make small changes; I wasn't locked into doing the steps exactly the way I was told. My observation is that although books are organized by the hierarchy of chapters, sections, and so forth as any book is, they often fail to impart a useful *conceptual hierarchy* to the student. This is because text books are trying to squeeze a great deal of detailed information between pages. Students typically only see a blur of sections. Regardless of the book's hierarchy, students' perception is often of a linear collection of sections; each as important as the last or the next.

Today, mathematics texts are covering more and more ground at shallower and shallower depths. The argument that is often used is “They need to know about such and such before they take course such and such.”[†] I much

[†] Strangely, physics courses are often six months to a year ahead of the mathematics they use.

prefer someone who has an understanding of the basics and can use this base to solve problems than someone who has memorized a long list of information for a test. To do this, I think you need to understand how to think about your subject. This is the intent of this book, to get you to see the how and the why of the *essential* concepts of calculus. This book is not meant to be encyclopedic; there are already a host of such references.

Regardless of my educational preferences, there is the looming issue of AI which promises to do much of the detailed work in many areas of endeavor. Going forward, we can see that there is going to be a greater need for humans to have a better understanding of the big picture of a subject. With this enhanced understanding we can ask better questions of AI and also better direct it. This preface was written, in part, through a collaboration with AI in August 2026. I have been surprised at how quickly AI has advanced in the last few years. However, I believe that there will still be a need for humans to understand the big picture of calculus, and to be able to ask good questions of AI. This book is meant to help you develop that understanding. To provide this understanding and remain an anti-tome book, there are a limited number of exercises. However, this book paired with a cheap book of exercises will provide a good understanding of the subject. One such source of exercises is "Schaum's Outline of Calculus" by Frank Ayres and Elliott Mendelson. The final appendix of this book was written in conjunction with AI. It provides a list of other sources for exercises; and more importantly, a way to use AI to generate exercises and solutions. This is a new way to learn and practice calculus and, ideally, a way to use AI to help rather than replace humans.

There is another aspect of learning that I think is underappreciated. A student needs to feel – at least to some extent – that they *could have* thought of the ideas being presented. Not that they would see the answer immediately, but more that they are a *player* in the subject. The feeling should be something like: "Yeah, I think I could have done that, looking at things this way, given enough time." Without this feeling, I'm afraid all human beings will learn how to mechanically manipulate symbols to get an "A" on a test and never engage with the understanding. This is similar to the way humans – and animals – play games. If the weaker animal doesn't win at least a third of the time, it loses interest in playing.

How do we keep the student as a player? A key idea is that new material should appear as a recognizable *variant* of a strategy the student already possesses. Consider first year algebra. To solve $ax + b = c$ one treats the equation as a balance and plays a game of adding, subtracting, multiplying, and dividing to isolate x on one side. To solve a quadratic, $ax^2 + bx + c = 0$, one manipulates until all the x 's sit inside a square, then takes a square root – and is back to a problem already understood. To solve a difference equation, one gets all the x 's under a difference operator on one side and applies the inverse operation, summation, to get back to an algebraic equation. To solve a differential equation, one does the same thing with a derivative and its inverse, the integral. The *strategic move* is always the same: isolate the unknown under one operation, then apply the inverse. What changes at each level is only *which* operation one is inverting. A student who sees this thread is playing the same game at every level, with new pieces.

The reason this works – the reason a small number of strategies can carry a student from arithmetic through calculus and beyond – is that mathematics reuses the same ideas at increasing levels of abstraction. What changes is the setting: numbers become vectors, vectors become functions, functions become elements of abstract spaces. But the game is the same. We might call this principle *same game, bigger board*. The student who has learned to isolate and invert in first year algebra is playing the same game when summation undoes differencing, or when integration undoes differentiation. The board got bigger; the move didn't change. Recognizing this is what keeps the student a player as the material grows more sophisticated.

To this end, we will introduce concepts through motivating examples. This means that the concepts may not appear in the order that you would see them in other calculus books, but it is hoped that you will see the point of the various definitions, theorems, and computational rules that arise. This is not to say that other texts don't use motivating examples. It is the case here that the examples will be of primary importance in creating and understanding the hierarchy of concepts of calculus. Major themes will be addressed straight away. In confronting these major themes we will see the need for a number of supporting ideas. Rather than build up to a major theme

by investigating related minor themes, the minor themes will be seen as ancillary to our main track. This is what is meant when we say that the concepts may appear out of order; in fact, most of these supporting ideas will be found in the appendices.

In this regard, this text may be considered a form of Test Driven Development. TDD is an idea from the software engineering world that attempts to produce software by starting with tests first and then writing code to make the tests pass. In a similar way, we will start with problems we wish to solve and then see what mathematical machinery is needed to solve them. We might call this Problem Driven Development (PDD). (If you are not a programmer, feel free to skip the analogy; the idea is simply problems first, machinery second.)

Another organizing principle is the recurring theme of discrete versus continuous. Here we develop a discrete as well as the traditional continuous calculus. This comparison is not done in a traditional course on calculus, but I think you will see that the discrete calculus also solves important problems and reinforces concepts from the continuous calculus. The discrete version has the advantage of not having to deal with the technical challenges that the continuous case forces us to consider.

More broadly, this book is organized around a small number of *strategic moves* that recur across all of calculus and, indeed, across most of the mathematics one encounters through an undergraduate education:

1. *Isolate and Invert.* Get the unknown under a single operation, then apply the inverse to peel it off. This is the thread that connects solving linear equations with solving difference equations and differential equations.
2. *Approximate, then take a limit.* Replace a hard continuous problem with a simple discrete one and ask what happens as the approximation is refined. Secant lines become tangent lines; piecewise constant sums become integrals. The discrete calculus developed in the early chapters makes this move especially visible.
3. *Change the representation.* Rewrite the problem in a form you already know how to handle. Completing the square turns a quadratic into a problem solvable by taking a square root. Substitution in integration turns an unfamiliar integral into a familiar one. The move is always the same: transform the problem so that an existing tool applies.
4. *Exploit linearity.* When an operation is linear – it distributes through addition and constant multiplication – you can break a problem into pieces, solve each piece, and add the answers. Distributivity in arithmetic is the same structural idea as the linearity of differentiation, integration, and summation.

These four moves cover a remarkable amount of ground. A student who holds these threads can walk into any course in the range from algebra through calculus and ask: “Which game is this? Which move applies here? What is the new tool for executing it?” They remain a player.

These convictions turn out to have distinguished company in the mathematics education literature. The idea that a tool should answer a question the student already has is what Harel calls *intellectual need*; the conviction that a student should feel they could have invented the material is what Freudenthal called *guided reinvention*; organizing a subject around recurring ways of thinking rather than topics is the *habits of mind* program of Cuoco, Goldenberg, and Mark; the strategic moves themselves stand in the tradition of Pólya’s heuristics; and the discontent with the standard ordering of calculus has company in Bressoud. The study method of the final appendix rests on a century of memory research. Full citations for all of these are collected in the References at the end of the book.

We start by looking at problems dealing with the differencing and the summing of discrete data. In the process we find that differencing and summing are essentially inverses of one another. Next, we look at an analog of differencing and summing applied to continuous data; what are called differentiation and integration and find a similar relationship between them.

I expect the reader to have a good understanding of algebra and algebraic notation. In particular, we take xy to mean x times y ; if there is confusion we will explicitly write $x * y$. Although trigonometric functions appear in a few sections, they are not necessary for an understanding of the rest of the material. Finally, I hope that your experience reading this work gives you an understanding of calculus that will last when you leave this book.

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Chapter 1: Difference Calculus

Motivational Problem for Differencing

Suppose there are several cars suspected in the use of an armed robbery, all of a particular make and model. Soon after the robbery, a high-speed chase ensued between the police and the getaway car. The car used had no license plates at the time of the crime; however, it's now believed that the getaway car has legitimate plates. There are not enough visual clues to distinguish the getaway car from the group of suspected cars. The police would like to be able to give enough evidence to a judge in order to get a search warrant for the owner of one of the cars. The one thing the police have going for them is that each car has a sensor which records the odometer reading every $1/10$ of a second. The odometer data is stored on an internal disk in feet. They want to examine the data in order to find out if there were high speeds occurring in any of these cars during the time of the getaway.

Each car would have a very long list of numbers representing odometer readings – meaning distance. How should we analyze the data to find velocity? Well, from physics we know the average velocity over a time interval is the distance traveled divided by the time. So, the average velocity over each time interval of $1/10$ th of a second is found by taking the difference in odometer readings over that interval and then dividing by $1/10$. Before discussing any more details, it would be convenient to have a notation to describe long lists of numbers in order to talk about operating on them.

Sequences

We'll define a *sequence* of numbers to be a list of numbers. It is more than a collection of numbers; it is an ordered collection. The distinction is the same as when one considers an ordered list of tasks to perform, a “to do” list, as opposed to an unordered collection of tasks. For instance, the list $\{1, 20, 3, 5, 1, 7\}$ is a sequence of numbers as is the list of positive integers in increasing order; or the list of odd numbers listed in increasing order. Since the sequence values are ordered we can talk about the 5th or the 10th element of a sequence (provided there are that many elements). The 5th positive integer is of course 5; the 10th odd number is 19. In order to talk about sequences in more generality we use the notation $\{x_n\}_{n=1}^N$ to represent the sequence: $x_1, x_2, x_3, \dots, x_N$; where the x values are numbers and the three dots, $\dots$, mean the intermediate values, in this case the sequence values between x_3 and x_N . Similarly, the notation $\{x_n\}_{n=0}^N$ is used to represent the sequence: $x_0, x_1, x_2, \dots, x_N$. All that changed here is the choice of how to start off the indexing. More generally, $\{x_n\}_{n=a}^N$ (a is an integer) is used to represent the sequence: $x_a, x_{a+1}, x_{a+2}, \dots, x_N$. If there is no value listed after the dots then we mean that there is no end to the sequence – like the sequence of odd numbers. In this case one would write: $\{x_n\}_{n=1}^\dagger$. The 5th and 10th values of such a sequence are x_5 and x_{10} respectively.

Sequences as Functions

Sequences can be thought of as functions on the integers; they take an integer as input and return a number. The functions that you are more familiar with, “continuous” functions like $f(x) = x^2$, take a number and return a number. For both sequences and continuous functions there is a distinction to be made between the function and the formula – if any – that defines it. See the appendix “Functions, Formulas, Graphs, and Notation” for a discussion of this. An example of a sequence that can be described by a simple formula is the even numbers. This sequence is $\{2, 4, 6, 8, \dots\}$. If we call this sequence x , then we can write the sequence values succinctly as $x_n = 2n$. In this case there is a formula that allows us to describe any element of the sequence simply. In cases where there is such a formula we will write the sequence using the formula. In the case of the even numbers we would write $\{2n\}_{n=1}^\infty$. Consider now the sequence of primes: $\{2, 3, 5, 7, 11, 13, 17, \dots\}$. Calling this sequence p , we would like

[†] The symbol ∞ is meant to represent infinity.

to write $p_n = \text{some-nice-formula-in } n$. However, there is no such nice formula. The sequence exists, but there is no known easily computable formula for it.

Differencing of Sequences

We know from the introduction that we need to “difference” the distance data to find the velocity. We define the difference symbol Δ (Greek letter Delta which corresponds to D – for difference), to operate on sequences by differencing successive elements of a sequence. Given a sequence, $\{x_n\}_{n=1}^{\infty}$, we define Δx_n by:

$$\Delta x_n = x_{n+1} - x_n \quad \text{for } n \geq 1$$

When using Δ on a sequence, x , we may replace x_n by its formula – if it exists. For instance, if $x_n = 5n + 1$, then we might write Δx_n as $\Delta(5n + 1)$. For this sequence, $\Delta x_n = x_{n+1} - x_n = (5(n + 1) + 1) - (5n + 1) = 5$. If p is the sequence of primes then what is Δp_n ? Well, there is no formula for p_n so we can’t say – meaning we can’t write down a formula for this. However, given a particular n , say 7, $\Delta p_7 = p_8 - p_7$. Since the 8th prime is 19 and the 7th prime is 17, we have $\Delta p_7 = 19 - 17 = 2$.

We can think of Δ as taking a sequence as input and returning a new sequence as the output. If we are given a finite sequence, then one must be careful at the end point. For a finite sequence, $\{x_1, x_2, \dots, x_N\}$, $\{x_n\}_{n=1}^N$, in our notation, Δ is defined to act on this sequence by

$$\Delta x_n = x_{n+1} - x_n \quad \text{for } n \in [1, N - 1]$$

In other words, the differencing is only defined up to $N - 1$. That is, up to where the differencing makes sense.

Example 1.1: If our sequence is $\{1, 3, 7, 15, 22\}$, then applying the differencing operation Δ to this sequence yields a new sequence: $\{2, 4, 8, 7\}$.

Notice that this sequence has one fewer element than the original.

Example 1.2: If $\{x_n\}_{n=1}^{\infty}$ is the list of odd numbers listed in increasing order, then after applying Δ to the sequence we obtain a new sequence: $\{2, 2, 2, 2, \dots\}$.

Sample Difference Formulas

We will soon want a stock of difference formulas on hand; in the next chapter we will be running Δ in reverse, and the more differences we recognize on sight, the easier that game becomes. Here are a few difference formulas for sequences of successive powers, along with the difference of a constant sequence. Let $w_n = c$ (a constant sequence), $x_n = n$, $y_n = n^2$, and $z_n = n^3$, then

$$\Delta w_n = \Delta c = c - c = 0$$

$$\Delta x_n = \Delta n = (n + 1) - n = 1$$

$$\Delta y_n = \Delta n^2 = (n + 1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

$$\Delta z_n = \Delta n^3 = (n + 1)^3 - n^3 = n^3 + 3n^2 + 3n + 1 - n^3 = 3n^2 + 3n + 1$$

Notice that the right hand side of the above formulas is a polynomial of one less power than the left hand side.

Exercise 1.1: Show that $\Delta 2^n = 2^n$; that is, the sequence $\{2^n\}_{n=0}^{\infty}$ is its own difference. (Much later, in the chapter on Differential Calculus, we will meet a famous function with the analogous property.)

Exercise 1.2: Compute the difference sequence of $\{0, 1, 4, 9, 16, 25\}$. Then compute the difference sequence of *that* sequence – the “second difference”. What do you notice?

Back to finding the maximum velocity

We can express the average velocity (in feet per second) of the car over the n^{th} interval $[n/10, (n+1)/10]$ as: $v_n = \Delta x_n / (1/10) = 10 \Delta x_n$. So, we have a new sequence, $\{v_n\}$ which represents an approximation to the velocity. More generally, if h is the length of the time interval, then the sequence of approximate velocities may be computed as:

$$v_n = \Delta x_n / h$$

To find the maximum velocity we could write a program which walks the sequence, $\{v_n\}$, looking for the maximum. At each point one compares the maximum value found so far with the current data point. If the data point has a larger value, we change the maximum value to this value. Such a program might take an array (representing the sequence of velocities) and return the maximum value. We offer some code written in the programming language equivalent of Esperanto[‡].

```
find-max(v) {
  // The current maximum - so far.
  vmax := v[0]

  // The number of values to examine.
  N := length(v)

  // Loop over the rest of the values and compare...
  for i in [1,N-1] do

    // Update current maximum if bigger value found.
    if (vmax < v[i]) then
      vmax := v[i]
    end if
  end for

  // Return the maximum.
  return vmax
}
```

Notice the work involved in finding the maximum velocity: the loop examines all N values, making $N - 1$ comparisons along the way.

After running this program on the data from the cars the police crime lab is able to identify a car which hit speeds of 80 mph. (Since the odometer data is in feet and the time steps are in seconds, the program reports velocities in feet per second; the lab converts to miles per hour by multiplying by $\frac{3600}{5280}$ – so 80 mph is about 117 ft/sec.) Unfortunately, the manufacturer informs them that while the dash-board odometer reading the owner sees is correct, the *historical* odometer data stored on disk is somewhat inaccurate. This “historical” odometer data was designed as a tracking system, originally intended to defeat attempts by used car dealers to illegally modify odometer readings, and it promised to help the service departments maintain the cars as well. But the manufacturer worried about privacy: no one wants their every move recorded, and lawsuits – or simply bad press – could follow. The engineering department found an ingenious compromise. They would alter the stored data so that it was a bit fuzzy – still useful to the service departments, who needed only general driving patterns along with velocity and acceleration over short intervals. There was a secondary benefit: a small signal could be embedded in the modified data to act as a fingerprint, making it easier to spot if someone had tampered with the recorded data.

[‡] We use the characters `//` to denote the beginning of a comment line.

This left the police with a problem. They asked the auto manufacturer for some information about its process of “fuzzifying up” the data; after all, all the police wanted to know was whether one of the cars went over 80 mph on the day of the robbery. The auto maker said the details were a Trade Secret, but agreed to say roughly what the process was. At time step n the odometer value, x_n , is modified by multiplying it by a factor which depends on n . Let us call this factor f_n , then the value stored on the disk is $d_n = x_n f_n$. Although the auto maker would not tell what f_n was, they would say that it varied in a range from 0.5 to 2.0 back to 0.5. It varied in this way over a period greater than 10,000,000 time steps. This meant the period was over 10 days of continuous driving.

What the police have recorded on disk was not the odometer reading, x_n , but $d_n = x_n f_n$. They needed a way to relate the differencing of $x_n f_n$ to the differencing of x_n (Then it is a simple matter to get the velocity – divide by h).

The difference of the disk data is $\Delta d_n = \Delta(x_n f_n) = x_{n+1} f_{n+1} - x_n f_n$. We would like to relate this difference to the difference of the actual distance values: Δx_n . This is accomplished by adding and subtracting the term $x_n f_{n+1}$.

$$\begin{aligned}\Delta(x_n f_n) &= x_{n+1} f_{n+1} - x_n f_n \\ &= x_{n+1} f_{n+1} - x_n f_{n+1} + x_n f_{n+1} - x_n f_n \\ &= (x_{n+1} - x_n) f_{n+1} + x_n (f_{n+1} - f_n) \\ &= \Delta x_n f_{n+1} + x_n \Delta f_n\end{aligned}$$

Or,

$$\Delta(x_n f_n) = \Delta x_n f_{n+1} + x_n \Delta f_n$$

This gives us the relationship we were after. Since $x_n f_n = f_n x_n$ we also have

$$\Delta(x_n f_n) = \Delta(f_n x_n) = \Delta f_n x_{n+1} + f_n \Delta x_n$$

Let us formalize what we have so far as a theorem. We state the result with different names for the sequences.

Theorem(Discrete Product Rule). *If $\{x_n\}_{n=0}^{\infty}$ and $\{y_n\}_{n=0}^{\infty}$ are sequences then the difference of their product is related to the individual differences by:*

$$\Delta(x_n y_n) = \Delta x_n y_{n+1} + x_n \Delta y_n$$

Or,

$$\Delta(x_n y_n) = x_{n+1} \Delta y_n + \Delta x_n y_n$$

Getting back to our problem, we wish to find Δx_n from $\Delta(x_n f_n)$ (the difference of the recorded odometer values), and we have the relation: $\Delta(f_n x_n) = \Delta(x_n f_n) = \Delta x_n f_{n+1} + x_n \Delta f_n$. Ideally, we would like to solve this equation for Δx_n . This is the same game as solving $3x + 2 = 8$ in algebra: get the unknown by itself under one operation, then undo that operation – a move we’ll call *isolate and invert*. Something like:

$$\Delta x_n f_{n+1} = \Delta(x_n f_n) - x_n \Delta f_n$$

So that

$$\Delta x_n = (\Delta(x_n f_n) - x_n \Delta f_n) / f_{n+1}$$

Unfortunately, the formula for Δx_n involves x_n , f_n , and Δf_n which are all unknown. But we are not necessarily looking for an exact solution; we would be happy with a statement which said $v_n = \Delta x_n / h \geq 80$ for some value of n . This is actually more incriminating than saying $v_n = 80$.

Since f changes over a very long scale in comparison with x , it is essentially constant for the purposes of our calculations. That is, for all intents and purposes over the short time span of the robbery f_n can be treated as a constant equal to some value, say F . Replacing a slowly varying quantity by a constant is an approximation –

the first half of the *approximate, then take a limit* move we will use throughout this book. If a sequence, f_n , is constant then $\Delta f_n = 0$. Therefore, the last equation becomes[†]

$$\Delta x_n \approx \Delta(x_n f_n)/F$$

We still don't know F . Here it is a simple matter for the police to run the car for a short distance and compare the odometer tracking data with the actual distance traveled. Because f_n has been virtually constant since the robbery (a few hours before) they are able to run an experiment to find F . They find that it was approximately 0.8 for the suspect car during the robbery. This means their maximum velocity calculation must be corrected by dividing by 0.8 for the suspect car. In this case $v_n = \Delta x_n/h \approx \Delta(x_n f_n)/(Fh) \approx 80/0.8 \text{ mph} = 100 \text{ mph}$. The police now have enough evidence to get a search warrant for the premises of the owner of this car.

Exercise 1.3: Find a similar formula expressing the difference of the product $x_n y_n z_n$ in terms of the differences: Δx_n , Δy_n , and Δz_n .

Summary

We started with a list of odometer readings and a question about velocity, and ended up inventing the difference operation. Along the way we made the idea of a list precise with sequences and the notation $\{x_n\}_{n=1}^N$, and we saw that differencing a sequence produces a new sequence – one that measures how fast the original changes.

Here is what is now in our toolkit:

1. The definition of the difference operation: $\Delta x_n = x_{n+1} - x_n$.
2. Sample difference formulas: $\Delta c = 0$, $\Delta n = 1$, $\Delta n^2 = 2n + 1$, and $\Delta n^3 = 3n^2 + 3n + 1$.
3. The Discrete Product Rule: $\Delta(x_n y_n) = \Delta x_n y_{n+1} + x_n \Delta y_n$.
4. The *isolate and invert* strategy: to recover an unknown from an equation, get it by itself under one operation and then undo that operation.
5. The first half of the *approximate, then take a limit* strategy: replace a slowly varying quantity (the fuzzing factor f_n) by a constant. The second half – refining the approximation with a limit – arrives in the chapter on Differential Calculus.

In the next chapter we turn the difference operation around and ask the reverse question: given the differences, how do we recover the original sequence?

[†] The symbol $\approx$ means approximately equal.

Chapter 2: Summation Calculus

Telescoping Sums

Here is a question left over from the police work of the last chapter. The crime lab, double-checking its analysis, asks: how far did the suspect car actually travel on the day of the robbery? What the lab has on hand is the velocity record $\{v_n\}$ it computed – one value for every $1/10^{\text{th}}$ of a second, on the order of a million values for the day. Since distance is velocity times time, the distance traveled over the n^{th} interval is $v_n h$ (with $h = 1/10$), and the total distance is the sum of all these small distances. A computer can add a million numbers, of course – the loop would look much like the find-max program of the last chapter. But a million additions to produce one number should make us pause: is there a shortcut?

There is, and it comes from remembering where the velocities came from. Each v_n was computed from a *difference* of odometer readings, $v_n = \Delta x_n / h$; so $v_n h = \Delta x_n$, where $\{x_n\}_{n=0}^N$ is the odometer data (this time starting the count at 0, the morning reading). The distance traveled over the i^{th} interval is $x_{i+1} - x_i$, and the total distance for the day is the sum of all of these. In mathematical terms, this would be written:

$$\underbrace{(x_1 - x_0)}_{\text{first interval}} + \underbrace{(x_2 - x_1)}_{\text{second interval}} + \underbrace{(x_3 - x_2)}_{\text{third interval}} + \cdots + \underbrace{(x_N - x_{N-1})}_{\text{Nth interval}} = \underbrace{(x_N - x_0)}_{\text{total distance}}$$

It seems like we have taken something obvious and produced a deep formula. If you look at the left hand side of the formula you will see that after adding the first and second distances the x_1 's cancel. If we then add the third distance the x_2 's cancel. And so on, all the intermediate x 's cancel except the first and the last. It isn't that mysterious after all. We say that such a sum is a telescoping sum; the sum acts like a collapsible telescope. And there is the crime lab's shortcut, in full: because the summands are differences, the million additions collapse to a single subtraction – the evening odometer reading minus the morning one, $x_N - x_0$.[†]

Fundamental Theorem of Difference Calculus

Previously, we introduced a notation to describe sequences and their differences; now, since we are talking about sums, it is convenient to use a notation for summation. The symbols $\sum_{n=1}^N x_n$, will mean the sum: $x_1 + x_2 + \cdots + x_N$. In our new notation the above telescoping sum becomes:

$$\begin{array}{ccccc} \text{Discrete Sum} & & \text{Telescoping Difference} & = & \text{Difference at Ends} \\ \underbrace{\sum_{n=0}^{N-1}} & & \underbrace{\Delta x_n} & = & \underbrace{x_N - x_0} \end{array}$$

Note: The Greek letter $\sum$ (sigma) corresponds to the letter S – for sum.

There is nothing special about starting at 0; the sum telescopes from any starting index. For any integers $a \leq b$ we have $\sum_{n=a}^b \Delta x_n = x_{b+1} - x_a$, and we will use this general form freely in what follows.

Although the above formula seemed rather obvious from the way we derived it, we will refer to this formula as *The Fundamental Theorem of Difference Calculus*, or FTDC. The reason why it is important is that it relates something that is computationally hard to do, sum N things, with something that is computationally easy, subtract *two* things. This is very useful if we have a sum of a large number of things and then recognize that we are summing

[†] Strictly speaking, the lab holds the “fuzzed” disk data of chapter 1, $d_n = x_n f_n$, not the odometer data itself; each velocity was recovered as $v_n \approx \Delta d_n / (F h)$. The shortcut is the same: the million additions collapse to the single subtraction $(d_N - d_0) / F$, which approximates the true difference $x_N - x_0$.

the differences of a sequence. If we make this connection, the computation becomes easy. Suppose for instance, that someone asks you to sum a sequence of the form: $\sum_{n=0}^{N-1} v_n$. If you can find another sequence, $\{x_n\}_{n=0}^N$ such that $\Delta x_n = v_n$, then you can sum this series easily by the Fundamental Theorem of Difference Calculus since $\sum_{n=0}^{N-1} v_n = \sum_{n=0}^{N-1} \Delta x_n = x_N - x_0$. The tricky part, of course, is finding the sequence $\{x_n\}_{n=0}^N$ whose difference is $\{v_n\}_{n=0}^{N-1}$.

In later chapters we will introduce a notion of “infinitesimal” differences and “continuous” sums which will have a similar relationship. The corresponding theorem is the “Fundamental Theorem of Calculus”. It is the cornerstone of calculus and it is a generalization of the simple observation above.

Linearity of Summation and Differencing

We write down a basic property of both the summing and differencing operations, $\sum$, and Δ . One of the most *important properties* of the two operations is that they both are *Linear Operations* in that they distribute through both constant multipliers and sums.

1. $\sum_{n=1}^N (a x_n + b y_n) = a \sum_{n=1}^N x_n + b \sum_{n=1}^N y_n$.
2. $\Delta(a x_n + b y_n) = a \Delta x_n + b \Delta y_n$.

From this it follows that $\sum_{n=1}^N a x_n = a \sum_{n=1}^N x_n$ and $\Delta(a x_n) = a \Delta x_n$. This can be seen by taking $y_n = 0$.

This is the *exploit linearity* move of the preface making its first formal appearance: break a problem into pieces, handle each piece, and add the answers. It allows us to more easily compute sums and differences of sequences in terms of their component parts. We will need this property almost immediately, to split differences like $\Delta(n^2 - n)$ into pieces we already know.[†]

Applications of the Fundamental Theorem

As an example of how this might be useful consider the problem of summing the integers from 1 to N : $\sum_{n=1}^N n$. It would be nice if we could find a simple formula for this sum in terms of N . If we could recognize this sum as a sum of a telescoping difference then we could easily find the answer. That is, if we could find a sequence $\{x_n\}$ such that $\Delta x_n = n$, then $\sum_{n=1}^N n = \sum_{n=1}^N \Delta x_n = x_{N+1} - x_1$. We call x_n an *anti-difference* of the sequence $\{n\}_{n=1}^N$. But, can we find such a sequence $\{x_n\}$? Well, from the last chapter we know that $\Delta n^2 = 2n + 1$. We also know that $\Delta n = 1$. Therefore, $\Delta(n^2 - n) = \Delta n^2 - \Delta n = (2n + 1) - 1 = 2n$.[‡]

Finally, if we divide by 2 we have $\Delta((n^2 - n)/2) = n$.^{*} Therefore, if $x_n = (n^2 - n)/2$ then $\sum_{n=1}^N n = \sum_{n=1}^N \Delta x_n = x_{N+1} - x_1 = ((N+1)^2 - (N+1))/2 - (1^2 - 1)/2 = N(N+1)/2$. Summarizing:

$$\sum_{n=1}^N n = N(N+1)/2$$

So the sum of the first 1000 numbers is $(1000 * (1000 + 1))/2 = 500,500$.[§] One can extend this technique to finding the sum of squares of the first N integers. The difference of the sequence n^3 is a polynomial of degree 2. So n^3 is an imperfect anti-difference in that its difference produces a multiple of n^2 plus lower-order terms – a term in n and a constant term. But, as above, if we add combinations of n^2 and n to n^3 we may rid ourselves of these extraneous terms. That is, we would try to find an *anti-difference* of n^2 as a linear combination of n^3 , n^2 , and n . Once we have this anti-difference we can easily find the formula for the sum of n^2 . Let us flesh this out. First,

[†] The theme of linearity will continue in the next chapters when we talk about the continuous analogs of differencing and summing.

[‡] Using the linearity of Δ .

^{*} Linearity of Δ again.

[§] For clarity we will occasionally write “ $a * b$ ” instead of the conventional “ $a b$ ” to indicate the multiplication of a and b .

find a linear combination of n , n^2 , and n^3 whose difference is n^2 . That is, find c_1 , c_2 , and c_3 such that

$$\Delta(c_1 n + c_2 n^2 + c_3 n^3) = n^2$$

Since Δ is linear it distributes over the sum and through multiplication by the c 's. That is,

$$\Delta(c_1 n + c_2 n^2 + c_3 n^3) = c_1 \Delta n + c_2 \Delta n^2 + c_3 \Delta n^3 = c_1 + c_2(2n + 1) + c_3(3n^2 + 3n + 1)$$

As a polynomial in n this is

$$(c_1 + c_2 + c_3) + (2c_2 + 3c_3)n + 3c_3n^2$$

We need to find c_1 , c_2 , and c_3 so that this is n^2 . For this to happen the first two terms should be 0 and the last coefficient should be 1. This gives us three equations:

$$c_1 + c_2 + c_3 = 0$$

$$2c_2 + 3c_3 = 0$$

$$3c_3 = 1$$

Starting at the last equation and working backwards we have $c_3 = 1/3$, $c_2 = -3c_3/2 = -1/2$, and $c_1 = -(c_2 + c_3) = 1/6$. So, an anti-difference is $1/6 n - 1/2 n^2 + 1/3 n^3$. Now we can compute the sum using the FTDC:

$$\begin{aligned} \sum_{n=1}^N n^2 &= \sum_{n=1}^N \Delta\left(\frac{1}{6}n - \frac{1}{2}n^2 + \frac{1}{3}n^3\right) \\ &= \left(\frac{1}{6}(N+1) - \frac{1}{2}(N+1)^2 + \frac{1}{3}(N+1)^3\right) - \left(\frac{1}{6} * 1 - \frac{1}{2} * 1^2 + \frac{1}{3} * 1^3\right) \\ &= (2(N+1)^3 - 3(N+1)^2 + (N+1))/6 - 0 \\ &= (N+1)[2(N+1)^2 - 3(N+1) + 1]/6 \\ &= (N+1)[((N+1)-1)(2(N+1)-1)]/6 \\ &= (N+1)(N)(2N+1)/6 \end{aligned}$$

Therefore, the formula is $\sum_{n=1}^N n^2 = N(N+1)(2N+1)/6$.

An Aside: Falling Powers

The bookkeeping above – adding combinations of lower powers to kill off the straggler terms – suggests that the ordinary powers, n^k , are not quite the natural sequences for differencing. There is a family of sequences that plays the differencing game perfectly. Define the *falling power* $n^{\underline{k}}$ (read “ n to the k falling”) to be the product of k factors which start at n and step down by one:

$$n^{\underline{k}} = \overbrace{n(n-1)(n-2)\cdots(n-k+1)}^{k \text{ factors}}$$

For example, $n^{\underline{1}} = n$, $n^{\underline{2}} = n(n-1)$, and $n^{\underline{3}} = n(n-1)(n-2)$. Now compute a difference:

$$\Delta n^{\underline{2}} = (n+1)n - n(n-1) = n((n+1) - (n-1)) = 2n = 2n^{\underline{1}}$$

The same cancellation works for every k . The products $(n+1)^{\underline{k}}$ and $n^{\underline{k}}$ share the $k-1$ factors $n(n-1)\cdots(n-k+2)$; factoring these out leaves $(n+1) - (n-k+1) = k$. That is,

$$\Delta n^{\underline{k}} = k n^{\underline{k-1}}$$

with *no* straggler terms – compare $\Delta n^2 = 2n + 1$ and $\Delta n^3 = 3n^2 + 3n + 1$. As a one-line application, an anti-difference of $n = n^{\underline{1}}$ is $n^{\underline{2}}/2$, so by the FTDC

$$\sum_{n=1}^N n = \frac{(N+1)^{\underline{2}}}{2} - \frac{1^{\underline{2}}}{2} = \frac{(N+1)N}{2} - \frac{1*0}{2} = \frac{N(N+1)}{2}$$

in agreement with the formula found above. When we meet the rule of differentiation for power functions, $(x^k)' = kx^{k-1}$, in the chapter on Differential Calculus, it is the falling powers that it mirrors *exactly*; on the discrete board, the ordinary powers were the wrong pieces all along.

Exercise 2.1: Using the falling power rule, find an anti-difference of $n^2 = n(n-1)$ and use it, with the FTDC, to compute $\sum_{n=1}^N n(n-1)$.

Summation By Parts

Consider the sum $\sum_{n=1}^N n 2^n$. The linear-combination method of the Applications section is of no help here: that method hunted for an anti-difference of n^2 among linear combinations of the powers n , n^2 , n^3 , but no combination of the sequences we have differenced so far has $n 2^n$ as its difference. The trouble is that $n 2^n$ is a *product* of two very different sequences. We need a tool built for products.

Powerful applications of the FTDC result from equations that have a difference on one side. By summing both sides of the equation, the FTDC implies that the difference side is easy to compute. This means that there is a simple formula for the summation of the other side of the equation. We do have such a formula for products; it is the Discrete Product Rule from the last chapter. It states that:

$$\Delta(x_n y_n) = \Delta x_n y_{n+1} + x_n \Delta y_n$$

Summing both sides of this equation from 0 to N gives us:

$$\sum_{n=0}^N \Delta(x_n y_n) = \sum_{n=0}^N \Delta x_n y_{n+1} + \sum_{n=0}^N x_n \Delta y_n$$

By the FTDC the left hand side is just $x_{N+1}y_{N+1} - x_0y_0$. On the right hand side we have two complicated sums; moving one of them to the other side of the equation gives us a way to compare two seemingly unrelated sums.

$$\sum_{n=0}^N x_n \Delta y_n = (x_{N+1}y_{N+1} - x_0y_0) - \sum_{n=0}^N \Delta x_n y_{n+1}$$

This formula is referred to as “Summation by Parts”. It relates one sum to another. If one sum is known, we can find the other from this formula. Before unleashing it on the sum $\sum n 2^n$ that opened this section, we first test the new tool by re-deriving two results we already know – that is how one learns to trust a new tool.

Applications of Summation By Parts

Let’s try to apply summation by parts to the previous example of summing the first N integers. To use “Summation by Parts” we have to map our problem onto this formula. We can do this if we let $x_n = n$ and $\Delta y_n = 1$. We need to find a y_n such that $\Delta y_n = 1$; but, we already know that the sequence $y_n = n$ works. Using the Summation by Parts formula gives:

$$\sum_{n=1}^N \underbrace{x_n \Delta y_n}_{n * 1} = \underbrace{x_{N+1} y_{N+1}}_{(N+1) * (N+1)} - \underbrace{x_1 y_1}_{1 * 1} - \sum_{n=1}^N \underbrace{\Delta x_n y_{n+1}}_{1 * (n+1)}$$

We’re trying to use this formula to relate our complicated sum to a simpler sum. Instead, we end up with a term that is very similar to what we started with – the last term of the above equation. This last term can be broken into a piece that is *exactly* like the left hand side of the equation. Unfortunately, it seems like we are going backwards. We have

$$\sum_{n=1}^N \underbrace{x_n \Delta y_n}_{n * 1} = \underbrace{x_{N+1} y_{N+1}}_{(N+1) * (N+1)} - \underbrace{x_1 y_1}_{1 * 1} - \sum_{n=1}^N \underbrace{x_n \Delta y_n}_{n * 1} - \sum_{n=1}^N 1$$

But, just as you would when solving an equation in high school algebra for “x”, you collect all expressions of the unknown to one side and isolate it; everything on the other side is the solution. Our unknown is $\sum_{n=1}^N n * 1$; that is, $\sum_{n=1}^N n$. So, collecting our unknown on one side and placing the rest of the expressions on the other yields:

$$2 \sum_{n=1}^N n * 1 = (N+1) * (N+1) - 1 * 1 - \sum_{n=1}^N 1 = N^2 + 2N + 1 - 1 - N = N^2 + N$$

Solving for the sum we have:

$$\sum_{n=1}^N n = N(N+1)/2$$

Example 2.1: Find a formula for the sum: $\sum_{n=1}^N n^2$. We can use the summation by parts formula again by letting $x_n = n^2$ and finding a y_n such that $\Delta y_n = 1$. $y_n = n$ works as before. Therefore,

$$\sum_{n=1}^N n^2 * 1 = (N+1)^2(N+1) - 1^2 * 1 - \sum_{n=1}^N \Delta n^2(n+1)$$

This simplifies to:

$$\sum_{n=1}^N n^2 * 1 = (N+1)^3 - 1 - \sum_{n=1}^N (2n+1)(n+1)$$

Or,

$$\sum_{n=1}^N n^2 * 1 = (N+1)^3 - 1 - \left(\sum_{n=1}^N 2n^2 + \sum_{n=1}^N 3n + \sum_{n=1}^N 1 \right)$$

We have now, as before, an equation for the unknown value $\sum_{n=1}^N n^2$. Solving for this value gives:

$$3 \sum_{n=1}^N n^2 = (N+1)^3 - 1 - 3 \sum_{n=1}^N n - \sum_{n=1}^N 1$$

We already know how to sum $\sum_{n=1}^N n$, so we replace it with the formula we just worked out.

$$3 \sum_{n=1}^N n^2 = (N+1)^3 - 1 - 3N(N+1)/2 - N$$

Collecting terms on the right yields

$$3 \sum_{n=1}^N n^2 = N^3 + 3N^2 + 3N + 1 - 1 - \frac{3}{2}N^2 - \frac{3}{2}N - N$$

This becomes

$$3 \sum_{n=1}^N n^2 = N \left(N^2 + \frac{3}{2}N + \frac{1}{2} \right) = N(N+1) \left(N + \frac{1}{2} \right) = N(N+1)(2N+1)/2$$

Finally, we have

$$\sum_{n=1}^N n^2 = N(N+1)(2N+1)/6$$

Exercise 2.2: Find a formula for the sum: $\sum_{n=1}^N n^3$.

Now for the payoff – the sum that opened the Summation by Parts section, the one the linear-combination method could not touch.

Example 2.2: Find a formula for the sum: $\sum_{n=1}^N n 2^n$. Recall from the first chapter that $\Delta 2^n = 2^{n+1} - 2^n = 2^n$; the sequence 2^n is its own difference. So, in the Summation by Parts formula let $x_n = n$ and $y_n = 2^n$; then $\Delta y_n = 2^n$ and $\Delta x_n = 1$. The formula gives:

$$\sum_{n=1}^N \overbrace{n 2^n}^{x_n \Delta y_n} = \overbrace{(N+1) 2^{N+1}}^{x_{N+1} y_{N+1}} - \overbrace{1 * 2}^{x_1 y_1} - \sum_{n=1}^N \overbrace{1 * 2^{n+1}}^{\Delta x_n y_{n+1}}$$

The remaining sum telescopes for the same reason: $2^{n+1} = \Delta 2^{n+1}$, so by the FTDC, $\sum_{n=1}^N 2^{n+1} = \sum_{n=1}^N \Delta 2^{n+1} = 2^{N+2} - 2^2 = 2^{N+2} - 4$. Therefore,

$$\sum_{n=1}^N n 2^n = (N+1) 2^{N+1} - 2 - (2^{N+2} - 4) = (N+1) 2^{N+1} - 2 * 2^{N+1} + 2 = (N-1) 2^{N+1} + 2$$

As a check, for $N = 2$ the formula gives $1 * 8 + 2 = 10$, which agrees with $1 * 2 + 2 * 4 = 10$.

Exercise 2.3: Find a formula for the sum: $\sum_{n=1}^N n^2 2^n$. Hint: in the Summation by Parts formula let $x_n = n^2$ and $y_n = 2^n$; the sums that appear can be handled with the formula just derived for $\sum_{n=1}^N n 2^n$ and the telescoping of 2^{n+1} .

Summary

The *Fundamental Theorem of Difference Calculus* can be stated succinctly: The *Summation Operation* is the inverse of the *Difference Operation*.

This can be seen from the FTDC: $\sum_{n=0}^{N-1} \Delta x_n = x_N - x_0$, and writing it more crudely as $\sum \Delta x = x$. What makes this result useful is if for a given sum, $\sum_{n=0}^{N-1} v_n$, one can recognize v_n as the difference of another sequence x_n , then this sum is $\sum_{n=0}^{N-1} \Delta x_n$ and by the FTDC this is just $x_N - x_0$.

The process of recognition is helped greatly if there is a well developed stock of difference formulas. Then when confronted with summing a sequence, we may be able to recognize, or more easily deduce, a sequence which is an *anti-difference* and then easily compute the sum. Actually, finding differencing and anti-differencing formulas for sequences is much like multiplying and dividing numbers. Differencing is easier – like multiplying, while finding anti-differences is harder – like division. To do long division it helps to be fluent with multiplication, because trial and error multiplications are necessary. The same is true for anti-differencing, it helps first to be fluent with difference formulas for a number of sequences. To this end, one would

1. Develop difference formulas for a base collection of sequences.
2. Develop difference formulas of sequences formed in various ways from the base; e.g., products and *Linear Combinations* of sequences.

This program has been partially carried out in the last chapter. There, we had difference formulas for a few sample sequences: $x_n = c$, $x_n = n$, $x_n = n^2$, $x_n = n^3$. We also had a way of finding formulas for sequences formed by linear combinations and products of other sequences. This helped us above as we tried to find summation formulas for $x_n = n$ and $x_n = n^2$. We will see in the next two chapters this strategy carried out in more detail. It turns out that this process often yields simpler formulas when dealing with the continuous analogs of sequences (discrete functions) – continuous functions.

In fact, the next chapters replay this chapter's game on a bigger board: each of our discrete tools has a continuous counterpart.

Discrete	Continuous (coming soon)
Δ (difference)	d/dx (derivative)
$\sum$ (summation)	$\int$ (integral)
FTDC	Fundamental Theorem of Calculus
Discrete Product Rule	Product Rule
Summation by Parts	Integration by Parts

Chapter 3: Differential Calculus

A Motivating Problem for Differential Calculus

Let me be clear that there is not only one motivating problem for this subject. But I think the problem we introduce illuminates the parade of definitions you see in standard calculus books dealing with differential calculus. You would in the course of study of a standard book come across such a problem only after you had built up a certain amount of mathematical machinery. As mentioned in the preface, I prefer to see the problem first and see what motivated the machinery. It helps me, and I hope it helps you, get a better perspective of the subject.

Problem. Suppose one is to make a storage box from a flat square sheet of metal 12 inches by 12 inches. The box is made by doing the following:

1. Cut along the dotted lines; that is, cut out the corner pieces.
2. Fold the remaining four “flaps” up.

Now, the resulting “box” is a container without a lid. See the figure below.

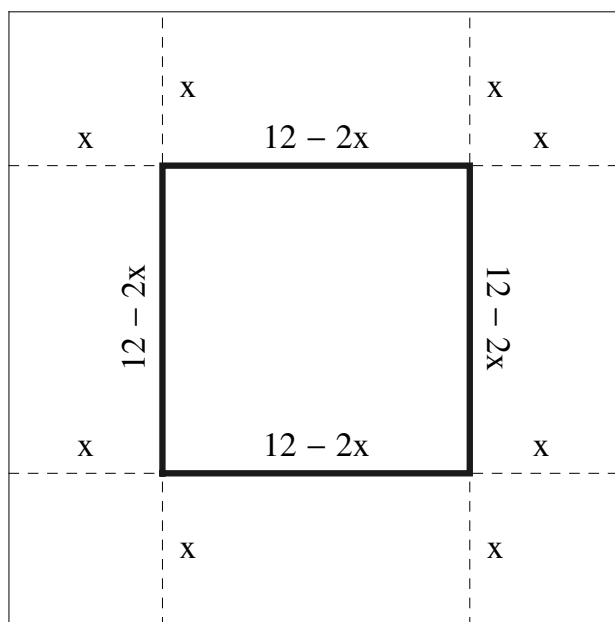


Figure 3.1: Metal Sheet

Question: What should the “flap distance” x be in order to make a box with maximum volume?

First, it seems clear that the volume is a function of the parameter x . What is this function? The volume of a box is the area of the base times the height. The area of the base is $(12 - 2x)^2$ while the height of the box is x . So, the volume function (in cubic inches) can be expressed as[†] $V(x) = x(12 - 2x)^2$. It is not clear from looking at this formula what the best choice of x should be. We could try many values of x and see which gives the largest value of V ; it's clear from the diagram that x must be in the range $[0, 6]$ (in inches). This is a valid approach, and though we may not get an exact best answer it could give us a value close to the best. For instance, we could plug in the input values $0, 1, 2, \dots, 6$, in the function V seeing which gave the biggest answer. But how would we know if we missed any input value to try? We could increase the sampling by plugging in the input

[†] See the appendix “Functions, Formulas, Graphs, and Notation”.

values $0, 1/2, 1, 1\frac{1}{2}, \dots, 6$ into V . Again, comparing the values of V after plugging in each of these inputs would perhaps give a better answer. But to find the best solution we would have to plug in all the numbers in the interval $[0, 6]$ into the function V to find the best solution. The problem here is that there are an infinite number of numbers in the interval, $[0, 6]$!

One of the difficulties with this problem is one of visualization. We can visualize a box and its construction but the function V , or for that matter any function, is a fairly abstract notion. It is not a number, it is a thing that takes a number as an input and produces a number as an output. One way to make a function more tangible is to draw its *graph*. When we do this we get back to things that we can see and touch. By drawing the graph of V , we make V more concrete, and we can see another way to find an input, x , which maximizes V . See the figure below.

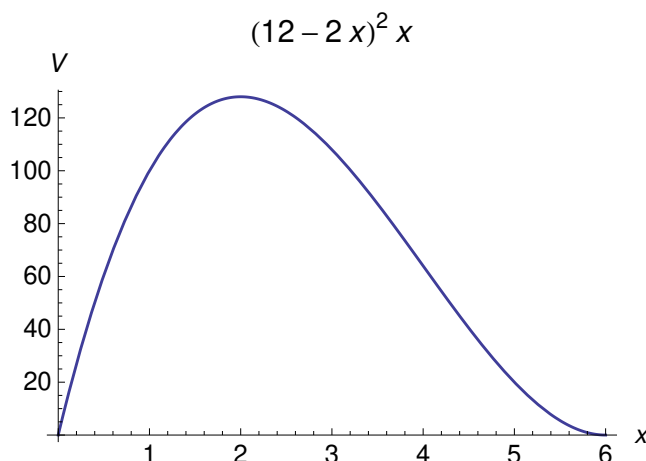


Figure 3.2: Box volume as a function of flap distance.

If you place a level[‡] against the graph at any point, the level is said to be *tangent* to the graph at that point. The line that passes through the level is called a *tangent line*. If the level is placed at the high point of the graph, the level lies flat. That is, the tangent line at that point is flat. See the figure below.

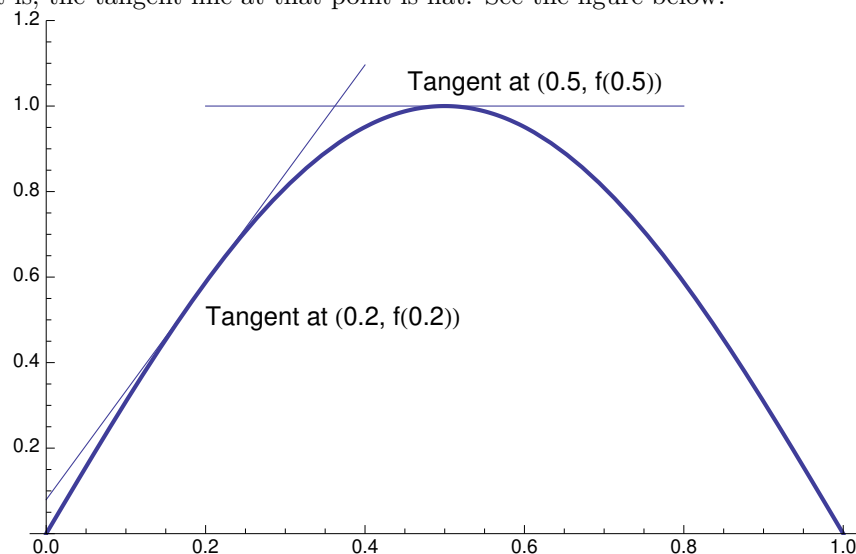


Figure 3.3: Graph of a function and sample tangent lines.

So, here is a strategy for finding the maximum: If someone gives us a graph of this function, place a level at some

[‡] That is, something with a straight edge like a level or a ruler.

point on the graph and move it along the graph. Notice that at a bump in the graph – where potentially there is a maximum or minimum value – the level is flat. If we mark the point or points where this occurs you have a set of input candidates for the maximum.* You need only compare the value of the function to maximize at these points and at the end points of the interval over which you are searching for a maximum. This, in general, should be a small number of points. See the figures below.

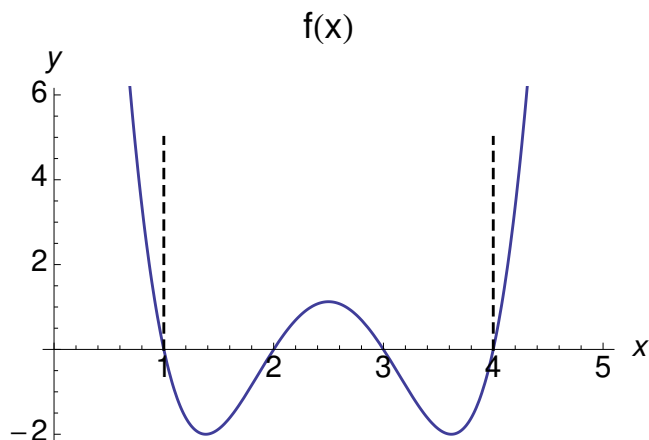


Figure 3.4: Find the maximum of $f(x)$ on the interval $[0, 4]$

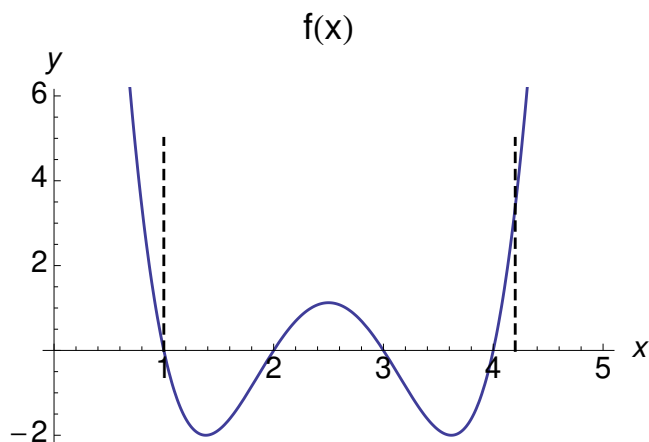


Figure 3.5: Find the maximum of $f(x)$ on the interval $[0, 4.2]$. Although this is the same function as the last figure, the change in the interval leads to a different solution.

The great thing about this method is that if we can identify these points we have reduced the comparison to a handful (hopefully) of points.

The problem with this method is that it is not much different from the last method. To draw a good approximation to the graph of this function over the interval of interest, $[0, 6]$, you need to sample the function at a large number of points on the interval $[0, 6]$. The more you sample, the better the approximation to the graph. So, the work is not really much different. To make matters worse, for some applications the interval could be huge. To draw the graph would be a great deal of work.

* We are quietly using a fact about continuous functions: a continuous function on a closed interval $[a, b]$ actually *attains* a maximum value somewhere on the interval. For the nice functions of this book this may be taken on faith.

Suppose we had a way to numerically measure the “steepness” of the level applied at any point of the graph *WITHOUT HAVING TO GRAPH IT*. In addition, this “steepness measure” had a numeric value of 0 when the level was flat. In effect, we would have a new function, which I will call V' such that $V'(x)$ is the “steepness” of V at x . If someone were to hand us this function given V , we could find an x which maximizes V by examining *only* points in the set, $\{x \mid V'(x) = 0\}$ (the places where V is flat), and the end points. To find a best x , we would plug in each of these inputs into V – hopefully a small number of inputs – and find a maximizing x . Notice again that instead of having to sift through an infinite number of input values we would only have to check a small handful.

But all this is predicated on easily finding the magical function V' given V . It turns out that the “measure of steepness” that we will introduce can be computed as $V'(x) = 12(12 - 8x + x^2)$. Now we need only find where the steepness is 0. But this is a quadratic and can easily be solved. In this case, $\{x \mid V'(x) = 0\} = \{2, 6\}$. If we add the end points of the interval $[0, 6]$, then the set of points to check becomes, $\{0, 2, 6\}$. Plugging either 0, or 6 into V gives 0. If we plug in 2 into V we get 128. Therefore, $x = 2$ gives us the biggest value for V and is therefore the solution to our problem.

Again the magic is that rather than compare an infinite number of input values we end up looking at only a handful of candidate inputs that have the potential to produce the maximum value of V provided:

1. We are able to easily find the steepness of the graph of the function to maximize as a function of the input, x . That is, we are able to find a steepness function that has a nice formula.
2. We are able to find the places where the steepness function is zero. That boils down to setting the steepness function to zero and solving for inputs. We then need only compare the values of the function to maximize on these inputs (and the end points of the interval).

Measure of Steepness

How do we measure steepness of these tangent lines, or lines in general? Road builders have a similar problem when describing the steepness of their roads. The standard way to describe road steepness is to refer to the road’s grade. This is the amount of “rise” of the road divided by its (horizontal) “run”*. The measure is *independent of the particular “run” length*. This can be seen from the graph below using simple geometry. This “rise” over “run” number is called the slope of a straight line. It has the property that it is zero when the line is flat.

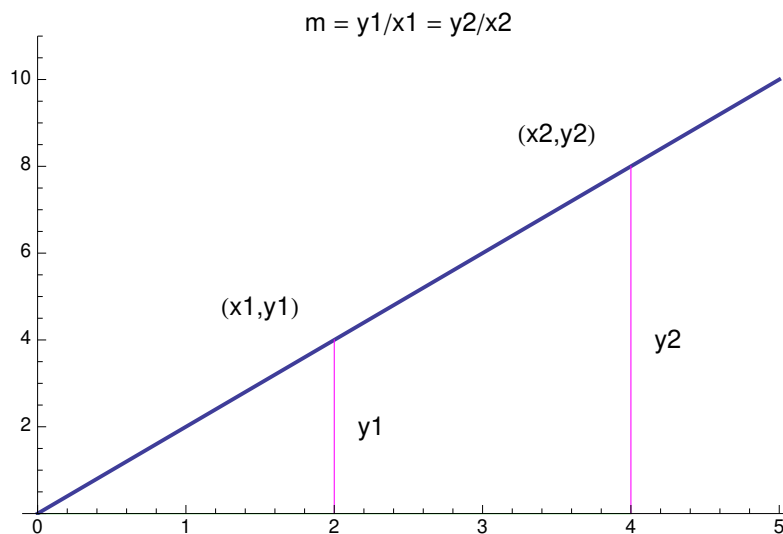


Figure 3.6: The grade gives the same value regardless of the amount of run.

So, our definition of steepness of a line is the slope of the line – see the appendix “Straight Line Functions”.

* This ratio is actually multiplied by 100, but this distracts from the main idea.

Therefore, the steepness at a point on the graph of a function f at x is the slope of the line tangent to the graph of f at x . Let's formalize this.

Definition Steepness. *The steepness of a function f at x is the slope of the line tangent to the graph of f at x (provided the tangent exists).*

Remember, we need to move from the geometric view of steepness (a tangent line) to an algebraic/mechanical one so we can compute efficiently. That is, we need to avoid having to graph our function in order to solve the maximization problem. What we have now done, in effect, is represent our *geometric* notion of a tangent line in an *algebraic* way as a single number. This is the *change the representation* move of the preface: we trade a picture for a number we can compute with. This is important enough to state formally.

Theorem(Slope Representation). *The steepness of a non-vertical straight line is captured by a single number - its slope. The line itself is pinned down by its slope together with any one of its points.*

proof: From the appendix on straight line functions we see that a non-vertical line with slope m and y -intercept b is the graph of the function $f(x) = mx + b$. Two lines with the same slope m differ only in b - a vertical shift - so they have the same steepness everywhere, and knowing any one point on the line determines the shift b .

The big question now is: How do we compute this measure of steepness for a given function?

Computing the Slope: Approximating Tangents

So, we know how to measure the steepness of a straight line - it is its slope. Although it's clear what we mean by tangent when we physically put a level against a graph, it is not so clear how to compute it algebraically in terms of the original function.

One way to get a handle on this idea of tangent is to work with approximate tangents to a function at a point, $(x, f(x))$ on the graph. In the figure below you see that the line which passes through the points $(x, f(x))$ and $(x + h, f(x + h))$ is approximately the same as the tangent line. The approximation gets better the closer h is to 0.

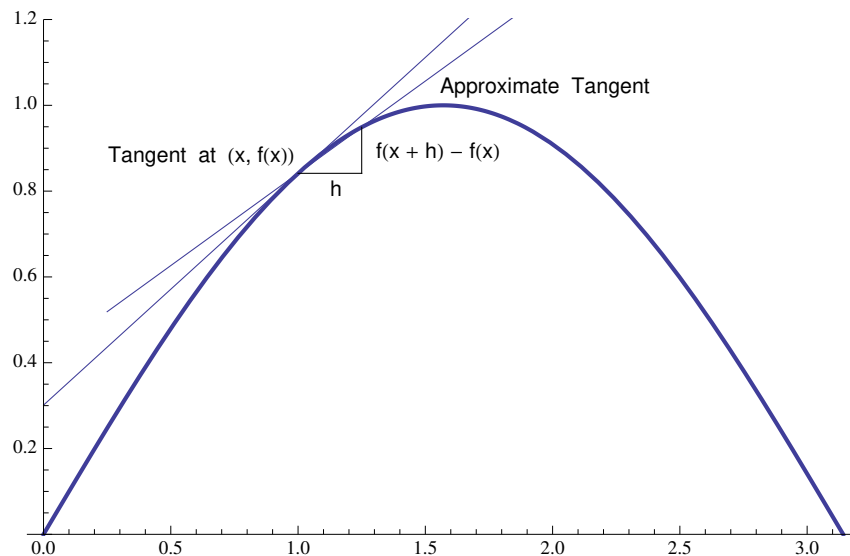


Figure 3.7: Tangent and approximating tangent.

Therefore, if we can find the slope of the approximating tangents then we will get an approximation of the slope of the tangent line. Computing the slope of the approximating tangent line is easy, we just need a “rise” and

a “run”. But the two points on the approximate line provide us with this. The vertical “rise” between the two points is $f(x+h) - f(x)$ while the horizontal “run” is h (we use the variable h to indicate that it is a *horizontal* increment). Therefore, the slope of the approximating tangent line is[†]

$$\frac{f(x+h) - f(x)}{h} \quad (1)$$

So, if the approximation to the slope gets better and better the closer h is to 0, then it must be best when h is 0. However, if we plug 0 in for h in the equation above we get $0/0$ – oops! – $0/0$ is not defined. What we need to do is track the values of (1) as h gets smaller and smaller and see where these values “tend”. This is the *approximate, then take a limit* move of the preface – the same move we used implicitly in chapter 1 when we replaced a slowly varying factor by a constant; this time we follow the approximation all the way to its limit.

For instance, given the sequence of numbers $1, 1/2, 1/3, \dots$, the 10th element in the sequence is $1/10$, the 103rd element in the sequence is $1/103$. In general, the n^{th} element of the sequence is $1/n$.

Question: Where is the sequence tending? Put another way, what is the *limit* of the sequence?

Answer: It seems clear that although this sequence never reaches 0 it is heading there.

Before continuing, let us introduce a notational convenience. To make it easier to talk about sequences we will use the notation $\{1/n\}_{n=1}^{\infty}$ to refer to the sequence $1, 1/2, 1/3, \dots$; and more generally, we use the notation from chapter 1 to refer to a sequence of values: $\{x_n\}_{n=1}^{\infty} = \{x_1, x_2, \dots\}$.

Now, referring to the sequence, $1, 1/2, 1/3, \dots$ ($\{1/n\}_{n=1}^{\infty}$ in our notation) we say that its limit is 0. This leads to a general operational definition of a limit.

Definition (Intuitive Limit). We say a sequence $\{x_n\}_{n=1}^{\infty}$ has the limit a if the sequence “tends to” a as n “tends to” ∞ . We write this with the following notation: $\lim_{n \rightarrow \infty} x_n = a$. We also say that x_n converges to a .

We leave for the time being what exactly “tends to” means. When Calculus was originally developed this notion was not rigorously understood. It was defined much the same way beauty is defined in many people’s minds: “I can’t define it, but I know it when I see it.” For a precise definition of limit see the appendix “Infinite Sequences and Limits”.

NOTE: In our definition of a limit, the limit point a is a number. It can’t be infinity or negative infinity. So given the sequence of numbers $1, 2, 3, \dots$, this sequence does not have a limit as it doesn’t tend to any number.

Now, going back to the approximating slopes we have our first major definition.

Definition (Slope of Tangent Line). The slope of the tangent line at a point $(x, f(x))$ of the function f exists if the limit

$$\lim_{n \rightarrow \infty} \frac{f(x+h_n) - f(x)}{h_n}$$

exists for every sequence $\{h_n\}_{n=1}^{\infty}$ such that h_n goes to 0 with $h_n \neq 0$ for any n ; and, every one of these limits is identical. In this case, the slope of the tangent line is this unique limiting value.

That is, if *every* sequence of approximating slopes tends to the same value, then the slope of the tangent line exists and is this value.

Discrete Connection: We may relate this “infinitesimal differencing” to the discrete differencing in chapter 1. Fix a step size $h \neq 0$ and sample f at the points $x, x+h, x+2h, \dots$; that is, define a sequence, named y , by $y_k = f(x+kh)$ and a sequence t , by $t_k = x+kh$. The approximate tangent slope $(f(x+h) - f(x))/h$ may be

[†] Some books will use Δx rather than h to indicate a small “run”. The approximating slope is then written as $\frac{f(x+\Delta x) - f(x)}{\Delta x}$.

written: $\frac{\Delta y_0}{\Delta t_0}$. When the step size is $h = 1$, this is *exactly* the difference operator Δ of chapter 1 applied to the samples of f (the “run”, Δt_0 , is just 1). The derivative is Δ run on a shrinking step: difference, divide by the step, and let the step tend to 0.

Note: This limit may not exist. There are many examples of functions where the limit does not exist at a particular point or a collection of points. Graphically this means the tangent line doesn’t exist or is not unique – how can this be? We list a few examples of functions with points on their graphs where the tangent line does not exist.

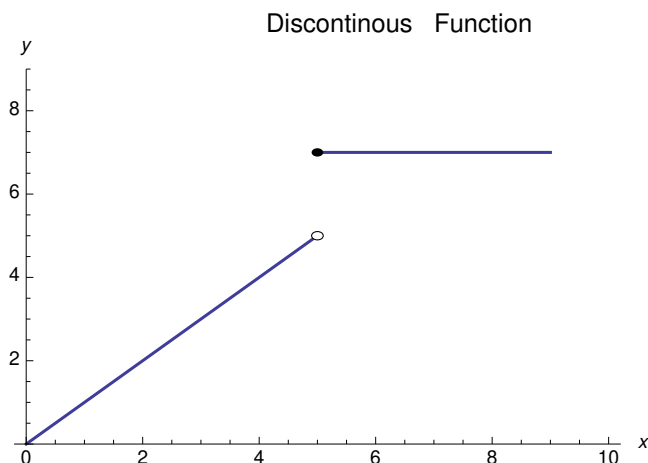


Figure 3.8: This function does not have a tangent line at $(5, f(5))$. All of the possible approximating slopes tend to ∞ – not a number.

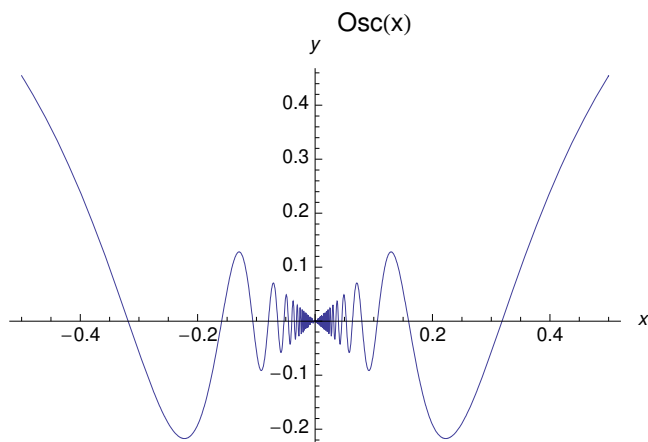


Figure 3.9: The function $\text{Osc}(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ has no tangent at $(0, \text{Osc}(0))$. That is, there are many tangents depending on the sequence of approximating tangents we choose – but no *UNIQUE* tangent.

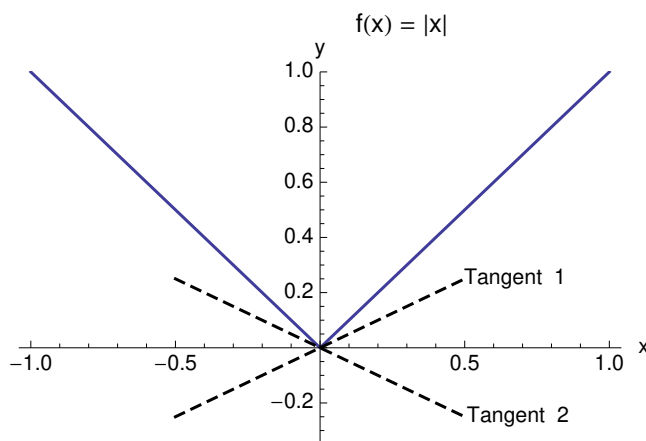


Figure 3.10: No *UNIQUE* tangent line at $(0, f(0)) = (0, 0)$. The two possible tangents according to our definition are the lines with slopes 1 and -1 that go through the origin. We note in the graphic that one may consider other lines as “tangents” – any line that can “teeter” on the graph at $x = 0$. This leads to the notion of a sub-differential – a set of “valid” tangents rather than just one. We will not consider this notion here, but it is interesting to note that the notion of a derivative is not necessarily set in stone.

Definition (Derivative). *The slope of the tangent line (when it exists) at a point $(x, f(x))$ is called the derivative of f at x and is denoted $f'(x)$. We take f' to mean the function whose value at any x is the derivative of f at x . The domain of this new function will be, in general, a subset of the domain of f .[‡] We call f' the derivative of f . We say that we differentiate f to get f' .*

We will also write $(f(x))'$ to denote $f'(x)$ when f is represented by an expression, or if it is more readable. For instance, $(g(x)h(x))'$ means the derivative of the function represented by the expression of g times h evaluated at x .

Recap

Recall that in our example problems we needed a function f' which would give the steepness of the tangent line at each point $(x, f(x))$ (now called the derivative). The idea was that f' would be easy to compute from f . From the last section, it seems things are getting more complicated. In the process of trying to find the slopes of the tangent lines, we introduced a fairly complicated notion of limit. Further, it seems we are back to having the same problem we had with sampling and graphing the function to find a tangent slope. In those methods one needed to sample the function at many points. In fact, for a completely accurate result all these methods require one sample f at an infinite number of points to find the slope of f at just one point x ! Worse, since we want the slope of the tangent line not just at a given point x , but at all points, things look grim.

The good news is that while the limiting process of finding tangent slopes (the derivative) is an infinite process, it turns out that for a large number of nice functions, f , one can write down a formula for f' at *any* point without doing an infinite amount of sampling.

Derivatives of Straight Line Functions

We start with straight line functions first because the tangent line at any point of a straight line function is the

[‡] See the appendix “Functions, Formulas, Graphs and Notation”.

straight line function itself. Therefore, we don't have to go through any complicated limiting process.

The simplest such straight line functions are the constant functions. A constant function is one which has only one output for all inputs. Such a function has the following formula:

$$f(x) = c$$

What is the slope of the tangent line at a given point x ? The graph of this function is a flat line whose height is c . Its tangent line is the line itself. We know a flat line has 0 as its slope. So, what is the value of the derivative function, f' for each x ? It is another constant function which is identically zero. That is, $f'(x) = 0$. Again, this says that the slope of the tangent line (derivative) at $(x, f(x))$ is just the slope of the graph of f at x , which has slope 0.

What about finding the derivative function for a general straight line function? See the appendix "Straight Line Functions" for more information about the formulas for straight line functions. In this case, the function has the formula: $f(x) = mx + b$. As in the constant function case the tangent line to each point $(x, f(x))$ is just the graph of f . Therefore, the slope of the tangent line (derivative) is just the slope of the graph of f which is m . So, $f'(x) = m$.

In each case we avoided the limiting process because the graph of the functions *were already* the tangents to the function. This is not so for more complicated functions.

Computing a Non-Trivial Derivative

From algebra class, the next functions studied after straight line functions are the quadratic functions. Let us choose the simplest quadratic, $f(x) = x^2$. Unlike straight line functions, the graph of this function is not the same as the tangent line at any point. To compute the derivative at a point x , we will have to use the approximating tangent lines to compute the answer. Let $\{h_n\}_{n=1}^{\infty}$ be a sequence of numbers which tend to 0 and have the property that $h_n \neq 0$ for all n . Let us compute, for a given n , the slope of the n th approximating tangent. This is given by $\frac{f(x+h_n)-f(x)}{h_n}$ which is

$$\frac{(x+h_n)^2 - x^2}{h_n}$$

Expanding the expression $(x+h_n)^2$ and simplifying yields:

$$\frac{2xh_n + h_n^2}{h_n}$$

Since h_n is not 0, we may cancel an h_n from the top and the bottom of the last expression. This yields

$$2x + h_n$$

What does this tend to as n tends to infinity? The limit is $2x$ since h_n goes to zero. Notice, that this is true for *every* such sequence $\{h_n\}_{n=1}^{\infty}$ that goes to zero and $h_n \neq 0$. In other words, we've found the derivative (slope of the tangent line) at *each point* x . The formula is $f'(x) = 2x$. Notice, too, that the expression $2x + h_n$ is the discrete answer: in chapter 1 we found $\Delta n^2 = 2n + 1$, which is the same expression with a step size of 1. The limit simply kills the step-size correction term, leaving $2x$. *Notice that in one calculation not only have we found a formula for the derivative at one particular point, but for all points!*

A Function without a Derivative

As an example of a function that is not differentiable at a point, consider the absolute value function, $f(x) = |x|$, from the previous figure. The figure demonstrates that there is not a *UNIQUE* tangent at $x = 0$. We now show that the derivative does not exist when $x = 0$ according to our definition. Let $h_n = 1/n$ and try to take the limit

of $\frac{f(0+h_n)-f(0)}{h_n}$. This is $\lim_{n \rightarrow \infty} (1/n - 0)/(1/n)$, since $h_n > 0$; which is $\lim_{n \rightarrow \infty} 1 = 1$. Now let's try a different sequence converging to zero. Let $h_n = -1/n$, then $\lim_{n \rightarrow \infty} \frac{f(0+h_n)-f(0)}{h_n} = (1/n - 0)/(-1/n)$, since $h_n < 0$; this is $\lim_{n \rightarrow \infty} -1 = -1$. Since different sequences converging to 0 provide different limiting slopes, the derivative does not exist for this f at 0.

In fact, one can show that if w is any sequence with $w_n > 0$ such that w_n goes to 0 as n goes to infinity then $\lim_{n \rightarrow \infty} \frac{f(0+w_n)-f(0)}{w_n} = 1$. And, if z is any sequence with $z_n < 0$ such that z_n goes to 0 as n goes to infinity then $\lim_{n \rightarrow \infty} \frac{f(0+z_n)-f(0)}{z_n} = -1$. For the derivative to exist at a given point we need the approximating slopes formed from *any* sequence h (with $h_n \neq 0$ such that h_n goes to 0 as n goes to infinity) to have the same limiting value. It is not enough to have it true for one sequence or a family of sequences, but for *all* sequences with the requisite properties.

However, we did find the derivative of $f(x) = x^2$; so, one function down, how many more to go? Well, there are an infinite number of functions and this was one of the easy ones. This may mean that given a function which has nice algebraic properties, we can do manipulations like we did above and get a formula for the derivative at all points of the function. This is indeed the case for many functions; the algebra just gets messier for more complicated functions. Can we do better than to do this limiting process again and again?

If we look at our function $f(x) = x^2$ with derivative function $f'(x) = 2x$ and then look at the related function $g(x) = x^2 + c$, where c is some constant, can we determine the derivative function, g' , from our knowledge of f and its derivative f' without going through the limiting process? Yes, the graph of the function g is just the graph of the function f shifted up by the amount c . That is, g has the same shape as f and so the slope of its tangent lines must be the same as f . We've just argued that $g'(x) = f'(x) = 2x$ without having to go through the limiting process. What about other functions which are simple modifications of f , can we find their derivatives so easily? For instance, consider the function $h(x) = 2x^2$. Can we relate the derivative h' to f' without going through a limiting process? The answer again is yes.

Differential Calculus: Computational Strategy

So, we see differential calculus comes down to finding the derivative function, f' , from a function f . Our plan to find the derivative of a function is: Compute the derivative of a base set of functions; we only have to do this once. Next, find formulas for computing the derivatives of functions composed in various ways from these base functions. The base functions and their compositions will be the only functions for which nice formulas will be found for their derivatives. That is to say, we are giving up on finding a formula for the derivative of an arbitrary function. This is not so bad since, hopefully, the pool of functions which we can differentiate will be big enough for the applications we consider. If not, then we will have to compute the derivative of a function the hard way; that is, use the limit process directly.

A Base – a Family of Functions and their Derivatives

We already saw how to find the derivative of the function $f(x) = x^2$ using the limit of tangent line approximations. We present formulas for derivatives of the family of *power* functions, $f(x) = x^n$ but omit the details.

Theorem(Derivative of Power Functions). *If f is the power function $f(x) = x^n$ with n an integer, then the derivative, f' , is given by*

$$f'(x) = nx^{n-1}$$

provided $x \neq 0$ when $n < 0$.

Notice that when n is zero the function is $f(x) = x^0 = 1$ and its derivative is $f'(x) = 0$. For a proof of this theorem see the applications section of the appendix on “Mathematical Induction”.

Compare this rule with the differences we computed in chapter 1: $\Delta n^2 = 2n + 1$ and $\Delta n^3 = 3n^2 + 3n + 1$. In each case the leading term follows the same pattern – $k n^{k-1}$ for the k^{th} power – while the straggler terms are step-size corrections, the very terms the limit kills off. The power function x^k plays the same role on the continuous board that n^k played on the discrete one, and its rule of differentiation is the discrete rule with the stragglers gone. (Recall from the aside on falling powers in the chapter on Summation Calculus that the discrete board has its own exact version of this rule: $\Delta n^{\underline{k}} = k n^{\underline{k-1}}$.)

Exercise 3.1: Show this result in the case when $n = 3$. That is, derive the derivative of the function $f(x) = x^3$ using the definition of the derivative.

Linearity of Differentiation

A common way of constructing new functions from others is through what is called a *linear combination*. Given functions $f_1, f_2, \dots, f_n$ a new function f is a linear combination of the functions $\{f_i\}_{i=1}^n$ with constants $\{c_i\}_{i=1}^n$ if

$$f(x) = c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)$$

As an example consider $f_i(x) = x^{i-1}$, for $i \in \{1, 2, 3\}$. Then the function

$$p(x) = 2 + 3x + 5x^2$$

is a linear combination of the functions $f_1(x) = 1$, $f_2(x) = x$, and $f_3(x) = x^2$. The coefficients of the combination (the c 's) are 2, 3, and 5.

We are able to form a much larger collection of functions from a set of base functions in this way. And, it turns out one can easily compute the derivative of the new function in terms of the derivatives of its component functions.

Theorem (Linearity of Differentiation). *If the functions f'_1 and f'_2 exist and $f(x) = c_1 f_1(x) + c_2 f_2(x)$, then the derivative, f' , of f exists and may be computed as*

$$f'(x) = c_1 f'_1(x) + c_2 f'_2(x)$$

Note that this means that $(cf)' = cf'$. This follows by letting $f_2(x) = 0$ in the above theorem. This result may be extended to an arbitrary linear combination of functions.

Corollary (Linearity of Differentiation). *If the functions: $f'_1, f'_2, \dots, f'_n$ exist and $f(x) = c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)$, then the derivative, f' , of f exists and may be computed as*

$$f'(x) = c_1 f'_1(x) + c_2 f'_2(x) + \dots + c_n f'_n(x)$$

This follows from the previous theorem and mathematical induction.[†] The theorem and corollary say that the operation of differentiation is linear. This is directly analogous to the linearity result for the difference operation, Δ , of chapter 1. In effect, differentiation (like differencing) passes through constants and sums. In the language of the preface, this is the *exploit linearity* move on the continuous board: to differentiate a complicated function, break it into pieces you can differentiate, and add the answers.

We show this for the case when f is the sum of two functions; that is $f(x) = c_1 f_1(x) + c_2 f_2(x)$. Let h be a sequence with the property that $h_n \neq 0$ and h_n tends to 0 as n tends to infinity. This sequence allows us to form approximating tangent lines whose slopes at $(x, f(x))$ are $\frac{f(x+h_n)-f(x)}{h_n}$. This is related to approximating tangent slopes of f_1 and f_2 as follows:

$$\begin{aligned} \frac{f(x+h_n)-f(x)}{h_n} &= \frac{(c_1 f_1(x+h_n) + c_2 f_2(x+h_n)) - (c_1 f_1(x) + c_2 f_2(x))}{h_n} \\ &= c_1 \left(\frac{f_1(x+h_n) - f_1(x)}{h_n} \right) + c_2 \left(\frac{f_2(x+h_n) - f_2(x)}{h_n} \right) \end{aligned}$$

[†] See the appendix “Mathematical Induction”.

Taking the limit of both sides of this equation as h_n goes to 0 gives[‡]

$\lim_{n \rightarrow \infty} \left(\frac{f(x+h_n) - f(x)}{h_n} \right) = c_1 f'_1(x) + c_2 f'_2(x)$. Since this limit exists and is the same for *all* such sequences h the derivative of f at x exists and is this value. That is, $f'(x) = c_1 f'_1(x) + c_2 f'_2(x)$.

Example 3.1: Let's use this theorem to find the derivative of the polynomial $p(x) = 2 + 3x + 5x^2$.

The theorem says that if we can *recognize* p as a linear combination of functions for which we know how to compute the derivative, then we can easily compute p' . We see that $p(x) = 2f_1(x) + 3f_2(x) + 5f_3(x)$, with $f_1(x) = 1$, $f_2(x) = x$, and $f_3(x) = x^2$. Therefore, $p'(x) = 2f'_1(x) + 3f'_2(x) + 5f'_3(x) = 2 \cdot 0 + 3 \cdot 1 + 5 \cdot 2x = 3 + 10x$.

The linearity theorem combined with the rule for differentiating powers allows us to compute the derivative of any polynomial!

Theorem(Polynomial Differentiation). Let $p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$, (n a non-negative integer) then

$$p'(x) = c_1 + 2c_2x + 3c_3x^2 + \cdots + nc_nx^{n-1}$$

This result follows easily because of the linearity of differentiation; we need only know how to find the derivative of each term. But each term is just a power function whose derivative we know.

Armed with our ability to compute derivatives for our base family – the polynomials, let's go back and solve the original problem we posed about the metal box container.

Application Revisited

Consider again the original problem of the metal box container. The problem was to maximize the volume of a box made from a piece of sheet metal. The volume of the box was a function of the “flap distance” x . A formula for the volume was $V(x) = x(12 - 2x)^2$.

We will solve this again using our knowledge of how to compute derivatives. We proceed by finding all solutions to $V'(x) = 0$ combined with the end points of our search and then determine a value which maximizes V .

Now, V is a polynomial since, after expanding the formula for V we have: $V(x) = x(144 - 48x + 4x^2) = 144x - 48x^2 + 4x^3$. But we know now how to compute V' when V is a polynomial. According to the last theorem it is $V'(x) = 144 - 2 \cdot 48x + 3 \cdot 4x^2$; or, $V'(x) = 144 - 96x + 12x^2$. Recall the point of having V' : *By finding those inputs, x , such that $V'(x) = 0$ we have a list of places where V is flat and therefore a potential maximum.* Solving the equation, $V'(x) = 0$, for x implies that x satisfies $144 - 96x + 12x^2 = 0$. Or, $12 - 8x + x^2 = 0$. Again, the two solutions are 2 and 6. To these values we merge the set of end points, $\{0, 6\}$, to give the candidate input set, $\{0, 2, 6\}$. Plugging in 0 and 6 yields 0. Plugging in 2 produces 128. Therefore, $x = 2$ is the input that gives the maximum value of V , the same result as before.

One may argue that with the advent of computers it might be easier to find this solution by brute force. This is a reasonable argument. However, what if someone asked to maximize the volume that could be produced from square sheets of metal other than 12 inches? One could resolve this problem using Calculus with a parameter representing the size of the square. The resulting formula would allow you to solve an entire *family* of problems. This is something the brute force method cannot do.

Exercise 3.2: A farmer has 100 feet of fencing and wants to enclose a rectangular pen against a straight river bank; the side along the river needs no fence. Express the enclosed area as a function of the side length perpendicular to the river, and use the candidate-input strategy of this chapter to find the dimensions giving the largest area.

[‡] Passing the limits through constant multipliers is discussed in the section “Computational Strategy for Limits” in the appendix “Infinite Sequences and Limits”.

Enlarging the Class of Differentiable Formulas

For our derivative technique to be useful for more complicated maximization problems we need to enlarge the class of functions, f , for which we can find a formula for f' . As we know from our strategy we need to do two things: Enlarge the set of base functions for which we know how to compute the derivative; and, find ways to combine these functions in a way that we can determine the derivative from the component functions. How else can we form more complex functions, f , from simpler functions, g , and h ? Here are three ways:

1. $f(x) = g(x)h(x)$
2. $f(x) = g(x)/h(x)$
3. $f(x) = g(h(x))$

It turns out one can find the derivative of functions formed in the ways enumerated above in terms of the derivatives of the functions used to compose them. This is just what we need: a function built up from simple pieces – like $(1+x^2)^7$, which we will meet shortly – would be hopeless to differentiate from the definition alone; the rules below let us differentiate combinations in terms of their parts. Here are the three corresponding theorems for computing their derivatives. We also restate the theorem on linear combinations for completeness.

Theorem(Linearity Rule). If $g'_1(x), g'_2(x), \dots, g'_n(x)$ exist and $f(x) = c_1g_1(x) + c_2g_2(x) + \dots + c_ng_n(x)$, then $f'(x)$ exists and

$$f'(x) = c_1g'_1(x) + c_2g'_2(x) + \dots + c_ng'_n(x)$$

Theorem(Product Rule). If $g'(x)$ and $h'(x)$ exist and $f(x) = g(x)h(x)$, then $f'(x)$ exists and

$$f'(x) = g'(x)h(x) + g(x)h'(x)$$

Theorem(Quotient Rule). If $g'(x)$ and $h'(x)$ exist, $h(x) \neq 0$, and $f(x) = g(x)/h(x)$, then $f'(x)$ exists and

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h^2(x)}$$

Theorem(Chain Rule). If $h'(x)$ and $g'(h(x))$ exist and $f(x) = g(h(x))$, then $f'(x)$ exists and

$$f'(x) = g'(h(x))h'(x)$$

To make the chain rule plausible, write $u = h(x)$ and $y = g(u) = f(x)$ and think of an approximating slope as a ratio of small changes; then $\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \frac{\Delta u}{\Delta x}$, and letting the step shrink, the two factors on the right tend to $g'(h(x))$ and $h'(x)$. A careful proof, however, must handle a subtlety this argument hides – the inner change Δu can be zero even when Δx is not, making the ratio $\Delta y/\Delta u$ meaningless – and is beyond the scope of this book.

We derive the product rule as it is similar to the product rule in the Discrete Calculus. Let h be a sequence with $h_n \neq 0$ such that h_n goes to 0 as n goes to infinity. Given that $f(x) = g(x)h(x)$, consider the slope of an approximating tangent to f at x based on the sequence h : $\frac{f(x+h_n)-f(x)}{h_n}$. There is a naming problem here. We have been using a sequence named h in our approximating slopes but the function f is composed of a function with the name h . What do we do? Well, we could press on and not let it bother us because a subscripted h is an element of the sequence h while $h(x)$ means the function h evaluated at x . Or, to be more clear we could change the name of either the sequence or the function h . We will change the name of the sequence. There is nothing special in the name of this sequence. All that is important is that its elements are not 0 and that the sequence tends to 0. So, let's rename the sequence h to z . An approximating tangent slope of f at x is $\frac{f(x+z_n)-f(x)}{z_n}$. We

now try to relate this to the approximating tangent slopes of g and h .[†]

$$\begin{aligned}\frac{f(x+z_n) - f(x)}{z_n} &= \frac{g(x+z_n)h(x+z_n) - g(x)h(x)}{z_n} \\ &= \frac{g(x+z_n)h(x+z_n) - g(x)h(x+z_n) + g(x)h(x+z_n) - g(x)h(x)}{z_n} \\ &= \left(\frac{g(x+z_n) - g(x)}{z_n} \right) h(x+z_n) + g(x) \left(\frac{h(x+z_n) - h(x)}{z_n} \right)\end{aligned}$$

Taking the limit as n tends to infinity of the equation above gives:

$\lim_{n \rightarrow \infty} \left(\frac{f(x+z_n) - f(x)}{z_n} \right) = g'(x) \lim_{n \rightarrow \infty} h(x+z_n) + g(x) h'(x)$. We are assuming that the limit of $h(x+z_n)$ exists and are using the properties of the limiting operation as described in the section “Computational Strategy for Limits” in the appendix, “Infinite Sequences and Limits”. It turns out that a differentiable function is a continuous function and this means[‡] that $\lim_{n \rightarrow \infty} h(x+z_n) = h(\lim_{n \rightarrow \infty} (x+z_n)) = h(x)$. Finally, since this limit exists for *all* such sequences z , the derivative of f exists and is equal to this value. That is, $f'(x) = g'(x)h(x) + g(x)h'(x)$.

The quotient rule need not be taken on faith either – it is the product and chain rules in disguise, and the reader owns every piece needed to unmask it. Write the quotient as a product: $f(x) = g(x)/h(x) = g(x)h(x)^{-1}$. The inner function of $h(x)^{-1}$ is h and the outer function is the power function z^{-1} , whose derivative, by the power rule, is $-z^{-2}$. So the chain rule gives

$$(h(x)^{-1})' = -h(x)^{-2} h'(x) = -\frac{h'(x)}{h^2(x)}$$

and then the product rule finishes the job:

$$f'(x) = g'(x)h(x)^{-1} + g(x) \left(-\frac{h'(x)}{h^2(x)} \right) = \frac{g'(x)h(x) - g(x)h'(x)}{h^2(x)}$$

which is exactly the quotient rule stated above. (This is the *change the representation* move in miniature: rewrite the quotient as a product, and tools we already trust take over.)

So, given a function f , we can find its derivative if we can recognize it as being either:

1. One of the base functions for which we know the derivative. For us, this is the set of all polynomials and power functions.
2. Composed (in one of the ways above) of functions for which one knows the derivatives.

Keep in mind, there may be many ways to go about computing derivatives. It’s really a kind of pattern recognition.

Example 3.2: Suppose $f(x) = (1+x+5x^2)(1-x^2)$. Find $f'(x)$.

We first check to see if this function is one of our base functions. Currently, we only recognize the set of polynomials as our base set. So, this is not a base function. Next, we check to see if f can be decomposed into one of the above patterns. Yes, f can be written as the product of two functions, g and h , where $g(x) = (1+x+5x^2)$ and $h(x) = 1-x^2$. Do we know how to compute the derivative of g or h ? Yes, each of these are polynomials and we have a formula for computing their derivatives. So, using the product rule we have

$$\begin{aligned}f'(x) &= g'(x)h(x) + g(x)h'(x) \\ &= (1+x+5x^2)'(1-x^2) + (1+x+5x^2)(1-x^2)' \\ &= (0+1+2*5x)(1-x^2) + (1+x+5x^2)(0-2x) \\ &= (1+10x)(1-x^2) - 2x(1+x+5x^2)\end{aligned}$$

Of course, another way to compute the derivative would be to multiply out the product of g and h . This would give us a polynomial and we could use the formula for differentiating polynomials.

[†] The term $g(x)h(x+z_n)$ is subtracted and added in an analogous way to the derivation of the product rule in the Discrete Calculus.

[‡] Also discussed in the section “Computational Strategy for Limits” in the appendix “Infinite Sequences and Limits”.

Example 3.3: Let $f(x) = 1 + x + x^2 - 2x$. Find $f'(x)$.

We can find the derivative in one of two ways. First, f is a polynomial isn't it? Well, no, not in the form our theorem on polynomial differentiation requires. Our theorems refer to polynomials as being a linear combination of *DISTINCT* power functions. In the case of f , the “ x ” term appears twice; first as x and then later as $-2x$. It is easy enough to turn this into a “canonical” polynomial, just add like powers. f becomes: $f(x) = 1 - x + x^2$. Now we can apply the theorem for computing derivatives of polynomials giving us: $f'(x) = -1 + 2x$. Another way to compute this is to use the linearity theorem which just distributes the derivative through sums and products with constants. It says $f'(x) = 1' + x' + (x^2)' - 2x'$. Applying the rule for differentiating powers and constants gives us $f'(x) = 0 + 1 + 2x - 2 = -1 + 2x$. This agrees with our previous computation.

These rules are usually easy to apply. The only one which is harder to recognize is the chain rule. Here is an example of how it is used. Let $f(x) = (1 + x^2)^7$. What is the derivative of f ? Again, in this case we could raise $(1 + x^2)$ to the 7th power to give us a polynomial and then write down its derivative. We can also recognize f as a composition of the function $g(x) = x^7$ with the function $h(x) = 1 + x^2$. That is, $f(x) = g(h(x))$. Applying the chain rule for the composition of two functions gives

$$\begin{aligned} f'(x) &= g'(h(x))h'(x) \\ &= 7(1 + x^2)^6(1 + x^2)' \\ &= 7(1 + x^2)^6(0 + 2x) \\ &= 7(1 + x^2)^6 2x \\ &= 14x(1 + x^2)^6 \end{aligned}$$

Notice that $g'(x) = 7x^6$ and therefore, $g'(h(x)) = 7h(x)^6$. Since $h(x) = (1 + x^2)$ it follows that $g'(h(x)) = 7(1 + x^2)^6$.

Exercise 3.3: Find $f'(x)$ for each of the following: (1) $f(x) = (1 + x^3)^{10}$; (2) $f(x) = x^2(1 + x)^5$; (3) $f(x) = \frac{x}{1 + x^2}$. In each case, first decide *which* rule applies.

Enlarging the Base

So far, our collection of functions for which we know the derivative is the collection of power functions, $f(x) = x^n$, for integer n , and all polynomials. Why enlarge it? Because nature keeps handing us functions that are not polynomials. Growth processes – money compounding in a bank account, a population of bacteria doubling again and again – force exponential functions on us, and the chapter ahead on differential equations will need their derivatives. We now enlarge this base collection to include formulas for computing the derivatives of logarithmic functions (and their inverses) as well as trigonometric functions. Along the way the next subsections help to explain the number e that you may have heard of (the story of the angle measure called radians is told alongside the trigonometric limits in the appendix “Infinite Sequences and Limits”). The *derivations* in the next two subsections may be skimmed on a first reading without harm to the essential aspects of Calculus; the *results*, however, will be used – the log and exponential functions in the chapter on differential equations, and the sine and cosine derivatives as soon as the chapter on Integral Calculus.

Derivatives of Log and Exponential Functions

Let's try to compute the derivative at a given point x of the log function base b , $\log_b$. The logarithm only makes sense for positive inputs, so throughout our computations, x is assumed positive. The derivative of this function at x is the limit – if it exists – of $\lim_{n \rightarrow \infty} (\log_b(x + h_n) - \log_b(x)) / h_n$. Here $\{h_n\}_{n=1}^{\infty}$ is a sequence which converges to 0 with the property that $h_n \neq 0$ for all n . This may be written as $(\log_b(x(1 + h_n/x)) - \log_b(x)) / h_n$. Using the property of logs, this may be written: $(\log_b(x) + \log_b(1 + h_n/x) - \log_b(x)) / h_n$; or, $\log_b(1 + h_n/x) / h_n$. Using the property of logs one more time we may write this as $\log_b(1 + h_n/x)^{1/h_n}$; or, $\log_b[(1 + h_n/x)^{x/h_n}]^{1/x}$. We need to find the limit of this as n goes to ∞ . Since the logarithm and the power function $z(s) = s^{1/x}$ are continuous

for $x > 0$, we may pass the limit through these functions[‡] yielding $\log_b [\lim_{n \rightarrow \infty} (1 + h_n/x)^{x/h_n}]^{1/x}$. It turns out that the limit inside the log (regardless of $x > 0$) is the same as the limit: $\lim_{n \rightarrow \infty} (1 + 1/n)^n$ – the reduction to this special sequence is a step we take on faith here. That this special limit *exists* is shown in the section “An Application: The Number e ” of the appendix “Infinite Sequences and Limits”, by way of the Monotone Sequence Theorem. Its limiting value is called e which is approximately 2.718281828459045. So the limit becomes $\log_b (e^{1/x}) = (1/x) \log_b(e)$. Therefore, we have the formula:

$$(\log_b(x))' = (1/x) \log_b(e)$$

Notice the messy constant $\log_b(e)$ tagging along; it depends on our choice of base b . The number e is precisely the base for which this constant equals 1, so that the derivative comes out clean – we will make the same kind of choice later when we pick radians as our angle measure. If we choose the base of the logarithm to be e , then

$$(\log_e(x))' = (1/x) \log_e(e) = 1/x$$

The function, $\ln$, is the name used to denote $\log_e$.

We can use this knowledge to compute the derivative of $\ln$'s inverse, the function e^x . By definition, $\ln(e^x) = x$, taking the derivative of this equation yields $(\ln(e^x))' = x' = 1$. Or,

$$1 = (\ln(e^x))' = \underbrace{\frac{1}{e^x}}_{h'(g(x))} \underbrace{(e^x)'}_{g'(x)}$$

The last equality follows from the chain rule with $h(x) = \ln(x)$ and $g(x) = e^x$. Rewriting, this becomes:

$$(e^x)' = e^x$$

We repeat this procedure for b^x , where b is any positive number. Taking the derivative of the identity $\log_b(b^x) = x$, gives $(\log_b(b^x))' = x' = 1$. Or,

$$1 = (\log_b(b^x))' = \underbrace{\frac{1}{b^x} \log_b(e)}_{h'(g(x))} \underbrace{(b^x)'}_{g'(x)}$$

The last equality follows from the chain rule with $h(x) = \log_b(x)$ and $g(x) = b^x$. Rewriting, this becomes: $(b^x)' = \frac{1}{\log_b(e)} b^x$. Since, $1/\log_b(e) = \log_e(b) = \ln(b)$, we may write this last expression as

$$(b^x)' = \ln(b) b^x$$

We finish this section with a formula for computing the derivative of the function e^{ax} . This is really the composition of functions $h(x) = e^x$ and $g(x) = ax$; that is, $e^{ax} = h(g(x))$. By the chain rule the derivative is $h'(g(x)) g'(x)$. Since $h'(x) = (e^x)' = e^x$ and $g'(x) = a$ it follows that $(e^{ax})' = e^{ax} a = a e^{ax}$. This formula will be of use when we get to the chapter on differential equations. We list our main results:[†]

1. $\ln'(x) = \frac{1}{x}$
2. $(e^x)' = e^x$
3. $(e^{ax})' = a e^{ax}$

[‡] A continuous function, f , has the following property: If $\lim_{n \rightarrow \infty} x_n = x$ then $\lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n) = f(x)$. We say in this case that the limit passes through f . See again, appendix C for a discussion of continuous functions.

[†] The domain of $\ln'$ is the same as $\ln$ which is $\{x \mid x > 0\}$.

Derivatives of Trig Functions

Why should the trigonometric functions join our base collection? Because they are the functions that *oscillate* – and nature is full of oscillation: a weight bobbing on a spring, a pendulum swinging, alternating current in a wire. At the close of the chapter on differential equations we will meet the equation of the swinging pendulum, $x'' + x = 0$, and check that sine and cosine are the functions that solve it; to be ready for that we need their derivatives now.

Let us attempt to compute the derivative of the sine function, $\sin$, at a point x . Again, let $\{h_n\}_{n=1}^\infty$ be a sequence which converges to 0 with the property that $h_n \neq 0$ for any n . For any n , the approximating slope to the tangent line is

$$\begin{aligned} \frac{\sin(x + h_n) - \sin(x)}{h_n} &= \frac{\overbrace{\sin(x) \cos(h_n) + \cos(x) \sin(h_n)} - \sin(x)}{h_n} \\ &= \sin(x) \frac{\cos(h_n) - 1}{h_n} + \cos(x) \frac{\sin(h_n)}{h_n} \end{aligned}$$

We need to be able to find the limits (if they exist) of $\frac{\sin(h_n)}{h_n}$ and $\frac{\cos(h_n)-1}{h_n}$. The picture below shows what is at stake in the first of these limits. When the angle θ is small, the vertical segment BC (whose length is $\sin(\theta)$) and the arc DC (whose length is θ , in radian measure) hug each other. In fact, one might conclude that in the limit their ratio $\frac{\sin(\theta)}{\theta}$ is 1.

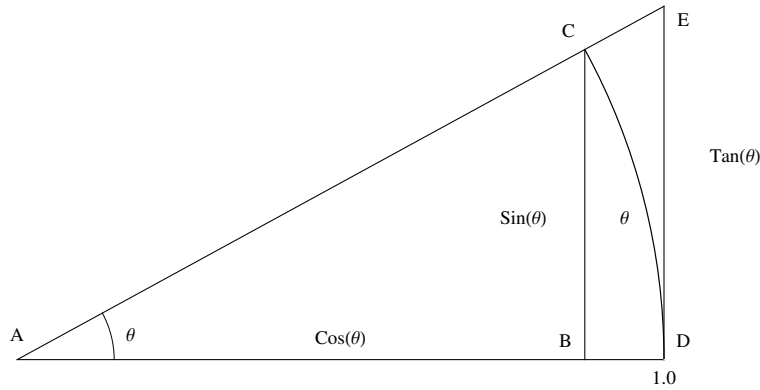


Figure 3.11: Comparison of triangle ABC, sector ADC, and triangle ADE.

This conclusion is correct. The precise argument – comparing the areas of the triangles ABC and ADE with the area of the sector ADC and squeezing – is a fine application of the Squeeze Limit Theorem, and it may be found, along with a discussion of radian measure, in the section “Two Trigonometric Limits” of the appendix “Infinite Sequences and Limits”. The radian is to angles what the base e was to logarithms: precisely the choice of measure that makes these limits – and with them the derivative formulas below – come out clean, with no messy constants tagging along. The second limit follows from the first (also shown in the appendix). The results are:

$$\lim_{n \rightarrow \infty} \frac{\sin(h_n)}{h_n} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{\cos(h_n) - 1}{h_n} = 0$$

Applying these limits to calculate the derivative of sine we have:

$$\begin{aligned} \lim_{n \rightarrow \infty} (\sin(x + h_n) - \sin(x)) / h_n &= \sin(x) \lim_{n \rightarrow \infty} \frac{(\cos(h_n) - 1)}{h_n} + \cos(x) \lim_{n \rightarrow \infty} \frac{\sin(h_n)}{h_n} \\ &= \sin(x) * 0 + \cos(x) * 1 \\ &= \cos(x) \end{aligned}$$

Therefore, we have shown that $\sin'(x) = \cos(x)$.

The derivative of the cosine function can be done now without much effort.

$$\begin{aligned}\frac{\cos(x + h_n) - \cos(x)}{h_n} &= \frac{\overbrace{\cos(x) \cos(h_n) - \sin(x) \sin(h_n)}^{\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)} - \cos(x)}{h_n} \\ &= \cos(x) \frac{\cos(h_n) - 1}{h_n} - \sin(x) \frac{\sin(h_n)}{h_n}\end{aligned}$$

Taking the limit we have $\cos'(x) = \cos(x) * 0 - \sin(x) * 1 = -\sin(x)$.

The other trig functions can be described in terms of sin and cos so we may find their derivatives through the derivatives of sin and cos. We list the derivatives of these trig functions and restate the results for sin and cos:

1. $\sin'(x) = \cos(x)$
2. $\cos'(x) = -\sin(x)$
3. $\tan'(x) = \sec^2(x)$
4. $\sec'(x) = \sec(x) \tan(x)$
5. $\csc'(x) = -\csc(x) \cot(x)$
6. $\cot'(x) = -\csc^2(x)$

To see how these formulas are derived, we compute the derivative of the tangent function, $\tan$. Since $\tan$ is the ratio of sin to cos we can use the quotient rule to compute its derivative. The quotient rule says $(f/g)' = (f'g - fg')/g^2$. In our case, $f = \sin$ and $g = \cos$. Plugging these into the quotient rule gives

$$\begin{aligned}\tan'(x) &= (\sin(x)/\cos(x))' \\ &= (\sin'(x)\cos(x) - \sin(x)\cos'(x))/\cos^2(x) \\ &= (\cos^2(x) + \sin^2(x))/\cos^2(x) \\ &= 1/\cos^2(x) \\ &= \sec^2(x)\end{aligned}$$

This is not valid for $x = \frac{\pi}{2}$; in fact, it is not valid for any x of the form $x = k \frac{\pi}{2}$ (k an odd integer). This is because $\tan$ is not defined at these values.

Just as was the case with the logarithmic functions, there are inverses of each of the trig functions. Some care must be taken to describe the domains of such inverses, but the approach to compute the derivatives of these inverses is the same as with the logarithms. We ask the reader to consult a calculus tome for a discussion and listing of these functions and their derivatives.

An Interpretation of the Derivative

Let d and v be the functions which give the distance and velocity, respectively, of a car as a function of time. Let's start by assuming that the car is traveling at a constant 30 mph. We know from simple physics that when the velocity is constant the distance function is given as $d(t) = v_0 t$, where v_0 is the velocity. This is simply the statement distance = rate $\times$ time. In our case this becomes: $d(t) = 30t$. What do the graphs of d and v look like?

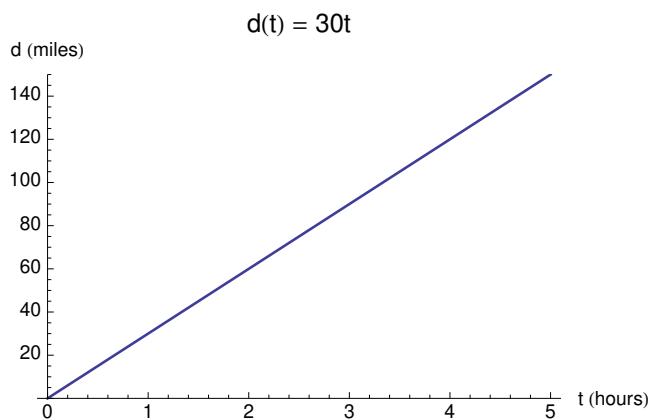


Figure 3.12: Distance graph when car is traveling 30 mph.

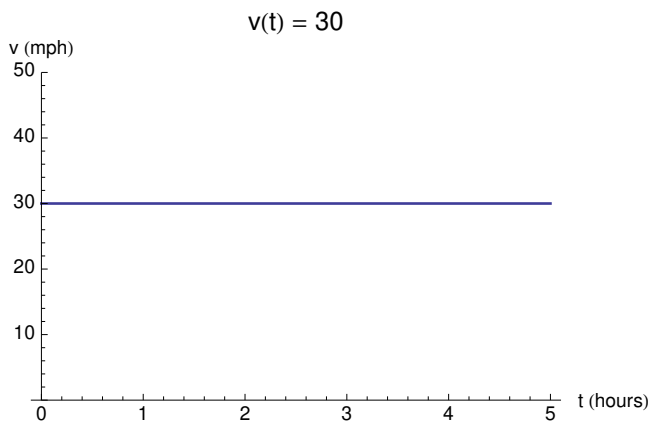


Figure 3.13: Velocity graph when car is traveling 30 mph.

Another way to say this is that the velocity = distance $\div$ time. This last statement translates geometrically as follows: The slope of the distance graph is 30. More generally, for a car traveling at a constant velocity v_0 , the slope of the distance graph is v_0 . That is, at any point on the distance graph, $(t, d(t))$, its slope is the velocity at time t , $v(t)$.

What about more general velocity and distance functions. Can we relate the distance function, $d(t)$, to the velocity function, $v(t)$ as we did above? The answer is yes. Take a look at the distance and velocity graphs below.

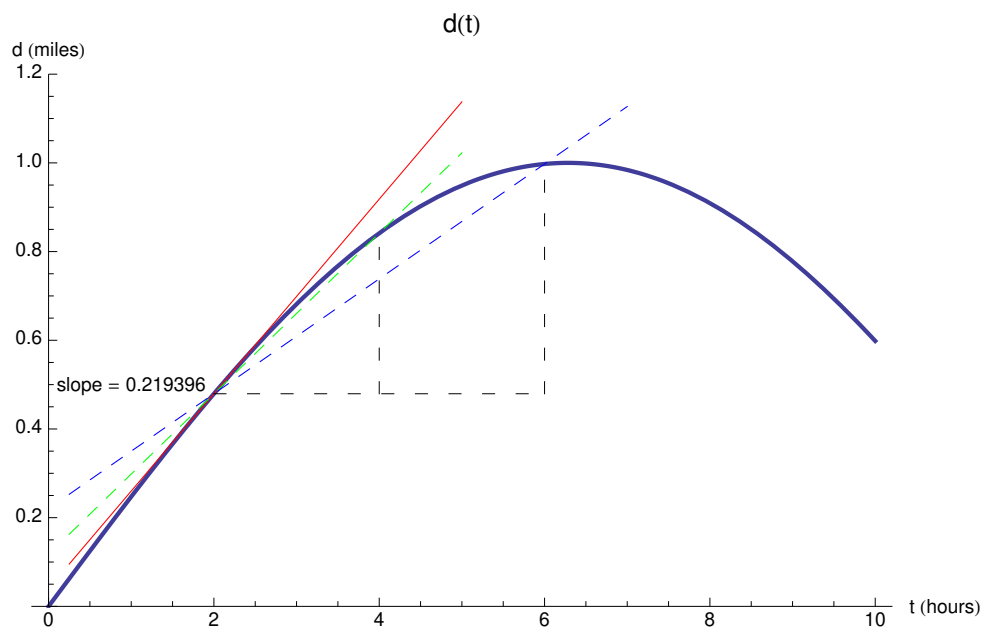


Figure 3.14: Distance function.

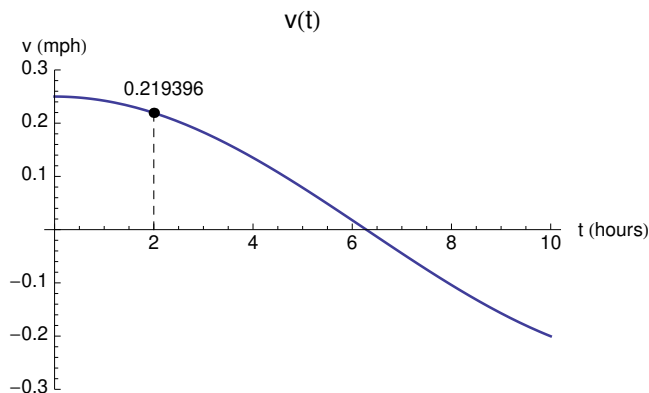


Figure 3.15: Velocity function.

The tangent line approximations to the slope at a point in the distance graph are changes in the distance function divided by the time interval. This gives the average velocity over that time interval. As the approximations get better, the intervals of time shrink. In the limit we get the velocity at that instant of time; that is

$$d'(t) = v(t)$$

Using the same arguments one can show the relation between velocity, v , and acceleration, a :

$$v'(t) = a(t)$$

An Application To Physics

If you throw a ball up in the air with an initial velocity v_0 (we take positive values to mean up), how high up will the ball travel? This is a maximization problem again; we're maximizing the height a ball travels. From physics, discounting air resistance, the distance as a function of time is: $d(t) = v_0 t - \frac{1}{2}gt^2$, where $d(t)$ is the distance in feet at time t in seconds and g is the constant 32 ft/sec^2 . To solve this problem we need to produce candidate inputs

for the maximum. This is done by finding the solution set, $\{t \mid d'(t) = 0\}$, and adding to this set the end points. There is only one end point, 0; we are not constrained by any end time condition. The solutions set $\{t \mid d'(t) = 0\}$ is the set of solutions to $d'(t) = (v_0 t)' - (\frac{1}{2}gt^2)' = v_0 - gt = 0$. There is only one solution which is, $t = v_0/g$. So, the set of input candidates is $\{0, v_0/g\}$. Plugging 0 in for t gives $d(0) = 0$ as it should. Plugging in v_0/g in for t gives $d(v_0/g) = v_0^2/g - gv_0^2/(2g^2) = v_0^2/g - v_0^2/(2g) = v_0^2/(2g)$. If v_0 is positive, then this time corresponds to the maximum height. Positive means throwing the ball up with a nonzero speed. So, if we throw the ball up with speed 80ft/sec, (about 55 mph) then the time which maximizes the height is $80/g$; this is $80/32 = 2.5$ seconds. Plugging in this time into the distance function gives a maximum height of $80^2/(2 * 32) = 6400/64 = 100$ feet.

Where did this distance function come from? Can we derive this formula ourselves with what we know? The answer is yes. The earth's gravitation pull is constant. This is what Galileo observed: acceleration is constant for any object. For our ball this means

$$a(t) = -g$$

To find the corresponding velocity we need to find a function, v , such that $v' = a$. Notice this is the opposite of differentiation. We need to find an *anti-derivative* of a .[†] This is the *isolate and invert* move of the preface once more: differentiation is the operation, and anti-differentiation is its inverse. A function which when differentiated is a constant, c , is ct . Therefore, $v(t) = -gt$ works. We're using the power rule in reverse. One can easily check its derivative is $-g$. But is this the only function which when differentiated gives $-g$? No, anti-derivatives are not unique. If you remember in an earlier section we argued that if h is a function with derivative, h' , then h_1 , defined by $h_1(x) = h(x) + c$ also has derivative h' . The idea was that the constant just shifted the graph of h up or down and didn't change h 's shape and therefore the slopes of the tangent lines were unchanged. So, we claim that $v(t) = v_0 - gt$ is also a solution to $v' = a$. In this case, $v(t)$ not only has the correct acceleration but also the correct initial velocity. Now, we also know that $d' = v$. So by the same arguments we need to find a function whose derivative is $v_0 - gt$. Using the power rule in reverse again, we see that d could be any one of the functions in the family $c + v_0 t - \frac{1}{2}gt^2$. The constant c is chosen so that the initial distance is right. Since our distance function is measured from the ground, initially our distance must be 0. That is, $d(0) = c + v_0 * 0 - \frac{1}{2}g * 0^2 = c = 0$. So, $d(t) = v_0 t - \frac{1}{2}gt^2$. We've just derived the distance function used in physics to describe an object under the influence of gravity – not a small feat.

Summary

We motivated the definition of the derivative through a maximization problem over an interval.

1. Finding the maximum potentially means sifting through infinitely many points – **this is hard/impossible**.
2. Found a *geometric* way to reduce this search to just a handful of points for smooth functions. The candidate solutions come from the end points of the interval and the points where the graph of the function is flat.
3. Defined the derivative as a measure of steepness of the graph - this measure is 0 at the flat points.
4. Found an *algebraic* way to compute the derivative (at *all* points) for a certain class of functions.
5. *Extended* this class to a much larger one by finding formulas for the derivative of linear combinations, products, division, or composition of functions we already know how to differentiate.
6. Now finding the maximum of a *nice* function can be found from a handful of candidate values: the end points and the points where $f'(x) = 0$ – **this is often easy**.

Finally, notice that this whole chapter has been the *same game, bigger board* theme of the preface at work – each tool of the Discrete Calculus has a continuous counterpart:

[†] This is analogous to finding an *anti-difference* of a sequence.

Discrete	Continuous
$\Delta x_n = x_{n+1} - x_n$	$f'(x) = \lim_{n \rightarrow \infty} (f(x + h_n) - f(x))/h_n$
Δn^k : leading term $k n^{k-1}$	$(x^k)' = k x^{k-1}$
linearity of Δ	linearity of differentiation
Discrete Product Rule	Product Rule
anti-difference (summation)	anti-derivative (integration – next chapter)

Chapter 4: Integral Calculus

Fundamental Theorem of Calculus

Keeping with the car theme, let $x(t)$ and $v(t)$ represent the distance and velocity of a car at time t . We found a connection between these two functions in the last chapter: the “instantaneous” difference of $x(t)$ over the “instantaneous” difference in time is $x'(t) = v(t)$. We now explore the relationship between these two functions over an *interval* of time. Recall that a similar discrete relationship led to the Fundamental Theorem of Difference Calculus. If $\Delta x_n = v_n$, then the sum over the “time” values, $0, 1, \dots, N-1$, of the difference, Δx_n , is $\sum_{n=0}^{N-1} \Delta x_n = \sum_{n=0}^{N-1} v_n = x_N - x_0$. We seek an analog of this discrete result.

The sense of the result we would expect is: The “continuous” sum of x' over the time interval $[t_b, t_e]$ is equal to $x(t_e) - x(t_b)$. For a start, set the discrete times $t_b = t_0 < t_1 < \dots < t_N = t_e$ and create a *sequence*, x , whose n^{th} value, x_n , is defined in terms of the *function*, x , by $x_n = x(t_n)$. Consider the following discrete sum:[†]

$$\begin{aligned} \sum_{n=0}^{N-1} x'(t_n) \Delta t_n &\approx \sum_{n=0}^{N-1} \frac{\Delta x_n}{\Delta t_n} \Delta t_n \\ &= \sum_{n=0}^{N-1} \Delta x_n = x_N - x_0 = x(t_e) - x(t_b) \end{aligned}$$

Note the approximation $x'(t_n) \approx \frac{\Delta x_n}{\Delta t_n}$ just says that the derivative of x at t_n is approximately the “rise” between x_n and x_{n+1} , $x_{n+1} - x_n$, divided by the “run”, $t_{n+1} - t_n$.

We now associate a physical meaning with the above sum and extend this discrete sum to a “continuous” one. We start with the simplest case: Let v be the constant function having value v_0 over the time interval $[t_0, t_1]$. Then the change in distance from time t_0 to t_1 is (from high school physics) $x(t_1) - x(t_0) = v_0(t_1 - t_0)$. Another way to interpret this is that the average velocity of the time interval $[t_0, t_1]$ is $v_0 = \frac{x(t_1) - x(t_0)}{t_1 - t_0}$. Suppose instead the car velocity is a piecewise constant function with two values over the interval. The velocity has the constant value v_0 over the interval $[t_0, t_1]$ [‡], and has the constant value v_1 over the interval $[t_1, t_2]$. Forgetting for the moment how the car manages to jump instantly from one velocity to the next, what is the relationship between the distance function and the velocity function? We know how to compute the change in the distance function over each sub-interval, so the total change is the sum of the changes.

$$\begin{aligned} x(t_2) - x(t_0) &= x(t_1) - x(t_0) + x(t_2) - x(t_1) \\ &= \Delta x_0 + \Delta x_1 \\ &= \frac{\Delta x_0}{\Delta t_0} \Delta t_0 + \frac{\Delta x_1}{\Delta t_1} \Delta t_1 \\ &= v_0 \Delta t_0 + v_1 \Delta t_1 \end{aligned}$$

Suppose instead the car velocity is a piecewise constant function, with velocities $v_0, v_1, \dots, v_{N-1}$ over intervals $[t_0, t_1], [t_1, t_2], \dots, [t_{N-1}, t_N]$. Again, let x_n be the sequence defined by $x_n = x(t_n)$. What is the change in the distance over the interval now? The answer is just the sum of all the changes produced by each interval of constant velocity.

$$\begin{aligned} x(t_N) - x(t_0) &= \Delta x_0 + \Delta x_1 + \dots + \Delta x_{N-1} \\ &= \frac{\Delta x_0}{\Delta t_0} \Delta t_0 + \frac{\Delta x_1}{\Delta t_1} \Delta t_1 + \frac{\Delta x_2}{\Delta t_2} \Delta t_2 + \dots + \frac{\Delta x_{N-1}}{\Delta t_{N-1}} \Delta t_{N-1} \\ &= v_0 \Delta t_0 + v_1 \Delta t_1 + v_2 \Delta t_2 + \dots + v_{N-1} \Delta t_{N-1} \end{aligned}$$

[†] The symbol $\approx$ means approximately equal.

[‡] The half open interval $[a, b)$ is the same set as $[a, b]$ excluding the point b .

This can be written in summation notation as:

$$x(t_N) - x(t_0) = \sum_{n=0}^{N-1} v_n \Delta t_n$$

Since the velocity has value v_n over the interval $[t_n, t_{n+1})$, the area under that portion of the velocity graph is $v_n \Delta t_n$ (height $\times$ width). The area under the velocity graph from t_0 to t_N is just the sum of these, which is the right hand side of the above formula.

That is, the area under the velocity graph (over an interval) is the change in distance (over the same interval).

In the figure below, the dotted lines are not part of the graph of the velocity function but have been added so that the distance traveled can be seen geometrically as the *area under the velocity graph*. The distance over the interval $[0, 8]$ is the sum of the areas of each of the boxes in the figure below. That is, $x(t_8) - x(t_0) = 30 * 2 + 60 * 2 + 40 * 2 + 80 * 2 = 420$.

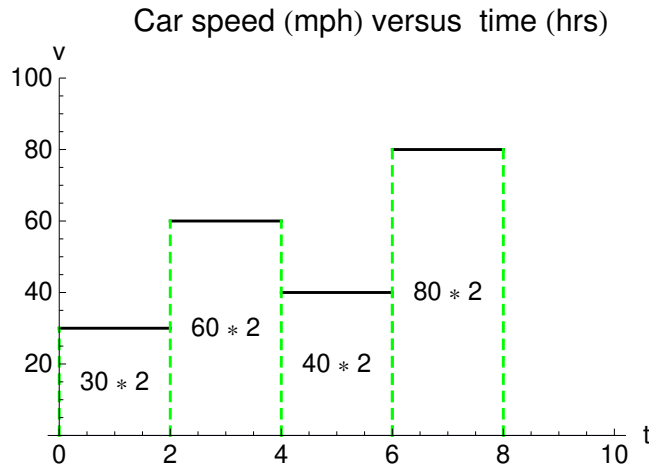


Figure 4.1: Piecewise constant velocity.

The corresponding graph of the distance function, x , appears below.

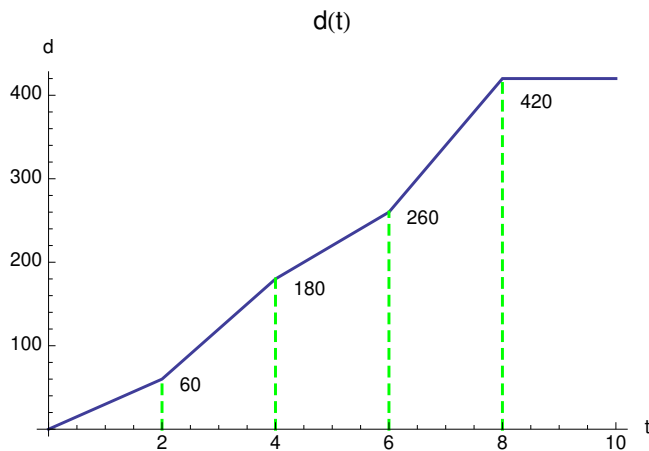


Figure 4.2: Piecewise linear distance function.

We repeat what we have discovered: No matter how complicated a piecewise constant velocity function is, the area under a portion of its graph represents the change in distance over that segment.

But what about velocity functions which aren't piecewise constant? How do we find the distance traveled? Well, we could approximate the velocity function by a piecewise constant function, v_{approx} . We will call the

corresponding distance function x_{approx} . The smaller the intervals of constancy, the better the approximation would be to the actual velocity function. Specifically, we divide the interval $[t_b, t_e]$ into N intervals, each with width $h_N = (t_e - t_b)/N$. Letting $t_n = t_b + n h_N$, for $n = 0, 1, \dots, N$, we define the piecewise constant approximation function to take the value $v(t_n)$ on the interval $[t_n, t_{n+1})$. That is, the approximation takes the value of the function v at the left end point of a given interval and stays at this value for the duration of the interval. In other words, the approximation is a piecewise constant function. For an example, see the figure below where N is 10.

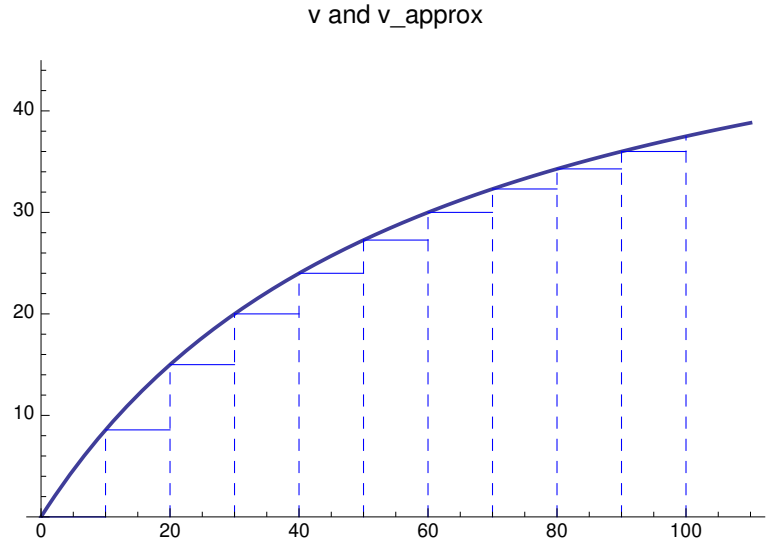
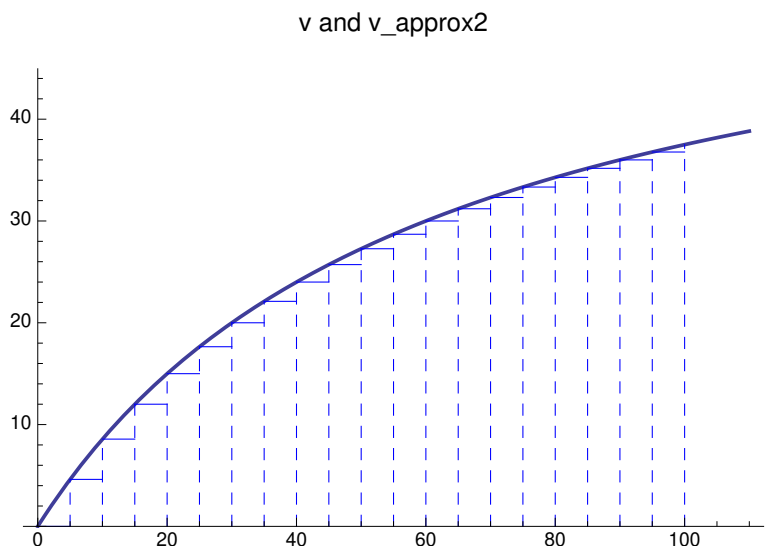


Figure 4.3: Piecewise constant approximation to velocity function, v .

From the figure, the distance traversed by an object with the piecewise constant function, v_{approx} is as before, the sum of the velocities $\times$ the corresponding width of the time interval, which is 10. That is, $x_{\text{approx}}(100) - x_{\text{approx}}(0) = v(0) * 10 + v(10) * 10 + v(20) * 10 + \dots + v(80) * 10 + v(90) * 10$. Note that each term in the sum is the area of one of the columns in the figure. So, the distance traversed when the velocity profile is v_{approx} is the sum of all the boxes under the curve. In summation notation, we may rewrite the expression for x_{approx} as

$$x_{\text{approx}}(100) - x_{\text{approx}}(0) = \sum_{n=0}^9 v(10n)10$$

Here is a piecewise constant function which does a better job of approximating v .

Figure 4.4: A better approximation to v .

The total distance traversed by an object with velocity v_{approx2} is again the area of its graph, which is the sum of all the rectangular dashed boxes. That is,

$$x_{\text{approx2}}(100) - x_{\text{approx2}}(0) = \sum_{n=0}^{19} v(5n)5$$

You can see that if we make more refined approximations to v , the value of the corresponding distance function gets closer and closer to the area under the graph of v and I hope you would agree that this distance function will get closer and closer to x . Therefore, the area under the curve of v is the limit of these refined sums. This is the same *approximate, then take a limit* move we used to define the derivative. If we divide the interval $[0, 100]$ into N intervals then the approximation would be $\sum_{n=0}^{N-1} v(h_N n)h_N$ where $h_N = (100 - 0)/N$. Just as when we defined derivatives, it would be useful for us to have a notation for the limit of this process.

Definition(Integral). Given a function f on an interval $[a, b]$. For any integer N we chop up the interval $[a, b]$ into N uniform pieces of length $h_N = (b - a)/N$. By the notation $\int_a^b f(t) dt$ we mean[†]:

$$\int_a^b f(t) dt \stackrel{\text{def}}{=} \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} f(t_n) \Delta t_n$$

where $t_n = a + n h_N$ is the left end point of the n^{th} interval.

The sum on the right represents an increasingly refined approximation to the area under the curve using stepwise approximations.[‡]

We say that $\int_a^b f(t) dt$ is the *integral* of f over the interval $[a, b]$. We call the process of computing the integral *integration*.

Note: The symbol, $\int$, is a stretched out ‘S’, the first letter of “Sum”, while dt is interpreted as a single symbol – not multiplication of d times t – and is an echo of the symbols Δt_n used in the approximating discrete sums.

[†] This definition only works for well behaved functions. For example, it works for continuous functions or a patch work of continuous functions. See the “Computational Strategy for Limits” section of the appendix “Infinite Sequences and Limits” for a definition of a continuous function.

[‡] The right hand side is really $\lim_{N \rightarrow \infty} S_N$, where $S_N \stackrel{\text{def}}{=} \sum_{n=0}^{N-1} f(t_n) \Delta t_n$. That is, this is really the limit of a sequence, something we have already discussed.

Note: We chopped $[a, b]$ into *uniform* pieces only for simplicity; the general partitions $t_0 < t_1 < \dots < t_N$ used earlier in this chapter work as well, provided all the widths shrink to 0. For the well behaved functions of this book every such scheme produces the same limiting value.

Note: The variable t (and its cousin dt) in the expression $\int_a^b f(t) dt$ are “dummy” variables. Their notational purpose is to represent the “continuous” looping variable used in this “sum”. Just as in programming languages, the name is not important. It is important, however, that it not be a name which causes confusion. For instance, if the interval over which we are integrating is $[a, t]$ then the integral, $\int_a^t f(t) dt$, doesn’t make sense. This is directly analogous to writing a “for loop” that confuses the looping variable with a variable used to determine the end of the loop. The confused code might look like this in the language ‘C’: `for(i = 0; i < i; i++) {...}`. This loop can be written correctly by using a different looping variable: `for(j = 0; j < i; j++) {...}`. The incorrectly written integral, $\int_a^t f(t) dt$, can be fixed in the same way: $\int_a^t f(s) ds$. To make the point clear, let us note that all of the integrals below mean the same thing:

$$\int_a^b f(t) dt = \int_a^b f(s) ds = \int_a^b f(u) du = \int_a^b f(x) dx$$

Our new shiny notation says that $\int_{t_b}^{t_e} v(t) dt = \text{the area under the graph of } v^\dagger$. This is also the limit of $x_{\text{approx}}(t_e) - x_{\text{approx}}(t_b)$ as the approximations get closer to x . Therefore, if we believe that the x_{approx} values at t_b and t_e approach the values of x at t_b and t_e , then we must have

$$\int_{t_b}^{t_e} v(t) dt = x(t_e) - x(t_b)$$

So, just as with the piecewise constant functions it is the case that the area under v is just the difference in the value of the corresponding distance function, x , at the endpoints.

We can also relate the point-wise relationship of v to x with the above interval relationship. Since $x'(t) = v(t)$, we may write the above as

$$\int_{t_b}^{t_e} x'(t) dt = x(t_e) - x(t_b)$$

It turns out that this formula is true even if x is not a distance function. That is, given any function F , with corresponding derivative, F' , we may write what is called *the Fundamental Theorem of Calculus*:

Theorem(Fundamental Theorem of Calculus). *If F is a function such that F' exists and is continuous over the interval $[a, b]$, then*

$$\int_a^b F'(t) dt = F(b) - F(a)$$

This is really the continuous analog of the Fundamental Theorem of Difference Calculus:

$$\begin{array}{ccc} \text{Discrete Sum} & \text{Telescoping Difference} & \text{Difference at End Points} \\ \underbrace{\sum_{n=0}^{N-1}} & \underbrace{\Delta F_n} & = \underbrace{F_N - F_0} \end{array}$$

We write the Fundamental Theorem of Calculus (FTC) again with suggestive markup to show the analogy with the above discrete equation:

$$\begin{array}{ccc} \text{Continuous Sum} & \text{Infinitesimal Telescoping Difference} & \text{Difference at End Points} \\ \underbrace{\int_a^b} & \underbrace{F'(t) dt} & = \underbrace{F(b) - F(a)} \end{array}$$

[†] For this statement to be true we are implicitly assuming that the velocity function is non-negative.

We will show how this result follows by breaking up the interval $[a, b]$ into N pieces (where N is large) and approximating the integral with a sum of telescoping differences of a related sequence. Then we can use the Fundamental Theorem of *Difference* Calculus (FTDC) to show the FTC.

To implement this program, note that each interval will have length $h_N = (b - a)/N$. Let F be a sequence (we use the same name as the function) whose n^{th} element, F_n , is the value of left end point of the n^{th} interval (intervals are labeled starting at zero). That is, $F_n = F(t_n)$ where t_n represents the left end point of the n^{th} interval; i.e., $t_n = a + n h_N$. Notice that $F_0 = F(a)$ and $F_N = F(b)$. In addition, note that $t_{n+1} = t_n + h_N$ ($\Delta t_n = h_N$) so that $F_{n+1} = F(t_{n+1}) = F(t_n + h_N)$. We also know that $F'(t_n) \approx \frac{\Delta F_n}{\Delta t_n}$.[†] We now perform a crude calculation to show why the Fundamental Theorem is true:

$$\int_a^b F'(t) dt \approx \sum_{n=0}^{N-1} F'(t_n) \Delta t_n \approx \sum_{n=0}^{N-1} \frac{\Delta F_n}{\Delta t_n} \Delta t_n = \overbrace{\sum_{n=0}^{N-1} \Delta F_n}^{\text{FTDC}} = F_N - F_0 = F(b) - F(a)$$

This approximation can be seen by performing a slightly more careful analysis:

$$\begin{aligned} \int_a^b F'(t) dt &\approx \sum_{n=0}^{N-1} F'(t_n) \Delta t_n \\ &= \sum_{n=0}^{N-1} \left(\frac{\Delta F_n}{\Delta t_n} + \text{error}(t_n) \right) \Delta t_n = \sum_{n=0}^{N-1} \Delta F_n + \overbrace{\sum_{n=0}^{N-1} \text{error}(t_n) \Delta t_n}^{\text{very small term}} \\ &\approx \sum_{n=0}^{N-1} \Delta F_n = F_N - F_0 = F(b) - F(a) \end{aligned}$$

Here, $\text{error}(t_n)$ is the correction term to make $\Delta F_n / \Delta t_n$ equal to $F'(t_n)$. The only hard part of our crude calculation is showing that $\sum_{n=0}^{N-1} \text{error}(t_n) \Delta t_n$ goes to zero as N gets large. If we call $\text{error}_{\max}$ the largest of the errors in all the intervals, then we may bound the “very small term” expression above by $\text{error}_{\max} \sum_{n=0}^{N-1} \Delta t_n$. This, in turn, is bounded by $\text{error}_{\max}(b - a)$.[‡] We need now only believe that the maximum error term will go to zero as N goes to infinity.

How do we use this theorem? For instance, suppose I want to compute the area under the curve of a function f over the interval $[a, b]$, how does this theorem help? Well, we need to find an F such that $F' = f$. Then

$$\int_a^b f(x) dx = \int_a^b F'(x) dx = F(b) - F(a)$$

We call F an *anti-derivative* of f . It turns out that F is not unique; but all anti-derivatives differ by a constant. It doesn't matter which one you use, you will get the same answer^{*}. Now we need to be able to find an anti-derivative of a function and we'll be done.

Let's see how to apply these ideas to a problem. Keeping with the car theme, suppose that you didn't have sensors in your car, but you were told that the velocity of the car over the interval $[0, 10]$ (in hours) would be $v(t) = t^2$ mph. How far will the car have traveled over this time period? With our new notation we can write down the answer $\int_0^{10} t^2 dt$. Although this answer is concise, the notation hides the real work involved in its computation. That is, to compute this value from its definition would be a great deal of work. But, the Fundamental Theorem of Calculus says we may avoid that work by evaluating a certain function x at 10 and 0 and taking the difference.

[†] The approximate derivative is obtained by dividing the “rise” in F , ΔF_n , by the “run”, Δt_n .

[‡] $\sum_{n=0}^{N-1} \Delta t_n = t_N - t_0 = b - a$ by the FTDC.

^{*} If G is another anti-derivative then $G(x) = F(x) + c$ for some constant c . So $G(b) - G(a) = (F(b) + c) - (F(a) + c) = F(b) - F(a)$.

It is a function that has the property that $x'(t) = v(t) = t^2$. Can we guess what functions have this property? We are looking for an *anti-derivative* of v . We know that from the chapter on differentiation that differentiation of power functions reduces their degree by 1. To find an anti-derivative we should do the opposite; that is, look for a function of degree one more than 2. Let's try $x(t) = t^3$. In this case, $x'(t) = 3t^2$. This isn't quite right, we're off by a factor of 3. If we let $x(t) = t^3/3$ then $x'(t) = t^2$ and we can find the distance traveled, it is $x(10) - x(0) = 10^3/3 - 0^3/3 = 1000/3 = 333\frac{1}{3}$ miles. That is,[†]

$$\int_0^{10} t^2 dt = t^3/3 \big|_{t=0} - t^3/3 \big|_{t=10} = 333\frac{1}{3}$$

We make a formal statement of the anti-derivative style FTC.

Theorem(FTC Anti-derivative Style). *If f is a function and F is an anti-derivative of f , then the following is true:*

$$\int_a^b f(x) dx = F(b) - F(a)$$

proof: This just says $\int_a^b f(x) dx = \int_a^b F'(x) dx = F(b) - F(a)$ (since $F' = f$). And the last equality is just the statement of the fundamental theorem of calculus.

So, in order to compute an integral of a function we need to find an anti-derivative of that function.

Exercise 4.1: Think about why all anti-derivatives of a function f differ by a constant. Hint: Let F and G be two anti-derivatives of f . Show that the derivative of $F - G$ is zero. What kind of functions have zero slope at every point?

Integral Calculus: Computational Strategy

We would like to be able to compute an *anti-derivative* of a function in order to be able to integrate it. We use the notation $\int f(x) dx$ to indicate an anti-derivative. That is, when we omit the boundaries of the integral we mean an anti-derivative; with the boundaries we mean the integral over an interval.

We proceed in the same manner as with the differential calculus. First, we find anti-derivatives for a base class of functions. Next, we find formulas for anti-derivatives of functions composed in some way from simpler functions. Just as with the differential calculus, we use the power functions as a base set.

A Base – a Family of Functions and their Anti-derivatives

Theorem(Anti-derivative of Power Functions). *An anti-derivative of the power function $f(x) = x^n$ is $F(x) = x^{n+1}/(n+1)$; valid for integer n , $n \neq -1$; valid for all x when $n \geq 0$ and for all $x \neq 0$ when $n < 0$. This may be written:*

$$\int x^n dx = x^{n+1}/(n+1)$$

Note that when $n = -1$ we already know an anti-derivative (only valid for $x > 0$): $\int 1/x dx = \ln(x)$.

We will need a slightly larger base than just the power functions. There is nothing new here – each of the formulas below is just a derivative formula from the chapter on differentiation read in reverse. Restating the formula for

[†] One last bit of notation, we use $f(t) \big|_{t=a}$ to mean $f(a)$. This is similar to notation you would see in other texts.

$1/x$ along with the others:

$$\begin{aligned}\int \cos(x) dx &= \sin(x) \\ \int \sin(x) dx &= -\cos(x) \\ \int e^x dx &= e^x \\ \int 1/x dx &= \ln(x) \quad (x > 0)\end{aligned}$$

Enlarging the Class of Functions to Integrate

Here is a result for the second stage of our program. It is a linearity theorem similar to the one for summation calculus.

Theorem (Linearity of the Anti-derivative). *An anti-derivative of the function f , where $f(x) = c_1 f_1(x) + c_2 f_2(x)$ is c_1 times an anti-derivative of f_1 plus c_2 times an anti-derivative of f_2 . This may be written:*

$$\int f(x) dx = c_1 \int f_1(x) dx + c_2 \int f_2(x) dx$$

Note that when $f_2(x) = 0$, the above implies $\int cf(x) dx = c \int f(x) dx$.

Corollary (Linearity of the Anti-derivative). *An anti-derivative of the function $f(x) = c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)$ is*

$$\int f(x) dx = c_1 \int f_1(x) dx + c_2 \int f_2(x) dx + \dots + c_n \int f_n(x) dx$$

This follows from the theorem on linearity and mathematical induction. This is the *exploit linearity* move played for the third time – first with Δ and $\sum$, then with the derivative, and now with the integral: break the function into pieces, integrate each piece, and add the answers. The corollary and the results for the base family can be combined to find an anti-derivative of *any* polynomial.

Theorem (Anti-derivative of a Polynomial). *An anti-derivative of the polynomial $p(x) = c_0 + c_1x + c_2x^2 + \dots + c_nx^n$, (n a non-negative integer) is*

$$\int p(x) dx = c_0x + c_1x^2/2 + c_2x^3/3 + \dots + c_nx^{n+1}/(n+1)$$

Example 4.1: Compute the integral: $\int_1^2 (1 + 2x + x^2) dx$.

We can use the last theorem to find an anti-derivative of $1 + 2x + x^2$; it gives $x + 2x^2/2 + x^3/3 = x + x^2 + x^3/3$. Therefore, by the fundamental theorem of calculus: $\int_1^2 (1 + 2x + x^2) dx = (x + x^2 + x^3/3) \big|_{x=2} - (x + x^2 + x^3/3) \big|_{x=1} = (2 + 4 + 8/3) - (1 + 1 + 1/3) = 26/3 - 7/3 = 19/3$.

Here is another result from stage 2 of our program.

Integration by Parts

Consider the integral $\int_0^{\pi/2} x \sin(x) dx$. Linearity is no help here – the integrand is a *product*, not a linear combination of functions we can already integrate. We faced exactly this predicament on the discrete board: the sum $\sum n 2^n$, a product of two very different sequences, defeated the linear-combination method, and Summation by Parts – built from the Discrete Product Rule – rescued us. We now build its continuous twin the same way, from

the product rule of differential calculus. The product rule states that $(f g)' = f' g + f g'$. We rewrite this as: $f g' = (f g)' - f' g$. Integrating both sides over the interval $[a, b]$ gives

$$\int_a^b f(x)g'(x) dx = \int_a^b (f(x)g(x))' dx - \int_a^b f'(x)g(x) dx$$

The first integral on the right hand side is easy to compute by the fundamental theorem. Since this integral is easy to evaluate by the FTC, we have a direct relationship between the other two integrals.

$$\int_a^b f(x)g'(x) dx = (f(b)g(b) - f(a)g(a)) - \int_a^b f'(x)g(x) dx$$

We call this result *Integration by Parts* which is the continuous analog of *Summation by Parts* introduced in chapter 2. It relates one integral to another. This is useful (just as it was for summing sequences) because if one of the integrals can be easily evaluated we get the value of the other one for free.

Example 4.2: Compute the integral $\int_0^{\pi/2} x \sin(x) dx$.

This is the integral that opened this subsection; the new tool cracks it. Notice it would be convenient to treat $\sin$ as the derivative of a function and then pass the derivative onto x . Let $g(x) = -\cos(x)$ and $f(x) = x$. Then $g'(x) = \sin(x)$. Integration by parts gives us:

$$\begin{aligned} \int_0^{\pi/2} \underbrace{f(x)}_x \underbrace{g'(x)}_{\sin(x)} dx &= \overbrace{((\pi/2) * (-\cos(\pi/2)) - 0 * (-\cos(0)))}^{f(\pi/2)g(\pi/2) - f(0)g(0)} - \int_0^{\pi/2} \underbrace{f'(x)}_{x'} \underbrace{g(x)}_{-\cos(x)} dx \\ &= - \int_0^{\pi/2} -\cos(x) dx = \int_0^{\pi/2} \cos(x) dx = \sin(\pi/2) - \sin(0) \\ &= 1 \end{aligned}$$

Note that the values 0 and $\pi/2$ are in radians. The value $\pi/2$ radians is 90 degrees. This means that $\sin(\pi/2) = 1$ and $\cos(\pi/2) = 0$.

Substitution

Consider the integral $\int \sin^2(x) \cos(x) dx$. None of the rules we have developed so far will crack it – the integrand is not a polynomial, nor is it a combination of our base functions that linearity or integration by parts can untangle. We need one more technique. It is the *change the representation* move of the preface: rewrite the integrand until a form we already know how to integrate appears. This technique uses the chain rule (from differential calculus) to compute an integral. The chain rule states that $f(u(x))' = f'(u(x))u'(x)$. Integrating both sides gives

$$\int_a^b f'(u(x))u'(x) dx = \int_a^b f'(u(x))' dx = f(u(b)) - f(u(a))$$

Successful use of the above techniques is based on how well one can pattern match. It takes practice, but it is not our focus here. We demonstrate the idea through a few examples.

Example 4.3: Compute $\int_a^b (x-a)^n dx$ when n is a non-negative integer.

Letting $u(x) = (x-a)$, we see that the integral can be written as $\int_a^b u(x)^n u'(x) dx$. In this case letting $f'(z) = z^n$ (which has $z^{n+1}/(n+1)$ as an anti-derivative), the substitution formula gives $\int_a^b f'(u(x))u'(x) dx = f(u(b)) - f(u(a)) = \frac{1}{n+1}(b-a)^{n+1} - \frac{1}{n+1}(a-a)^{n+1} = \frac{1}{n+1}(b-a)^{n+1}$.

Example 4.4: Compute $\int_a^b (a-x)^n dx$ when n is a non-negative integer.

Letting $u(x) = (a-x)$, we see that the integral can be written as $-\int_a^b u(x)^n u'(x) dx$. Setting $f'(z) = z^n$ as in the previous example the substitution formula gives $-\int_a^b f'(u(x))u'(x) dx = -(f(u(b)) - f(u(a))) = -(\frac{1}{n+1}(a-b)^{n+1} - \frac{1}{n+1}(a-a)^{n+1}) = -\frac{1}{n+1}(a-b)^{n+1}$.

Example 4.5: Compute $\int_0^1 \sin^2(x) \cos(x) dx$.

Letting $u(x) = \sin(x)$, and therefore $u'(x) = \cos(x)$, we see that the integral can be written as $\int_0^1 u(x)^2 u'(x) dx$. In this case $f'(z) = z^2$. Using the substitution formula gives $\int_0^1 f'(u(x))u'(x) dx = f(u(1)) - f(u(0)) = u(1)^3/3 - u(0)^3/3 = \sin(x)^3/3 \big|_{x=1} - \sin(x)^3/3 \big|_{x=0} = \sin^3(1)/3 - 0 = \sin^3(1)/3$. Note, as always the angle measure is in radians. So, the 1 in our answer, $\sin^3(1)/3$, means 1 radian, which is approximately 57 degrees.

Exercise 4.2: Compute the following integrals: (1) $\int_0^1 (1+x)^7 dx$; (2) $\int_0^{\pi/2} x \cos(x) dx$; (3) $\int_0^1 x e^{x^2} dx$. Hint: the first and last yield to substitution; the middle one to integration by parts.

Exercise 4.3: Compute the area under the graph of $\sin$ over the interval $[0, \pi]$. Then compute $\int_0^{2\pi} \sin(x) dx$. Why is the second answer 0 even though the graph over $[0, 2\pi]$ certainly encloses area? (Recall the footnote earlier in this chapter: the area interpretation assumed the function was non-negative.)

Can't always find Formulas for Anti-derivatives

As noted in the last chapter, not every function has a derivative. And even when it does, it may not be easy to find a formula for the derivative. However, if the function has a nice formula it is often the case that we can find a formula for the derivative. This is not the case with anti-derivatives. For nice continuous functions we always have an anti-derivative, but finding a formula is not always possible – even for relatively simple functions. That is, there are functions with nice formulas for which there is no formula for an anti-derivative. One example is the bell shaped functions associated with the subjects of probability and statistics. The graph of the function $f(x) = e^{-x^2}$ is bell shaped and its formula is simple. The area under the graph of this function exists and we can create a function which is the area under this curve from 0 to x . We can write the value down in our notation as: $A(x) = \int_0^x e^{-s^2} ds$. However, there is no formula for this function in terms of powers of x , logarithmic, exponential, or trig functions because there is no formula for an anti-derivative of e^{-x^2} . The integral (the area under the graph) exists; that is, the function $A(x)$ is a valid function – there is just no formula for it. See the appendix “Functions, Formulas, Graphs, and Notation” for other examples of functions without formulas.

Summary

The fundamental theorem of calculus can be stated succinctly: the integration operation is the inverse of the differentiation operation. This can be seen from the FTC: $\int_a^b f'(t) dt = f(b) - f(a)$, and writing it more crudely as $\int f' = f$. What makes this result useful is if for a given integral, $\int_a^b v(t) dt$, one can recognize v as the derivative of another function f , then this integral is $\int_a^b f'(t) dt$ and by the FTC this is just $f(b) - f(a)$.

The process of recognition is helped greatly if there is a well developed stock of differentiation formulas. Then when confronted with integrating a function, we may be able to recognize, or more easily deduce, a function which is an *anti-derivative* and then easily compute the integral. As with differencing and summing, finding formulas for derivatives and anti-derivatives of functions is much like multiplying and dividing numbers. Computing derivatives is relatively easier – like multiplying, while finding anti-derivatives is harder – like division. To do long division it helps to be fluent with multiplication, because trial and error multiplications are necessary. The same is true when computing anti-derivatives, it helps first to be fluent with the formulas for the derivatives of a number of functions. To this end, one would

1. Develop formulas for the derivatives of a base collection of functions.
2. Develop formulas for the derivatives of functions formed in various ways from the base.

This is exactly what we did in the last chapter.

Stepping back, this whole chapter was really the discrete game replayed on the continuous board – every move we made had already been rehearsed with sequences, differences, and sums. Here is the dictionary:

Discrete	Continuous
Δ	d/dx
Σ	$\int$
Fundamental Theorem of Difference Calculus (telescoping)	Fundamental Theorem of Calculus
Summation by Parts	Integration by Parts
anti-difference	anti-derivative

Chapter 5: The Derivative As Approximation

A Motivating Problem: The Concrete Wall

Here is the kind of question this chapter answers. Suppose you need to pour a concrete wall 100 ft long and 5 ft high. Builders size the depth of such a wall from its height: depth (in inches) = height (in inches) / 10. So if the wall's height is x inches, the volume of concrete needed, in cubic inches, is $V(x) = 12 \times 100 x (x/10) = 120 x^2$. For a 5 ft wall ($x = 60$ inches) that is $V(60) = 432,000$ cubic inches – about $9\frac{1}{4}$ cubic yards. But there is always error in measurement. Suppose, because of the ground conditions, the height measurement could be off by an inch. How much extra concrete should we order? We could simply compute $V(61)$ for this one case; but we would like an estimate for *any* small error h – and a way of thinking that transfers to problems where recomputing is not so easy. We don't want to guess high and waste money; we want an *estimate of the error*: how far off is V when x is off by a small amount h ?

In symbols, we want to understand the difference $V(x+h) - V(x)$ for small h , using only information at the height we actually measured, x . What do we know that connects the values of a function *near* x to information *at* x ? We know the derivative – and by the Fundamental Theorem of Calculus the connection is exact:

$$V(x+h) - V(x) = \int_x^{x+h} V'(s) ds$$

The question “how far off is the volume?” has become the question “how big is this integral?” We do not need its exact value – an estimate with a guarantee will do. So the first order of business is learning to estimate integrals we would rather not compute.

Our wall integral is one instance of a general situation: many times it is not clear – or even possible – how to evaluate an integral exactly. What one can often do is bound its value from above and below. For instance, if f is a continuous function[†] and non-negative on an interval $[a, b]$, then f attains a maximum, say M , and a minimum, say m on this interval. Therefore, the area under the graph of f (the integral $\int_a^b f(x) dx$) is bounded above by the area of the rectangle with height M and width $b - a$ and bounded below by the area of the rectangle with height m and width $b - a$. That is,

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

This result is true even if f takes on negative values. This is true since the integral is a limit of sums of samples of f along the interval, $[a, b]$, times small differences – which sum to $b - a$. But each of these sample values of f is bounded above and below by M and m respectively. Therefore, the sum is bounded above and below by $M(b-a)$ and $m(b-a)$. We can show something just a little more interesting with almost no extra work.

Mean Value Theorem

Again, consider the area under the graph of f from a to b ($a < b$)[‡]: $\int_a^b f(s) ds$. You can think of the graph as the outline of fine dust in a very narrow rectangular glass container. If we shift the dust around so that the dust is level with height, say h , we must have (since the area of the dust is conserved)

$$\int_a^b f(s) ds = h(b-a)$$

[†] Continuous functions are discussed in the section “Computational Strategy for Limits” in the appendix “Infinite Sequences and Limits”. They have the property that when drawing their graph, one never need take the pen off the page.

[‡] The following argument works for non-negative f , but can be extended to all *continuous* f (dropping the non-negativity assumption).

If m and M are the minimum and maximum values that f takes on in the interval $[a, b]$, then $m \leq h \leq M$. It turns out that continuous functions have the property that they attain all intermediate values. That is, if f is a continuous function which hits the values m and M over the interval $[a, b]$, then for any $m \leq h \leq M$ there exists a value $\mu \in [a, b]$ such that $f(\mu) = h$. Collecting these ideas we have:

Theorem(Intermediate Value Theorem). *If f is a continuous function on the interval $[a, b]$ attaining values m and M with $m \leq M$, then for any intermediate value h with $m \leq h \leq M$ there exists a value $\mu \in [a, b]$ such that $h = f(\mu)$.*

Theorem(Mean Value Theorem). *If f is a continuous function on the interval $[a, b]$, then there exists a value, $\mu \in [a, b]$, such that*

$$\int_a^b f(s) ds = f(\mu)(b - a)$$

Note: Many texts call this result the Mean Value Theorem *for integrals*, reserving the name “Mean Value Theorem” for the tangent-line statement, $f(b) - f(a) = f'(\mu)(b - a)$, derived from it below.

If f' is also continuous then the theorem applies to f' :

$$\int_a^b f'(s) ds = f'(\mu)(b - a)$$

By the Fundamental Theorem of Calculus this can be written

$$f(b) - f(a) = f'(\mu)(b - a)$$

which is

$$f(b) = f(a) + f'(\mu)(b - a) \tag{1}$$

This is exactly the kind of statement the wall problem handed us: taking $f = V$, $a = x$, and $b = x + h$, it says $V(x + h) - V(x) = V'(\mu)h$ for some μ between x and $x + h$ – the troublesome integral has been traded for a single value of the derivative.

This can be written in a way that has an immediate geometric interpretation:

$$f'(\mu) = \frac{f(b) - f(a)}{(b - a)}$$

This says that when f' is continuous there is a point $\mu \in [a, b]$ where the slope of the tangent line is the same as the slope of the line passing through the points $(a, f(a))$ and $(b, f(b))$.

Here is another way to view this. Consider the interval $[x - h, x + h]$ for some $h > 0$. Substituting $x + h$ for b and x for a in (1) gives $f(x + h) = f(x) + f'(\mu_1)h$ – for some $\mu_1 \in [x, x + h]$. Next substituting x for b and $x - h$ for a in (1) gives $f(x) = f(x - h) + f'(\mu_2)h$, or $f(x - h) = f(x) - f'(\mu_2)h$ – for some $\mu_2 \in [x - h, x]$. That is, we may write $f(x + h)$ as $f(x + h) = f(x) + f'(\mu)h$ regardless of the sign of h , for some $\mu \in [x - |h|, x + |h|]$. If h is small, then $f'(\mu)$ is close to $f'(x)$; therefore, an approximation to $f(x + h)$ is

$$f(x + h) \approx f(x) + f'(x)h \tag{1a}$$

Derivative as Best Linear Approximation

Equation (1a) amounts to a restatement of the definition of a derivative of f at x . This equation says that we may approximate $f(x+h)$ with a straight line; that is a *rectilinear* function – also called an *affine* function (most books say simply *linear*). This can be thought of as starting with $f(x)$ and adding a *Linear Approximation* that is a linear function of the change in x , h .

In what sense is the tangent line the *best* line? Among all lines passing through the point $(x, f(x))$, it is the one whose error goes to zero faster than h itself. Recall from the Mean Value Theorem that $f(x+h) - (f(x) + f'(x)h) = (f'(\mu) - f'(x))h$, and the factor $f'(\mu) - f'(x)$ itself shrinks to zero as h does. Any other line through $(x, f(x))$, say with slope $s \neq f'(x)$, has error approximately $(f'(x) - s)h$ – proportional to h itself.

This turns out to be a way to describe the derivative in a way that is extensible to “higher order” functions; e.g., so-called *multi-variable* and *vector-valued* functions. By this we mean functions of the form: $f : R^n \rightarrow R$ and $g : R^n \rightarrow R^m$. It is the same game on a bigger board. For $f : R \rightarrow R$ the derivative is a single number – a slope. For $f : R^n \rightarrow R$ it is a vector, called the *gradient*; and for $g : R^n \rightarrow R^m$ it is a matrix, called the *Jacobian*. In every setting the answer to the question “what is the derivative?” is the same: it is the linear map that best approximates the change in the function.

Such functions come under the general heading of “Advanced Calculus”, and are beyond the scope of this book.

Approximation Application

Back to the concrete wall that opened this chapter. There we found $V(x) = 120x^2$ and asked how much extra concrete a small error h in the measured height costs – that is, for the difference $V(x+h) - V(x) = \int_x^{x+h} V'(s) ds$. The Mean Value Theorem was built to size exactly this kind of integral, and equation (1a) is its answer.

Equation (1a) tells us that if there is a small error in the height then the approximate change in the volume is given by $V(x+h) \approx V(x) + V'(x)h$. Let's say that we are concerned with an error of 1 in when the height is 5 ft (60 inches). The magnitude of the change is $|V'(x)h|$. It is easy to see that $V'(x) = 240x$. When $x = 60$ we have $V'(60) = 14400$; if, in addition, $h = 1$, $|V'(x)h|$ becomes 14400 cubic inches. This is 0.309 cubic yards. Therefore, we should order an extra $1/3$ of a yard of concrete.

And the chapter's opening promise – an estimate with a *guarantee* – can be kept with the bounding idea we started from. Since $V'(s) = 240s$ is increasing, on the interval $[60, 61]$ the integrand of $\int_{60}^{61} V'(s) ds$ lies between $V'(60) = 14,400$ and $V'(61) = 14,640$; so, by the very first inequality of this chapter, the extra volume is bracketed between those two numbers. Our estimate is not just plausible, it is fenced in – indeed the exact value, $V(61) - V(60) = 14,520$ cubic inches, sits between the fences.

Exercise 5.1: Use the tangent line approximation (1a) with $f(x) = x^5$ to estimate $(2.01)^5$ without a calculator. The true value is $32.80804\dots$ – how far off is the estimate?

Exercise 5.2: In the concrete wall problem, suppose the height measurement could be off by as much as 2 inches instead of 1. Estimate the extra concrete needed. Is your answer exactly twice the 1 inch answer? Compare with the exact extra amount, $V(62) - V(60)$.

Taylor's Theorem

The tangent line approximation leaves two questions hanging: exactly how big is the error we make when we replace $f(x+h)$ by $f(x) + f'(x)h$, and can we shrink it by adding a quadratic correction – or a cubic one, and so on? The answer to both questions is a result called Taylor's Theorem[†]. Let us first describe a notation for differentiation. We have used f' to denote the derivative of f . We will use the notation f'' to denote the derivative

[†] This section stands by itself and may be skipped if you wish.

of the function f' . Similarly, we use the notation f''' to denote the derivative of f'' . In general, we use the notation $f^{(n)}(x)$ to denote the n^{th} derivative of f evaluated at x .

Now, if f' is a continuous function we know by the mean value theorem that

$$f(b) = f(a) + f'(\mu)(b - a)$$

But we could invoke the theorem for any $x \in [a, b]$ because $[a, x]$ is an interval where the theorem applies. In this case the conclusion of the theorem is

$$f(x) = f(a) + f'(\mu_1)(x - a) \quad (2)$$

Note that we changed μ to μ_1 because the interval $[a, x]$ is different than the interval $[a, b]$. All that the Mean Value Theorem says is that there is *some* value in $[a, x]$; it may not be the same one that applied on the interval $[a, b]$. However, for the time being, let us assume that it is the same; that is, we assume that $\mu_1 = \mu$.

If f'' is also continuous on $[a, b]$, then we may apply this formula to f' (it's a function which satisfies the requirements of the theorem) giving

$$f'(x) = f'(a) + f''(\mu_2)(x - a)$$

for some value μ_2 which depends on the interval $[a, x]$. Why is this different from μ_1 ? Because f' is a different function from f so we should not expect μ_2 to be the same as μ_1 . Let us assume, however, that $\mu_2 = \mu_1$, which we are assuming is μ .

Integrating this function over the interval $[a, x]$ we have

$$\int_a^x f'(s) ds = \int_a^x f'(a) ds + \int_a^x f''(\mu)(s - a) ds$$

Since the integration is linear, we may place the constants outside of the integral[‡].

$$\int_a^x f'(s) ds = f'(a) \int_a^x 1 ds + f''(\mu) \int_a^x (s - a) ds$$

Using the FTC on the left hand and first term of the right hand side gives

$$f(x) - f(a) = f'(a)(x - a) + f''(\mu) \int_a^x (s - a) ds$$

As an anti-derivative of $(s - a)$ is $\frac{(s-a)^2}{2}$, we have

$$f(x) = f(a) + f'(a)(x - a) + f''(\mu) \int_a^x \left(\frac{(s - a)^2}{2} \right)' ds$$

Using the FTC on the last term, we may write

$$f(x) = f(a) + f'(a)(x - a) + f''(\mu) \frac{(x - a)^2}{2} \quad (3)$$

Now, just as we plugged f' into equation (2), we can do the same with equation (3). If the function $f^{(3)}$ is continuous, then plugging f' into (3) gives

$$f'(x) = f'(a) + f''(a)(x - a) + f^{(3)}(\mu_3) \frac{(x - a)^2}{2}$$

[‡] Anything that does not depend on the integration variable, s , is a constant with respect to the integration. Therefore, it may be pulled outside of the integral.

Again, we need a new “ μ ” because we are using a new function, f' . As before, we integrate this equation over the interval $[a, x]$ giving

$$f(x) - f(a) = f'(a)(x - a) + f''(a)\frac{(x - a)^2}{2} + \int_a^x f^{(3)}(\mu_3)\frac{(s - a)^2}{2} ds$$

As an anti-derivative of $(s - a)^2/2$ is $\frac{(s - a)^3}{3 \cdot 2}$, the above becomes

$$f(x) - f(a) = f'(a)(x - a) + f''(a)\frac{(x - a)^2}{2} + f^{(3)}(\mu_3)\int_a^x \frac{(s - a)^2}{2} ds$$

Which is

$$f(x) = f(a) + f'(a)(x - a) + f''(a)\frac{(x - a)^2}{2} + f^{(3)}(\mu_3)\frac{(x - a)^3}{3 \cdot 2} \quad (4)$$

We can keep plugging f' into these successively more complicated equations (equations (2), (3), and (4)) provided f has enough derivatives and they are continuous. The pattern that emerges is called Taylor's Theorem:

Theorem(Taylor's Theorem). *If f is a function such that $f^{(n)}$ exists and is a continuous function on the interval $[a, b]$, then for any $x \in [a, b]$ [†]*

$$\begin{aligned} f(x) = & f(a) + f'(a)(x - a) + f^{(2)}(a)\frac{(x - a)^2}{2!} + f^{(3)}(a)\frac{(x - a)^3}{3!} \\ & + \dots + f^{(n-1)}(a)\frac{(x - a)^{n-1}}{(n - 1)!} + f^{(n)}(\mu(x))\frac{(x - a)^n}{n!} \end{aligned}$$

where μ is a function such that $\mu(x) \in [a, x]$.

Notice what this theorem says: from the values of f and its derivatives at the *single* point a , we can rebuild f across the whole interval. This is the same game we have been playing since the discrete chapters: reconstruct the whole from its rates of change at a single point.

Our derivations for Taylor's Theorem in the cases $n = 2$ and $n = 3$ are not correct since we assumed at each stage that the various “ μ ” were the same. A mathematically correct derivation can be found in the appendix on Taylor's Theorem. The derivation has the same structure as our derivation above but uses a generalization of the Mean Value Theorem to prove the result. One can also show without much work that the formula is true when x is less than a :

Corollary. *If f is a function such that $f^{(n)}$ exists and is a continuous function on the interval $[a, b]$, with $c \in [a, b]$, then for any $x \in [a, b]$*

$$\begin{aligned} f(x) = & f(c) + f'(c)(x - c) + f^{(2)}(c)\frac{(x - c)^2}{2!} + f^{(3)}(c)\frac{(x - c)^3}{3!} \\ & + \dots + f^{(n-1)}(c)\frac{(x - c)^{n-1}}{(n - 1)!} + f^{(n)}(\mu(x))\frac{(x - c)^n}{n!} \end{aligned}$$

where μ is a function such that $\mu(x)$ is a value between x and c .

Taylor's Theorem says that we may approximate a function f with n continuous derivatives, near the point c , with a *polynomial* of degree $n - 1$. The error in the approximation is the term $f^{(n)}(\mu(x))\frac{(x - c)^n}{n!}$.

Exercise 5.3: Use Taylor's Theorem with $c = 0$ and $n = 3$ to show that $\cos(x) \approx 1 - x^2/2$ with an error of size at most $|x|^3/6$. How good is the approximation when $|x| \leq 0.1$?

[†] The notation $n!$ means the product of all the integers from 1 to n .

Power Series Representation

In summation notation Taylor’s formula may be written:

$$f(x) = f(a) + \sum_{k=1}^{n-1} f^{(k)}(a) \frac{(x-a)^k}{k!} + f^{(n)}(\mu(x)) \frac{(x-a)^n}{n!}$$

If a function has an infinite number of derivatives and the last term in the above formula goes to zero in a sufficiently “well behaved” manner (which is beyond the scope of this book to describe) as n goes to ∞ , then we may write

$$f(x) = f(a) + \sum_{n=1}^{\infty} f^{(n)}(a) \frac{(x-a)^n}{n!}$$

One function that has an infinite number of derivatives is the exponential function, $f(x) = e^x$. We know that $f'(x) = e^x$, and consequently $f^{(n)}(x) = e^x$. It is also a function that is sufficiently “well behaved”. Letting $a = 0$ we may write[‡]

$$e^x = e^0 + \sum_{n=1}^{\infty} e^0 \frac{x^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

This says that e^x can be thought of as an “infinite polynomial”. Many other such functions can be represented in this way, for instance the trig functions, sin and cos. Such representations can be used to find (among other things) an approximation to a function, its derivative, or its integral.

Finally, notice that the value e itself can be approximated for a given N by

$$e = e^1 = \sum_{n=0}^{\infty} \frac{1^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \approx 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{N!}$$

Summary

This chapter recast the derivative: not just a slope, but the best *linear approximation* to a function near a point.

1. Bounded an integral above and below by rectangles, then “leveled the dust” to get the Mean Value Theorem: $f(b) = f(a) + f'(\mu)(b-a)$.
2. Read the Mean Value Theorem as an approximation: $f(x+h) \approx f(x) + f'(x)h$ – the tangent line approximation.
3. Saw in what sense the tangent line is the *best* line through $(x, f(x))$: its error dies faster than h itself, while any other line’s error is proportional to h .
4. Answered the problem that opened the chapter: used the approximation to estimate how a measurement error propagates to the concrete order.
5. Extended the tangent line to higher degree polynomials – Taylor’s Theorem – with an explicit formula for the error, made rigorous in the appendix “Taylor’s Theorem”.

And once more it is the *same game, bigger board* theme of the preface at work: the “best linear approximation” view of the derivative is the one that survives every enlargement of the playing field.

Function	Derivative (best linear approximation)
$f : R \rightarrow R$	a number (the slope)
$f : R^n \rightarrow R$	a vector (the gradient)
$f : R^n \rightarrow R^m$	a matrix (the Jacobian)

[‡] The notation $0!$ is defined to be 1. In this context we also adopt the convention that $x^0 = 1$ for *all* x – including $x = 0$ – so that the $n = 0$ term of the series makes sense at $x = 0$.

Chapter 6: Introduction to Difference Equations

Motivational Problem

If a person buys a house they will almost surely need a loan from a bank to pay for it. Once you borrow the money, the bank will have you spread the payments out over 15 to 30 years. You're not simply paying back the money to the bank; it charges interest on the use of its money. So, each payment can be thought of as consisting of two pieces: a payment for the principal borrowed and a second piece which is interest on the outstanding debt. We now describe what happens at any given month.

At month n let x_n be the outstanding debt we owe to the bank. The bank has decided that at the end of each month we pay p dollars for the loan. Also, during the month, we are charged interest (for that month) on the outstanding debt as of the beginning of that month. The rate of interest is given as an annual percentage rate (APR) and so must be divided by 12 to get the monthly interest rate, $r = \text{APR}/12$. At the beginning (month zero) the outstanding debt is known, it is just the loan amount; we'll call it L .

We would like to know what the outstanding debt is at any given month. Our strategy is similar to what one would do in an algebra class: rather than writing down $x = \text{answer}$, one first found a relation that x satisfied and then solved for x . We will do the same thing here. Rather than write down a formula for the outstanding debt at month n , x_n , we first find a relation that x_n satisfies and then solve for x_n . Such a relation can often be stated as a change or a *difference* from one time step to the next. This change is often easy to describe. For the problem at hand, the change in debt from one month to the next is the amount of new debt accumulated by interest less the monthly payment. Mathematically, this can be written as:

$$\underbrace{\text{Debt at month } n+1}_{x_{n+1}} = \underbrace{\text{Debt at month } n}_{x_n} + \underbrace{\text{Interest on debt over the month}}_{rx_n} - \underbrace{\text{Monthly payment}}_p$$
$$x_0 = L$$

This can be rewritten as:

$$x_{n+1} = (1+r)x_n - p$$
$$x_0 = L$$

Comparison with the Solution Strategy of Algebra

Recall that in high school algebra one solved “word” problems in a two step procedure:

1. Label the unknown *number*, x , and involve it in an algebraic equation.
2. Treat the equation as a “balance scale” and apply the operations of addition/subtraction and/or multiplication/division equally on both sides in an attempt to isolate x on one side – its value is then on the other side of the equation.

Here, the unknown labeled, x , is not a number, it is a *sequence* of numbers; so, what does it mean to solve for x ? Ideally, we would like a *formula* for the n^{th} element of the sequence x in terms of this n . Then, when someone asked for the debt at month 17, we could just plug 17 into this formula and out would pop the debt for that month.

Basic Solution Strategy

The equation we wrote down for the debt as a function of the month was an example of a *difference* equation: each term in the sequence is a function of previous terms. The difference and summation calculus of the first two chapters will play an important role in solving such equations. In the equation above, a given element of the

sequence is a function of the previous element. Such an equation is called a first order *difference equation*. The form of this relation between previous elements determines how easy it is to solve, depending on what the meaning of “solve” is. Before we start, let us be clear: The solution to any difference equation is not a number; that is, the unknown we are looking to find is an entire *sequence* – which we have said is a *function* on the integers.[†] Again, we are doing something very different from high school algebra – we are solving for a *function*.

Let’s start with a simpler difference equation. In fact, let’s start with the simplest first order difference equation relating a given value of a sequence with its previous value:

$$\begin{aligned}x_{n+1} &= x_n \\ x_0 &= L\end{aligned}$$

We can solve this difference equation by using the equation to write down the solution term by term. That is, we know $x_0 = L$. The first equation tells us that $x_1 = x_0$; and since we know x_0 , we know x_1 . Continuing, we have $x_2 = x_1$; and since we know x_1 , we know x_2 , it is just $x_1 = x_0 = L$. We can continue and produce each value of the sequence. It is tedious, but effective. Every difference equation can be solved by doing this sort of computation. In theory, any element of the sequence could be produced this way. In practice, one would compute as many elements of the sequence as one needed.

A better solution though, would be to write down a *formula* for any given element of the sequence. In our present example, we find that the formula for the n^{th} element of the sequence is $x_n = L$. Having a formula means one can, on request, produce any element of the sequence without having to compute *any* of the previous elements. The question is

“When presented with a difference equation, can we write down a formula for the n^{th} term of the sequence satisfying such an equation?” The answer is no in general. We will, however, be able to find a solution (in this sense) to our motivational problem.

The problem with finding a formula, in general, is that the n^{th} value of the sequence which satisfies such an equation is tangled up with the previous values. Because of this, it is not a simple matter to write down a formula for the n^{th} value. We introduce a technique which will allow us to solve a larger array of equations. Consider the previous simple difference equation; we know by the Fundamental Theorem of Difference Calculus that it is easy to compute the sum of telescoping differences. Therefore, we may rewrite the last equation as:

$$\begin{aligned}x_{n+1} - x_n &= 0 \\ x_0 &= L\end{aligned}$$

This is

$$\begin{aligned}\Delta x_n &= 0 \\ x_0 &= L\end{aligned}$$

Summing both sides of the first equation from 0 to $N - 1$ we have:

$$\sum_{n=0}^{N-1} \Delta x_n = \sum_{n=0}^{N-1} 0$$

After summing (using the FTDC on the left hand side):

$$x_N - x_0 = 0$$

Or, $x_N = L$. We were able to untangle the connections of the sequence by summing them out. This left an element of our choosing, N , and the 0^{th} term plus whatever the sum was on the right hand side of the equation. Notice

[†] Described in the sub-section “Sequences as Functions” in the first chapter.

that this is the same game we have been playing since algebra: we isolated the unknown under the operator Δ , then inverted Δ by applying its inverse, summation – the *isolate and invert* move of the preface.

If we modify the difference equation so that it becomes:

$$\begin{aligned}x_{n+1} &= x_n + b_n \\ x_0 &= L\end{aligned}$$

This is

$$\begin{aligned}\Delta x_n &= b_n \\ x_0 &= L\end{aligned}$$

Summing both sides of the first equation from 0 to $N - 1$ gives:

$$\sum_{n=0}^{N-1} \Delta x_n = \sum_{n=0}^{N-1} b_n$$

After summing (using the FTDC):

$$x_N - x_0 = \sum_{n=0}^{N-1} b_n$$

Or, $x_N = L + \sum_{n=0}^{N-1} b_n$. If $b_n = b$, that is, b_n is constant, then $x_N = L + bN$.

Solution of the Motivational Problem

We are moving toward the solution of our motivational problem. To get a step closer, consider the following difference equation:

$$\begin{aligned}x_{n+1} &= a x_n \\ x_0 &= L\end{aligned}$$

A solution to this difference equation is a sequence which has L for its initial value and then multiplies the current value by a to get the next. It is not too hard to guess that the solution[†] is $x_n = a^n L$. Let us check this is indeed a solution[‡].

We need to check two things. First, that $x_0 = L$. This is so, since $x_0 = a^0 L = 1 * L = L$. We also need to check $x_{n+1} = a x_n$. But $x_{n+1} = a^{n+1} L = a(a^n L) = a x_n$. So this is a solution. Remember, a solution to a difference equation is a *sequence*, not a number.

Now let's look at an equation that has the same form as our original problem:

$$\begin{aligned}x_{n+1} &= a x_n + b_n \\ x_0 &= L\end{aligned}$$

It is not as easy to guess a solution to this. Can we also use the differencing technique we used earlier? Well, if we rewrite the equation so that the left hand side is a difference as before, then:

$$\begin{aligned}x_{n+1} - x_n &= (a - 1)x_n + b_n \\ x_0 &= L\end{aligned}$$

This does not help us untangle the sequence of unknowns, x_n . If we try to sum this equation, the left hand side reduces to x_N and x_0 . However, the right hand side contains the unknowns for which we are trying to solve. For this technique to work we need to get all of the x_n 's on the left as part of *some* telescoping difference (not necessarily a difference in x_n alone), leaving the right hand side containing the rest of the *known* quantities.

[†] The pattern emerges quickly. $x_1 = a x_0 = a L$; $x_2 = a x_1 = a a L = a^2 L$; $x_3 = a x_2 = a a^2 L = a^3 L$.

[‡] In fact, it is *the* solution.

Summing the left hand side will reduce to two terms; from there we can solve for x_N . To start, let's rewrite the equation as a difference of the unknowns (not yet a telescoping difference):

$$\begin{aligned}x_{n+1} - a x_n &= b_n \\ x_0 &= L\end{aligned}$$

For this to work, the left hand side itself need not be a telescoping difference – it is enough that some known *multiple* of it is. So we play the *change the representation* move: find a known sequence $\{y_n\}_{n=0}^\infty$ so that, after multiplying our equation through by a suitable term of y , the left hand side becomes the difference of a product,

$$\Delta(y_n x_n) = y_{n+1} x_{n+1} - y_n x_n$$

– the very quantity whose expansion is the Discrete Product Rule of chapter 1. Now compare the shape we want with the shape we have:

$$\overbrace{y_{n+1} x_{n+1} - y_n x_n}^{\text{what we want}} \quad \text{versus} \quad \overbrace{x_{n+1} - a x_n}^{\text{what we have}}$$

The leading terms tell us which multiplier to use: multiplying our equation by y_{n+1} makes the leading terms agree. This gives:

$$\begin{aligned}y_{n+1} x_{n+1} - (a y_{n+1}) x_n &= b_n y_{n+1} \\ x_0 &= L\end{aligned}$$

Comparing the trailing terms then forces the choice of the sequence y : the left hand side is *exactly* the telescoping difference $\Delta(y_n x_n)$ provided

$$a y_{n+1} = y_n$$

Notice that nothing was pulled out of a hat. The multiplier y_{n+1} and the condition on y were both *forced* on us by matching our equation against the one difference-of-a-product formula we own. With this choice of y we have a telescoping difference on the left hand side

$$\begin{aligned}y_{n+1} x_{n+1} - y_n x_n &= b_n y_{n+1} \\ x_0 &= L\end{aligned}$$

Such a multiplier sequence y is called a *summing factor* – and when we solve differential equations in the next chapter, we will meet its continuous twin, the *integrating factor*.

The unknowns now sit under a telescoping difference, and summing is its inverse – the *isolate and invert* move finishes the job. Summing from 0 to $N - 1$ gives us $y_N x_N - y_0 x_0 = \sum_{n=0}^{N-1} b_n y_{n+1}$. Solving for x_N gives

$$x_N = \left(\frac{y_0 L + \sum_{n=0}^{N-1} b_n y_{n+1}}{y_N} \right)$$

So, we're done provided we know how to find the sequence y . But the sequence y satisfies its own difference equation: $y_{n+1} = a^{-1} y_n$. From what we discovered above, the solution is $y_n = a^{-n} y_0$. There are no further restrictions so we may choose y_0 to be anything; let's set it to 1. Therefore, we may write down a solution for the N^{th} element of the sequence x to be

$$x_N = a^N L + a^N \sum_{n=0}^{N-1} a^{-(n+1)} b_n$$

When b_n has the same value, say b , for all n this formula simplifies to

$$x_N = a^N L + b a^{N-1} \sum_{n=0}^{N-1} a^{-n}$$

Which simplifies to

$$x_N = a^N L + b a^{N-1} \left(\frac{a^{-N} - 1}{a^{-1} - 1} \right)$$

provided $a \neq 1$ (we used the following fact about geometric series: $\sum_{n=0}^{N-1} z^n = \frac{z^N - 1}{z - 1}$, for $z \neq 1$; used above with $z = a^{-1}$). This can also be written as:

$$x_N = a^N L + b \left(\frac{1 - a^N}{1 - a} \right)$$

The two terms of this solution each tell a story. The first term, $a^N L$, solves the equation with no payments ($b = 0$) and carries the initial condition; the second term solves the full equation starting from zero. Because the equation is *linear*, the two pieces simply add – this is the *exploit linearity* move (*superposition*).

Applying this formula to the motivational problem with $a = (1 + r)$, and $b = -p$ yields a formula for the debt owed at month N :[‡]

$$x_N = (1 + r)^N L - p \left(\frac{1 - (1 + r)^N}{-r} \right)$$

which is

$$x_N = (1 + r)^N L - p \left(\frac{(1 + r)^N - 1}{r} \right)$$

Example 6.1: If a 30 year loan is taken out on \$100,000 at an annual interest rate of 6% with monthly payment \$600, what is the outstanding debt after 5 years?

The monthly rate is $r = 0.06/12 = 0.005$, $L = 100,000$, $p = 600$, and $N = 5 * 12 = 60$. Plugging these values into the solution to our difference equation gives

$$x_{60} = (1.005)^{60} * 100000 - 600 * \left(\frac{1.005^{60} - 1}{0.005} \right)$$

After evaluating the expression on the right hand side we obtain $x_{60} = 93,023$. This is the outstanding debt after 5 years of monthly payments. During this time you will have paid $5 * 12 * 600 = 36,000$ dollars to the bank.

The graph of the solution over the full 30 years tells the story at a glance.

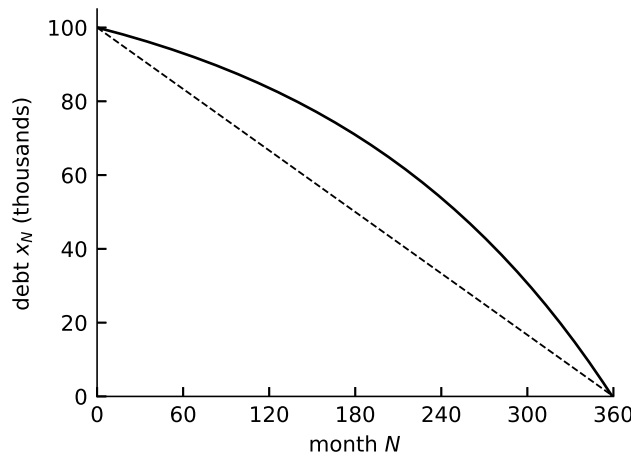


Figure 6.1: Outstanding debt x_N for the loan of the example ($L = \$100,000$, $r = 0.005$, $p = \$600$). The dashed line shows what a straight-line payoff would look like. Notice how slowly the debt falls in the early years: at first nearly all of each payment goes to interest.

[‡] Note $a \neq 1$ so we may use the just derived formula.

Exercise 6.1: Solve the difference equation $x_{n+1} = 2x_n + 1$, $x_0 = 0$, using the formula of this section. (The answer, $x_N = 2^N - 1$, counts the moves needed to solve the Tower of Hanoi puzzle with N disks.)

Exercise 6.2: You deposit L dollars in a savings account earning monthly interest rate r , and add p dollars at the end of each month. Write down the difference equation satisfied by the balance and solve it. How does the solution compare, term by term, with the loan formula above?

Summary

We are able to solve a simple difference equation by first rewriting the tangling of x 's as a telescoping difference and then use the Fundamental Theorem of Difference Calculus to “sum away” the tangles. This leaves one element of the sequence, x_N , on one side of the equation and the sum of the known values on the other side of the equation.

In fact, this whole chapter was first year algebra replayed with a new kind of unknown – the same game on a bigger board. Here is the dictionary:

Algebra	Difference Equations
unknown is a number x	unknown is a sequence $\{x_n\}$
find an equation x satisfies	find an equation $\{x_n\}$ satisfies
do the same to both sides (balance scale)	multiply both sides by a summing factor
undo multiplication by dividing	undo Δ by summing (the FTDC)
isolate x ; a number remains	isolate x_N ; a formula remains

The strategic moves were the ones we have played since algebra: *change the representation* (make the left hand side a telescoping difference), then *isolate and invert* (sum away the difference).

What's Left

A difference equation is effectively any equation which relates a given element of a sequence with previous values of the sequence. Obviously, they can be much more complicated than the ones we've examined. A careful study of the subject classifies difference equations into various categories and then provides recipes for solutions in some cases. The main solution technique is to “sum away” the differences; that is, use the Fundamental Theorem of Difference Calculus.

Fibonacci Sequence

As an example we consider a more complicated difference equation. The Fibonacci sequence occurs in many contexts, one of which is the modeling of rabbit populations. The sequence is defined through a recurrence relation (a difference equation) given by:[†]

$$F_{n+2} = F_{n+1} + F_n, \quad n \geq 0$$

$$F_0 = 1$$

$$F_1 = 1$$

This is categorized as a second order difference equation because a given element in the sequence is a function of the previous two values.

How might one solve it? Recall that earlier in this chapter the first order equation $x_{n+1} = ax_n$ had a *geometric* solution: $x_n = a^n L$. A second order equation is the same game on a bigger board, so we try the same *shape*

[†] Some texts index the sequence differently, starting with $F_0 = 0$ and $F_1 = 1$; this produces the same list of numbers shifted by one place.

of solution – a change of representation: try a solution of the form $x_n = \lambda^n$, where the number λ is left to be determined. Substituting into the recurrence and dividing by λ^n forces $\lambda^2 = \lambda + 1$, a quadratic whose roots, $(1 \pm \sqrt{5})/2$, involve the famous *golden ratio*. Because the recurrence is *linear*, any combination $A\lambda_1^n + B\lambda_2^n$ of the two solutions is again a solution, and the two initial values pin down A and B .

Exercise 6.3: Carry this out: find A and B so that $F_n = A\lambda_1^n + B\lambda_2^n$ satisfies $F_0 = F_1 = 1$, and thereby obtain a formula for the n^{th} Fibonacci number. Check your formula by computing F_2 and F_3 .

Chapter 7: Introduction to Differential Equations

Motivational Problem

The population of a species of bacteria, represented by $p(t)$, is the number of bacteria on a given surface. The evolution of the population is modeled by the assumption that its “instantaneous” growth rate is proportional to the amount of bacteria currently present. The growth is impeded by a chemical on the surface that kills bacteria at a rate of b bacteria per second. These facts may be collected into an equation:

$$\begin{aligned}p'(t) &= r p(t) - b \\ p(0) &= P\end{aligned}$$

Here, r is the proportionality coefficient and has units 1/second. With this choice the units of the two sides of the equation match: the left hand side is the limit, $\lim_{h_n \rightarrow 0} \frac{p(t+h_n) - p(t)}{h_n}$ (where $h_n \rightarrow 0$), which has units of bacteria per second; on the right hand side, $r p(t)$ has units 1/second $\times$ bacteria – bacteria per second again – matching b , which is given in bacteria per second.

This equation is called a differential equation which you can think of as an infinitesimal difference equation. To see why, chop time into small steps of length h and let $x_n = p(nh)$. Over one step the population changes by approximately h times its rate of change:

$$\begin{aligned}p(t+h) &\approx p(t) + h(r p(t) - b) \\ x_{n+1} &= x_n + h(r x_n - b)\end{aligned}$$

The second line is a difference equation of exactly the kind we solved in the last chapter; the differential equation is what remains as $h \rightarrow 0$. This is the *approximate, then take a limit* move run in reverse. How do we solve this and what does a solution mean? The solution is a *function* p which satisfies the above equations. We will find an explicit formula for the solution.

Basic Solution Strategy

Let's start with a simpler differential equation. In fact, let's start with the simplest differential equation that we can think of.

$$\begin{aligned}x'(t) &= 0 \\ x(0) &= P\end{aligned}\tag{1}$$

To solve this equation we must find a *function* such that its derivative is 0. That is, the tangent at each point is flat. We know x must be a constant function. Since the initial value is P , the function must be $x(t) = P$. Another way to solve this equation is to use the Fundamental Theorem of Calculus. We do this by integrating out the derivative. Integrating (1) from 0 to t gives

$$\begin{aligned}\int_0^t x'(s) ds &= \int_0^t 0 ds \\ x(t) - x(0) &= 0\end{aligned}$$

Therefore, $x(t) = x(0)$, and since $x(0) = P$, we have $x(t) = P$.

As with difference equations, there is no technique for writing a formula for the solution of a general differential equation. Again, as with difference equations, there are limited classes of equations for which we can write down

a solution. However, unlike difference equations there is a question of existence and uniqueness of a solution. It turns out there are differential equations which don't have solutions, and others for which the solution is not unique. When we say that a solution doesn't exist, we don't just mean that we can't find a formula to express the solution; we mean that there are some equations for which there is no function which satisfies the differential equation. And, there are differential equations which have more than one solution.[†]

The way to find formulas for the solutions to these problems is to use the Fundamental Theorem of Calculus to integrate away the derivative. This is analogous to the way we solved difference equations: we used a summation operator to get rid of the telescoping differences. That is, we used the Fundamental Theorem of Difference Calculus to find solutions.

Solution of the Motivational Problem

Let's look at an equation that is closer to our motivational problem. Consider the simpler problem:

$$\begin{aligned} p'(t) &= r p(t) \\ p(0) &= P \end{aligned} \tag{2}$$

If we integrate both sides we can use the Fundamental Theorem of Calculus to integrate away the derivative on the left hand side of the equation. However, we don't know what the integral of the right hand side is since it involves the unknown function, p . We need a way to combine all the p 's together and view them as the derivative of some function of p . Then, the Fundamental Theorem of Calculus can be used to eliminate the derivative. We then will have an equation which does not have derivatives, and therefore, we can use algebra to find p . To do this divide both sides by p – assuming $p(t) > 0$, which is reasonable for a population; we now have all the p 's (the unknowns) on one side of the equation. (The *integrating factor* method we use below for the full problem avoids this division entirely.)

$$\begin{aligned} \frac{p'(t)}{p(t)} &= r \\ p(0) &= P \end{aligned}$$

The left hand side is the derivative of $\ln(p)$.[‡] Integrating both sides from 0 to t we find:

$$\ln(p(t)) - \ln(p(0)) = r(t - 0)$$

Or,

$$\ln(p(t)) = \ln(p(0)) + r t$$

The inverse function of $\ln(x)$ is e^x ; therefore, exponentiating both sides of the equation gives us

$$p(t) = e^{\ln(p(0)) + r t} = e^{\ln(p(0))} e^{r t} = p(0) e^{r t} = P e^{r t}$$

To solve our motivational problem, we try to do the same thing; that is, arrange the p 's on one side in such a way that they are the derivative of some expression involving p . Then, use the Fundamental Theorem of Calculus to integrate away the derivative. We then have an ordinary equation for which we solve for p .

To do this we employ a technique analogous to what was done for the motivational problem in the chapter on difference equations. First move the p on the right hand side to the left hand side – grouping all the p 's together. Next, multiply both sides by some (as yet unknown) function $y(t)$. Our motivational problem becomes:

$$y(t)p'(t) - ry(t)p(t) = -by(t) \tag{3}$$

[†] A discussion of these issues is beyond the scope of this book.

[‡] This follows from the chain rule.

The goal is to choose y so that the left hand side is the derivative of the product, $(y(t)p(t))$. The derivative of this product is

$$(y(t)p(t))' = (p(t)y(t))' = p'(t)y(t) + p(t)y'(t) = y(t)p'(t) + y'(t)p(t)$$

To match the left hand side of (3) we need y to satisfy $y'(t) = -ry(t)$. But this is an equation we know how to solve, the solution is $y(t) = Ce^{-rt}$, for some constant C . Since there are no further restrictions on y , we take C to be 1. Such a multiplier function y is called an *integrating factor* – the continuous twin of the *summing factor* we met in the chapter on difference equations. So, equation (3) becomes

$$(e^{-rt}p(t))' = -be^{-rt}$$

Integrating both sides of this equation from 0 to t gives:[†]

$$\int_0^t (e^{-rs}p(s))' ds = \int_0^t -be^{-rs} ds$$

It is not clear what the anti-derivative of e^{-rs} is however. We do know that the derivative of e^{ax} is ae^{ax} from the subsection “Derivatives of Log and Exponential Functions” in the chapter on Differential Calculus. From this we can guess that the anti-derivative of e^{ax} is $\frac{e^{ax}}{a}$; which can be verified by taking its derivative.

Now, the integral above can be evaluated to give

$$e^{-rt}p(t) - e^{-r0}p(0) = \frac{b}{r}e^{-rt} - \frac{b}{r}$$

Solving for p we obtain (since $e^0 = 1$ and $p(0) = P$)

$$p(t) = Pe^{rt} + \frac{b}{r}(1 - e^{rt})$$

This may be written:

$$p(t) = \frac{b}{r} + (P - \frac{b}{r})e^{rt}$$

This solution has the same structure as the loan solution of the last chapter. The constant piece, $\frac{b}{r}$, is an *equilibrium* solution of the full equation (if $P = \frac{b}{r}$, then $p'(t) = 0$ and the population never moves); the exponential piece, $(P - \frac{b}{r})e^{rt}$, solves the no-harvest equation $p'(t) = rp(t)$. Because the equation is *linear*, the two pieces simply add – the *exploit linearity* move (*superposition*) again.

Notice that if $P - \frac{b}{r} < 0$ then $p(t)$ will eventually become zero. Let us assume that this is the case and find the time to extinction. That is, find t^* such that $p(t^*) = 0$. The equation to solve for t^* (put t^* on one side and everything else on the other) is

$$p(t^*) = 0 = \frac{b}{r} + (P - \frac{b}{r})e^{rt^*}$$

After some algebra this becomes

$$e^{rt^*} = -\frac{\frac{b}{r}}{P - \frac{b}{r}} = \frac{\frac{b}{r}}{\frac{b}{r} - P}$$

Taking the $\log_e$ (that is, $\ln$) of both sides and solving for t^* gives

$$t^* = \frac{1}{r} \ln \left(\frac{\frac{b}{r}}{\frac{b}{r} - P} \right) = \frac{1}{r} \ln \left(\frac{b}{b - rP} \right)$$

So, in the case that $P < \frac{b}{r}$ (which is the same as $rP < b$) the population of bacteria dies out at this t^* . That makes sense since initially the rate of change of the population is $p'(0) = rp(0) - b = rP - b < 0$; so the population starts out decreasing. A smaller population makes $p'(t) = rp(t) - b$ even more negative, and the decline feeds on itself. So, once the population starts to shrink it becomes a death spiral for the bacteria.

The whole family of solutions can be seen in one picture.

[†] We're using integration to collapse the infinitesimal telescoping differences; that is, we use the Fundamental Theorem of Calculus to get a simple difference. Notice we used a different integration variable, s , to avoid confusion with t .

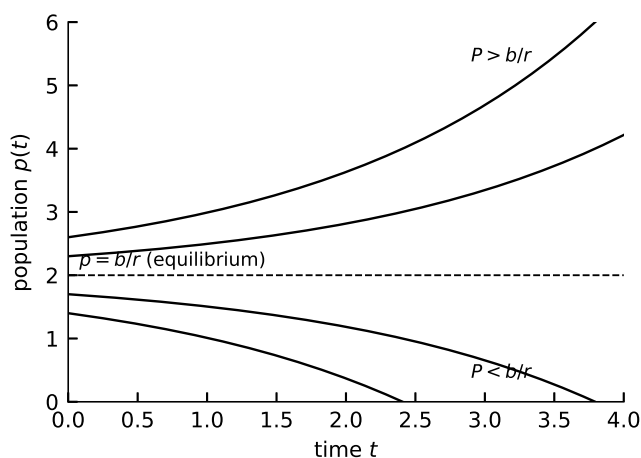


Figure 7.1: The family of solutions $p(t) = \frac{b}{r} + (P - \frac{b}{r})e^{rt}$ for several starting populations P . The dashed line is the equilibrium $p = b/r$: populations starting above it grow without bound, populations starting below it spiral down to extinction.

It is interesting to note that the formula for $p(t)$ becomes negative for $t > t^*$. Obviously, this doesn't make physical sense – the model no longer holds after the critical time t^* . Notice also that our model predicts that there are a non integral number of bacteria at almost any given time. The model could be corrected by removing the fractional part of the predicted population size by either truncating or rounding the result. One could think of this extra step as part of an improved model.

Keep in mind that models are always an approximation to the truth. For instance, although the model that we have may work over short intervals of time, there is no population that satisfies this equation for too long because as the population rises other factors in the environment inhibit growth.

Radioactive Decay

We apply this technique to a problem in physics.

A radioactive substance decays at a rate proportional to the amount present. If one starts with Q pounds of the substance, in 1,000 years there will be $Q/4$ pounds. When will there be $\frac{1}{2}Q$ pounds?

The equation is similar to the previous problem:

$$\begin{aligned}a'(t) &= -r a(t) \\ a(0) &= Q\end{aligned}$$

Here, $a(t)$ is the amount in pounds at time t in years; and, r is taken to be a positive constant, whose units must be $1 / \text{years}$ for the equation to make sense. We don't know what r is but we know that in 1,000 years there will be a quarter of Q left. Solving the equation gives $a(t) = Q e^{-rt}$. When t is 1,000 the equation reads $a(1000) = Q e^{-1000r} = \frac{1}{4}Q$. We may solve this equation for r by taking the $\log_e$ (that is, $\ln$) of the last equation. This gives us $\ln(Q) - 1000r = \ln(\frac{1}{4}Q) = \ln(1) - \ln(4) + \ln(Q) = \ln(Q) - \ln(4)$ (since $\ln(1) = 0$). Or, $-1000r = -\ln(4)$. So, $r = \frac{\ln(4)}{1000}$. To find the time t when $a(t) = \frac{1}{2}Q$, we need to solve: $a(t) = Q e^{-rt} = \frac{1}{2}Q$ for t (the only unknown in the equation is t as we now know r). Taking the $\ln$ of both sides gives us $\ln(Q) - rt = \ln(\frac{1}{2}Q) = \ln(1) - \ln(2) + \ln(Q) = \ln(Q) - \ln(2)$. So $t = \ln(2)/r = 1000 \ln(2)/\ln(4) = 500$.

That is, the time it takes the initial amount to decay to $\frac{1}{2}Q$ pounds is 500 years. Notice that the answer is independent of the initial amount Q . In other words, the “half-life” of this radioactive material is an intrinsic attribute of this substance.

Exercise 7.1: (Newton’s Law of Cooling) A cup of coffee at temperature $T(t)$ sits in a room at constant temperature A . It cools at a rate proportional to the difference between its temperature and the room’s: $T'(t) = -k(T(t) - A)$, with $T(0) = T_0$ and k a positive constant. Solve for $T(t)$ using an integrating factor. What does your solution predict as t grows large?

Exercise 7.2: A radioactive substance has a half-life of 100 years. How long until only 10% of the original amount remains? (First use the half-life to find r , as in the subsection above.)

Summary

The strategy of this chapter is the one we have been playing since first-year algebra: *isolate and invert*. We isolate the unknown function under the derivative operator d/dt – multiplying by an *integrating factor* when needed to make the left hand side a single derivative – and then invert d/dt by applying its inverse, integration, in the form of the Fundamental Theorem of Calculus. This is precisely what we did in the last chapter with Δ , summation, and the Fundamental Theorem of Difference Calculus. The whole book culminates in the following dictionary:

Discrete	Continuous
$x_{n+1} = a x_n$	$p'(t) = r p(t)$
geometric growth, a^n	exponential growth, e^{rt}
Δ	d/dt
summation	integration
FTDC	FTC
summing factor	integrating factor

The two columns are tied together by the correspondence $a = 1 + r$: for small r we have $(1 + r)^n \approx e^{rn}$, which is why the solution of the loan problem and the solution of the population problem look so alike.

Exercise 7.3: Make the correspondence concrete: for the loan of the last chapter the monthly rate was $r = 0.005$ and the term was $N = 60$ months. Compare $(1.005)^{60}$ with $e^{0.005 \times 60} = e^{0.3}$. How close are they?

And just as the Fibonacci sequence hinted at a second order discrete story, there is a continuous second order story – $x'' + x = 0$, the equation of the swinging pendulum. Its full treatment awaits a course on differential equations, but you can already verify its heart – and cash the promise made when the trig derivatives were introduced:

Exercise 7.4: Using the derivatives of sine and cosine from the chapter on Differential Calculus, check that $x(t) = \sin(t)$ and $x(t) = \cos(t)$ both satisfy $x'' + x = 0$. There is the oscillation the trig functions were built for. Then check that *any* combination $A \sin(t) + B \cos(t)$ is also a solution – the *exploit linearity* move (superposition) once more, just as it was for the two Fibonacci solutions λ_1^n and λ_2^n .

What’s Left

A differential equation is effectively any equation which involves a derivative. Obviously, they can be much more complicated than the ones we looked at. A careful study of the subject classifies differential equations into various categories and then provides recipes for solutions in some cases. The main solution technique is to “integrate away” the derivatives; that is, use the Fundamental Theorem of Calculus.

Appendix A: Functions, Formulas, Graphs, and Notation

In the main text we freely wrote formulas like $f(x) = x^2$ and drew graphs of functions – so you may well ask: are the formula, the graph, and the function all the same thing? For the patchwork functions and sequences of this book the distinction matters, and this appendix pins it down. As mentioned earlier in this book functions are rather abstract things. Functions are not numbers or a collection of numbers; they are rules which assign one number to another. As such, we are tempted to identify anything tangibly associated with a function *as the function*. The two things that people often confuse with a function are formulas – usually used to define a function – and the graph of a function. There is a third problem which comes from the way mathematicians and others notate functions.

Before we clarify these concepts let us emphasize that although functions are not formulas, the kinds of functions which make calculus computationally tractable are the ones which are defined by formulas or are made from a patch work of formulas.

Functions are NOT Formulas

You will often hear the terms *function* and *formula* used interchangeably. One may hear “This is a function that describes such and such.” Or, “Here is a formula for computing such and such.” Many people think they are the same thing. In fact, mathematicians a few centuries back *did* use the two terms interchangeably. They were driven to separate these notions through the problems they faced over the last few centuries – just as we are being driven to introduce the concepts of Calculus by a particular set of problems. A few definitions will help draw this distinction.

Definition(Function). A function, f , takes as input a number from a set called the domain of f , and produces one number as output. The set of outputs of f (using input values from the domain) is called the range of f .

A simple example is the function which squares its input. If we call this function f , then f is defined by $f(x) = x^2$. We can take the domain of this function to be any set where the squaring operation makes sense. One usually takes the domain to be the largest possible set one can. In this case, the domain is usually taken to be all real numbers. As a second example, consider the function which takes the square root of its input. If we call this function g , then g is defined by $g(x) = \sqrt{x}$. We can take the domain of this function to be any set where the square-root operation makes sense. Since taking square roots of a number is only valid for non-negative numbers, we can take the domain of this function to be any subset of the non-negative numbers. One usually takes the domain to be the non-negative numbers. Lastly, consider the family of functions, $f(x) = x^n$ with integer n , $n < 0$. The domain of these functions is taken to be the set of all numbers except 0; that is, all places where the function makes sense.

A formula doesn't have much of a formal definition, but we can define it roughly as:

Definition(Formula). A formula is an “algebraic” implementation of a function. That is, a formula is an “algebraic rule” which takes a numeric input and produces a numeric output.

Examples of formulas are:

1. $1 + 2x$
2. $x^2 + 3x + 2$
3. $\sin^2(x) + x + \tan(x)$
4. $\frac{1+x^2}{2x-7}e^x + 2\tan(x)$

These formulas can be used to define functions. Listed below are the associated functions from the formulas above.

1. $f_1(x) = 1 + 2x$
2. $f_2(x) = x^2 + 3x + 2$
3. $f_3(x) = \sin^2(x) + x + \tan(x)$
4. $f_4(x) = \frac{1+x^2}{2x-7}e^x + 2\tan(x)$

It may seem like we're quibbling over trivial matters. To drive home the point that they are different we present below functions whose definitions go beyond a single formula in various ways – patchworks of formulas, restricted domains, or no formula at all. In some examples, a graph picturing the function is given.

Example 1. f is the absolute value function defined on the real line. There is a simple formula for this function. $f(x) = |x|$. But it may be defined as a compilation of even simpler formulas as:

$$f(x) = \begin{cases} x & \text{if } x \geq 0; \\ -x & \text{otherwise.} \end{cases}$$

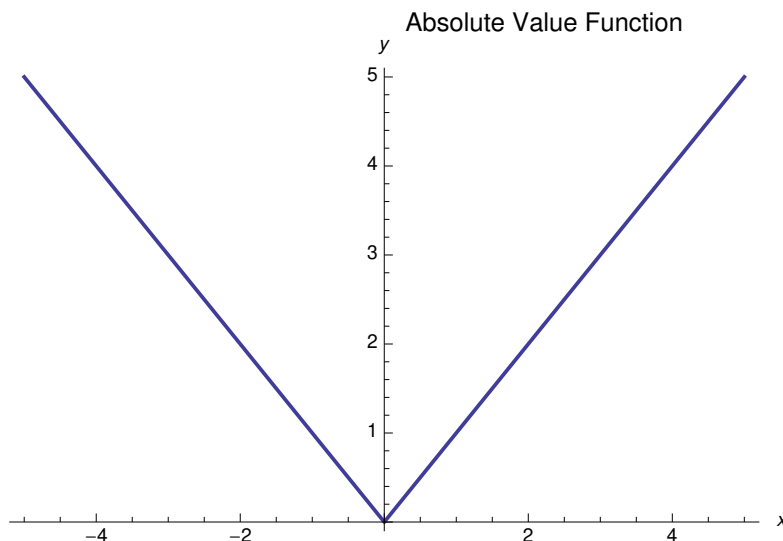


Figure 1: The absolute value function $f(x) = |x|$.

Example 2. g is referred to as a piecewise linear function. It is defined on the real line as:

$$g(x) = \begin{cases} 1 & \text{if } x < 0; \\ 1 + x & \text{if } 0 \leq x < 3; \\ 4 + (3 - x) & \text{if } 3 \leq x < 5; \\ 2 & \text{if } x \geq 5. \end{cases}$$

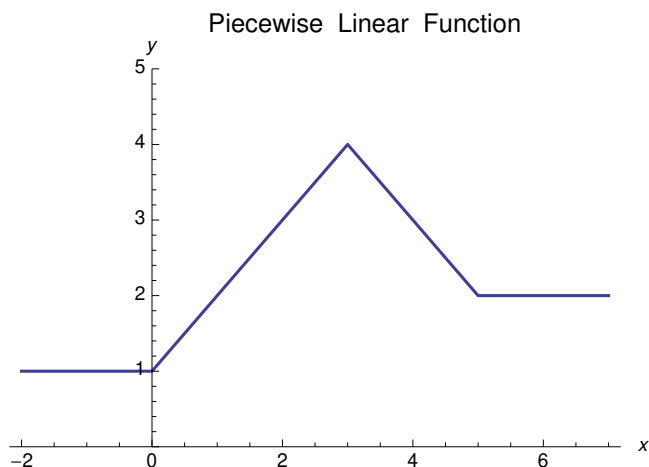


Figure 2: A sample piecewise linear function.

Example 3. h is defined only on the interval $[0, 10]$. That is, its domain is the set $[0, 10]$. This is an example of a discontinuous function; it makes a sudden jump when $x = 5$.

$$h(x) = \begin{cases} x & \text{if } 0 \leq x < 5; \\ 7 & 5 \leq x \leq 10. \end{cases}$$

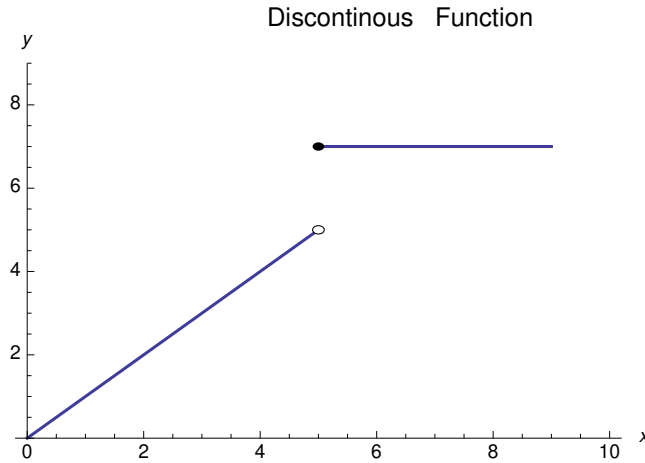


Figure 3: A discontinuous function.

Example 4. d is defined only on the input set $\{1, 2, 3\}$ (its domain) producing values in the output set $\{4, 10, 21\}$ (its range).

$$d(x) = \begin{cases} 10 & \text{if } x = 1; \\ 21 & \text{if } x = 2; \\ 4 & \text{if } x = 3. \end{cases}$$

Example 5. f is defined only on the set of positive integers producing the even numbers.

$$f(n) = 2n$$

Example 6. f is defined only on the set of positive integers producing the prime numbers.

$$f(n) = p_n \quad \text{where } p_n \text{ is the } n^{\text{th}} \text{ prime}$$

Several of the functions above, while not given by a single formula, are patchworks of formulas (Examples 3 and 4). Example 5 is given by a simple formula, $f(n) = 2n$, but on a restricted domain. Example 6 is a function for which one can't write down a formula; nor a (finite) patchwork of formulas as there is no known formula for expressing the sequence of prime numbers.[†]

Let us also try to remove the confusion about functions and their descriptions. The function f defined by $f(x) = x^2$ can also be defined by $f(z) = z^2$. Why? Because both descriptions tell us to do the same thing to their inputs. That is, the input variable is a dummy variable, not unlike a dummy variable used in a “for loop” of a programming language. For instance, consider the two “for loops” below:

```
sum = 0; for(i = 0; i < 10; i++) { sum += i; }
```

```
sum = 0; for(j = 0; j < 10; j++) { sum += j; }
```

The results don't depend on i or j . In the same way, be careful not to get too attached to the dummy variables used in function definitions. We state this one more time with emphasis:

[†] Actually, a relatively concise formula was produced in the 1960s but it is not a practical means of computing primes.

NOTE: The following all define a function named f that squares its input:

$$f(x) = x^2, \quad f(z) = z^2, \quad f(a) = a^2, \quad f(c) = c^2$$

Functions are NOT Graphs

Functions are abstract: a function is not a number but a rule that takes in one number and produces another; it is difficult to get a picture of this. Graphs of functions give us a picture. However, it is worth mentioning again that graphs are NOT functions. A function has a graph, but a graph may not correspond to a function. To make the distinction clearer we need to define what we mean by a graph.

Definition(Graph). A graph is a set of pairs of real numbers. That is, a graph is a subset of $\{(x, y) \mid x, y \in R\}$.[‡]

Definition(Graph of a Function). The graph of a function, f , is the set $\{(x, f(x)) \mid x \in \text{Domain}(f)\}$.

This leads to a question: Given a graph, G , is there a corresponding function, f , for which the graph of f is G ?

The answer, in general, is no. Consider, for example, a graph that is a circle. The graph of a function has the property that for any given input, x , (provided it's in the function's domain) only one output is produced. What does this mean for the graph of a function? It means that if we take a vertical line through x , it should only hit the graph of the function at one point. Because otherwise, if it hit the graph at more than one point, there would be more than one value produced by f – and this can't be. This criterion is called the *vertical line test*: a graph is the graph of a function exactly when every vertical line hits it at most once. So, in the case of the circle graph, we know that this is not the graph of a function because a vertical line hits more than one point for at least one value of x . In fact, a vertical line hits two points of the circle at all but the bounding x values.

Function Notation

In many respects mathematics is a triumph of notation. We use symbols to name and manipulate abstract ideas. It is often not as precise as mathematicians would like, however. There are cases where the short hand of notation is confusing or misleading. One case in particular is the way people refer to functions. In this book you will see functions notated as variables are notated; for example, we might call a function f . In some texts there is a concern that this simple name may confuse a function name with the name of a variable that has a numeric value. In order to add information (in this case to indicate that f is a function) some people use the notation $f(x)$ to try to make things clear. The sense which is to be conveyed by this notation is that f is a function because it takes inputs. The symbol x is often used because it conveys the idea of “variable input”; that is, f can be fed any number as input. This leads to confusion because the usual interpretation of $f(x)$ is f evaluated at x .[†] Other people will use the notation $f(\cdot)$ to more clearly indicate that f is a function. The “dot” serves as a place holder indicating that f feeds on something; that is, it is a function.

Reiterating, we will use the notation $f(x)$ to mean: evaluate the function f at x .

[‡] The symbol R stands for the set of all real numbers.

[†] The only place where $f(x)$ has the sense of f being fed an arbitrary input is when we *define* the function f as in $f(x) = \text{value, or expression involving } x$.

Appendix B: Straight Line Functions

Given a function, we know how to find the graph. What about the reverse? Namely, for a given graph can we find the function which has that graph? In appendix A it is shown that a graph doesn't necessarily correspond to a function. Since a function is just a mapping from one value to another and the graph of a function is just a visual representation of this fact, in some sense we are already done: the graph itself specifies the function. What I really mean is: can you find a *formula* for the function represented by a given graph? This is not possible in general even when the graph comes from a function, but it is for certain graphs. Perhaps the simplest graphs are straight lines, and in this case there are corresponding functions *and* we can find simple formulas for them. We make one restriction on the straight lines we consider: we assume that they are not vertical.

First, non-vertical straight lines are the graphs of functions. Why? Because they pass the vertical line test of Appendix A: a vertical line intersects a non-vertical straight line in at most one point.

Next, to find a formula for the corresponding function, let's start with the graph below. Pick an arbitrary point on the graph, (x, y) .

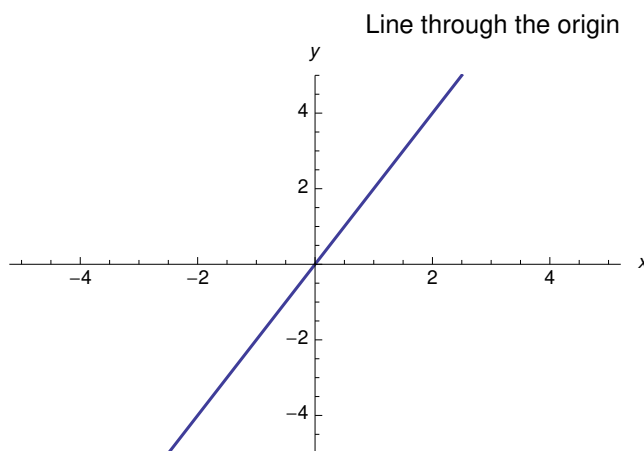


Figure 1: Straight line through the origin.

The strategy will be to examine the constraint of being on the line and see if this translates into a formula for y in terms of the x . Notice, in this case, y is a “rise” of the line and x is the corresponding “run”. Because the line is straight the ratio of the “rise” to the “run” is a constant we’ll call m . Therefore, $y/x = m$, or

$$y = mx$$

(When $x = 0$ the point in question is the origin, $(0, 0)$, itself, and the formula $y = mx$ holds there too.) Thus the function whose graph is the straight line passing through the origin is $f(x) = mx$. Such a function is called a *linear function*.

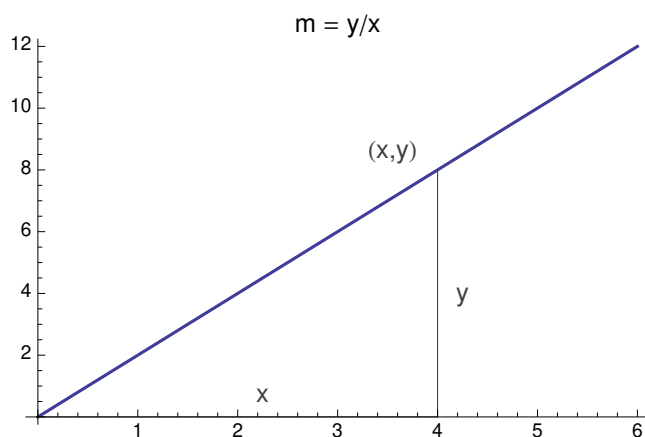


Figure 2: A linear function.

What about the graph in the figure below?

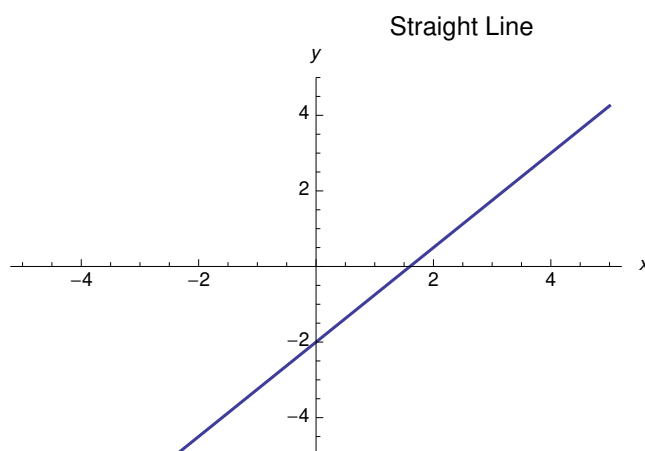


Figure 3: A straight line.

Can we find an equation for this more general line? Let's assume the slope of this line is m and it hits the y -axis at the point $(0, b)$. You should convince yourself this defines the line uniquely. To find the formula whose graph is the line, we take an arbitrary point on the line (x, y) as before and translate the geometric relationship of being on the line into an algebraic one. This is the *change the representation* move of the preface – we trade the geometric statement “lies on the line” for an algebraic one. Again, we will end up with y expressed as a formula in terms of x .

Notice the “rise” from the point $(0, b)$ to (x, y) is $y - b$ while the “run” is x . Since the slope of the line is m and is the ratio of any such “rise” to “run” we have $(y - b)/(x - 0) = m$. Or,

$$y = mx + b$$

(When $x = 0$ the point in question is $(0, b)$ itself, and the formula $y = mx + b$ holds there too.) So the function whose graph is a straight line is given by the function $f(x) = mx + b$. Such a function is called a *rectilinear function* (most books call these simply *linear*; we reserve that word for lines through the origin). In this formula, b is referred to as the y -intercept and m is called the slope. What we derived is a representation for non-vertical straight lines.

Theorem(Straight Line Representation). *A non-vertical straight line is the graph of a function of the form*

$$f(x) = m x + b$$

for some choice of m and b . The values m and b are unique.

Appendix C: Infinite Sequences and Limits

Every derivative in this book was computed by the *approximate, then take a limit* move, and each time the “limit” step was taken on faith. In this appendix we pay that debt by pinning down what it means for a sequence of approximations to head toward a value. That is, we will invent a definition for what we’ve been calling the limit of a sequence. The definition will come from looking at examples. I hope these examples will convince you that the definition given below is reasonable.

First to make life easier when discussing sequences let us define the *tail*, T_N , of a sequence $\{x_n\}_{n=1}^{\infty}$ at N to be the set $\{x_n \mid n > N\}$. You can think about forming the tail by visualizing the sequence as a list of numbers written out sequentially from left to right and then chopping the sequence at a location N ; this leaves the list $[x_1 \dots x_N]$ and the rest which we are calling the tail at N , T_N .

Example 1: Let $x_n = 1/n$. The tail of this sequence at 10, T_{10} , is $\{x_n \mid n > 10\} = \{1/11, 1/12, 1/13, \dots\}$.

You should draw a picture of the tail.

Notice that the sequence in the example seems to be “headed toward” 0. Question: Do you believe this, if so, how do we say this precisely?

We need to gain some experience with sequences in order to extract the essence of the notion of “heading towards”. To this end, let’s look at some examples to help us sharpen our idea of what this “heading toward” or “goes to” might mean. We will use the term *convergence* to mean that a sequence “goes to” a particular value. In this case we say the *limit* of the sequence exists. Our goal will be to give a precise definition of convergence.

Example 2: This example gets close and then “bounces away”, but eventually it “goes to” zero. We would say that this sequence converges to 0.

$$1/1, 100, 1/2, 100/2, 1/3, 100/3, 1/4, 100/4, \dots$$

Example 3: This example oscillates about 0, but gets smaller and smaller in absolute value, moving closer and closer to zero. In this case we would say the sequence converges to 0.

$$-1, 1/2, -1/3, 1/4, -1/5, 1/6, -1/7, 1/8, \dots$$

Example 4: This sequence has elements that get close but then “bounce away”. Therefore, this sequence does not converge to 0 as there is always an element in the tail of the sequence which is “far” away from 0. Nor can it converge to 1 (or to anything else) – every tail also contains elements $1/n$ which are far from 1.

$$1, 1, 1/2, 1, 1/3, 1, 1/4, 1, 1/5, 1, 1/6, 1, \dots$$

Example 5: This sequence has elements that get close but then “bounce away”. Therefore, this sequence does not converge to zero as there is always an element in the tail of the sequence which is “far” away from 0.

$$1, 0.0001, 1/2, 0.0001, 1/3, 0.0001, 1/4, 0.0001, 1/5, 0.0001, \dots$$

Example 6: This sequence has elements that tend to 1. The farther out one goes in the sequence, the tail gets close and closer to 1. In this case we would say this sequence converges to 1.

$$1 + 1, 1 + 1/2, 1 + 1/3, 1 + 1/4, 1 + 1/5, 1 + 1/6 \dots$$

In the examples where the sequence “goes to” a point we see that the tails get closer and closer to this point. In the examples where there wasn’t convergence we saw that getting close is a relative statement. In example 5, individual elements get arbitrarily close to 0, but every tail still contains elements a full 0.0001 away – the tail *as a whole* never gets closer than 0.0001 to 0. So, it is not enough for *elements* of a sequence to occasionally get closer to a limit point. For convergence we need the *tails* to get closer to the limit point; and, not just closer, I think you will agree that we need the tails to get *arbitrarily close* to the limit point. How do we precisely define the phrase *arbitrarily close*?

One way to say this is that a sequence converges to a number a if the sequence eventually gets within *every* prescribed “closeness” tolerance of a .

This leads to a definition of convergence for infinite sequences.

Definition. We say the sequence $\{x_n\}_{n=1}^{\infty}$ has the limit a , written $\lim_{n \rightarrow \infty} x_n = a$, if for every $\varepsilon > 0$ (closeness tolerance), there is a point in the sequence, N , (which may depend on ε) at which all the elements of its tail, T_N , are within ε of a .

We now have a precise definition of what it means for a sequence to have a limit, but how do we actually show a given sequence converges to a given limit point? Notice work has to be done for *every* $\varepsilon > 0$. The problem is that there are an infinite number of such ε ’s!

The only way to be able to handle an infinite collection of such ε ’s is to automate the process of finding an N for a given ε . That is, we need to be able to find a function for N in terms of ε . Then, given an ε , we use this function to produce an N and then argue that any element in the corresponding tail is within this ε of the limit value. In effect, we come up with one (or maybe a handful) of generic proofs which cover all possible tolerances.

Note: This is, in general, a quantum leap in difficulty from what you are probably used to seeing. Although I hope you have a good idea of the what and why of a limit, don’t be discouraged if you find the next example and the theorem after that hard. It took mankind a long time to figure out this definition and use it. You should also realize this definition is based on the examples we considered. The definition is a formalization of an agreement that was felt to embody convergence; that is, it is a human activity. Definitions are man made constructions based on concrete problems.

Worked Example. Prove that $\lim_{n \rightarrow \infty} x_n = 0$ when $x_n = 1/n$.

proof: We need to write a program to handle *any* ε as input. That is, if someone were to hand us a closeness tolerance, ε , we would need to produce an N such that all elements in the tail are within this ε of 0. This means all elements of the tail satisfy $|x_n - 0| < \varepsilon$, which is $|x_n| < \varepsilon$; or, $x_n < \varepsilon$, since all of the x_n are positive. What do we do?

Let’s first start with a particular ε and see if we can handle that. Then maybe we can see how to *automate* our procedure to handle *any* ε . If ε is 1/10, for instance, what would our choice be for N ? Well, there are many choices for N . We are asking for a place to cut the sequence so that all elements of its tail are within 1/10 of the limit point 0. We claim $N = 10$ works. Why? Because the tail in this case is: $T_{10} = \{1/11, 1/12, 1/13, \dots\}$. It’s clear that choosing N to be 100 or 137 or 1000 works as well. This is because each of their tails: $T_{100} = \{1/101, 1/102, 1/103, \dots\}$ or $T_{137} = \{1/138, 1/139, 1/140, \dots\}$ or $T_{1000} = \{1/1001, 1/1002, 1/1003, \dots\}$ all have the property that their elements are within 1/10 of the limit point 0. We only need to find *an* N that works, let’s

pick N to be 10. So, we have found an N for a *particular* ε . Can we automate this process so that we can find an N for *any* ε ? Notice that we found our N by finding the first N such that $1/(N+1)$ was less than $1/10$. If we do something like this for an arbitrary ε we will have handled the automation step. A convenient choice for N is[†] $N = \text{floor}(1/\varepsilon) + 1$. This is not always the *first* N that works – for $\varepsilon = 1/10$ it gives $N = 11$ rather than 10 – but it is a slightly generous choice which is always safe, and we only need *an* N that works. Try a few choices of ε and see if this makes sense. For this choice of N every element in the tail T_N is within $1/N$ of 0 which is within ε of 0.

We have done what was asked:

“Given any closeness tolerance $\varepsilon > 0$ find an N (which usually depends on ε) such that all elements in the tail of the sequence are within ε of the limit value.”

Proofs of this kind are probably different from what you are used to. You are asked to automate a proof of a generic example; i.e., that of a given ε . Further, the choice of N is not unique. In the example above letting $N = 100 * (\text{floor}(1/\varepsilon) + 1)$ is also a valid choice for N . In addition, the way you find this N (based on a given ε) will change from proof to proof.

One thing I think we all assume from our definition is that if a sequence converges to a number a , then a is unique. That is, a sequence can’t converge to two different numbers. This turns out to be true under our definition, but it requires a proof.

Theorem(Limits are Unique). *If the limit of a sequence exists, it is unique.*

proof: We will show that if the sequence x_n converges to a , $\lim_{n \rightarrow \infty} x_n = a$, and x_n converges to b , $\lim_{n \rightarrow \infty} x_n = b$, then a and b are equal.

We first sketch the outline of our argument. The key idea is that as we go farther and farther out in the sequence, the tail should be very close to both a and b ; in fact *arbitrarily close*. This suggests that, in turn, a and b must be very close; in fact *arbitrarily close*. But to be arbitrarily close means that they are the same number. Therefore, our strategy is to show that $|a - b|$ is less than any prescribed tolerance. This will then mean that $|a - b| = 0$; that is, they are the same number. We now show that when $|a - b| < \varepsilon$ for *every* $\varepsilon > 0$, then $a = b$. This is really the same as showing that when a number c satisfies $|c| < \varepsilon$ for *every* $\varepsilon > 0$, then $c = 0$.

claim: Suppose we have a non-negative number, c , which has the property that $c < \varepsilon$ for all $\varepsilon > 0$, then $c = 0$. One might say that c is arbitrarily close to 0. If c isn’t 0, then let $\varepsilon = c/2$. In this case c must be less than this ε , so $c < c/2$. Since c is positive the inequality is preserved after dividing by c yielding $1 < 1/2$. This is a contradiction; therefore c must be zero.

We now continue with our proof and show that $|a - b| < \varepsilon$ for all $\varepsilon > 0$. Thus showing a and b are equal.

Let $\varepsilon > 0$ be a given number, then since we know x_n converges to a , given an ε_1 we know that there exists an N_1 such that $|x_n - a| < \varepsilon_1$ for all x_n in the tail T_{N_1} . Similarly, since x_n converges to b , given an ε_2 we know there exists an N_2 such that $|x_n - b| < \varepsilon_2$ for all x_n in the tail T_{N_2} . If we let $N = \max(N_1, N_2)$, then the element x_{N+1} lies in both tails T_{N_1} and T_{N_2} , so

$$|a - b| = |a - x_{N+1} + x_{N+1} - b| \leq |a - x_{N+1}| + |x_{N+1} - b| < \varepsilon_1 + \varepsilon_2$$

Now we need to choose ε_1 and ε_2 so that their sum is less than or equal to ε . This will certainly be the case if both are chosen to be equal to $\varepsilon/2$. So, for an arbitrary $\varepsilon > 0$ the relation, $|a - b| < \varepsilon$ holds. Therefore, a is equal to b and we have shown that the limit of a sequence is unique.

[†] The floor function takes a number, x , and returns the greatest integer which is less than or equal to x . In effect, it “rounds down” to the nearest integer.

Computational Strategy for Limits

One of the things we would like to be able to do, just as was done when computing derivatives, is to come up with ways to find the limits of sequences formed from simpler sequences. When computing derivatives we had a basic set of functions and computed their derivatives from the definition. We then had rules for how to find the derivatives of functions that were composed of these basic functions. We would like to invoke a similar strategy when computing limits. To this end, we would like to compute the limit of a sequence by recognizing it as a “base” sequence or by recognizing it as a composition of base sequences and use some set of rules to determine the limit in terms of limits of its base component sequences. Hopefully, we could avoid using the definition directly to work out a given limit just as we were able to avoid using the direct definition of the derivative to compute derivatives of certain classes of functions.

In what follows we assume that $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are sequences with limits a and b respectively.

Here are some rules for combining sequences stated without proof:

Theorem(Linearity of Limits). *If $z_n = c_1x_n + c_2y_n$ then*

$$\lim_{n \rightarrow \infty} z_n = c_1a + c_2b$$

Theorem(Product of Limits). *If $z_n = x_n * y_n$ then*

$$\lim_{n \rightarrow \infty} z_n = a * b$$

Theorem(Quotient of Limits). *If $z_n = x_n/y_n$ and $b \neq 0$ and $y_n \neq 0$ (for all n) then*

$$\lim_{n \rightarrow \infty} z_n = a/b$$

Theorem(Squeeze Limit). *If $x_n \leq z_n \leq y_n$ and $a = b$ then*

$$\lim_{n \rightarrow \infty} z_n = a = b$$

To this we add a specific set of sequences for which we can compute the limit. We will need the notion of a *bounded* sequence: a sequence is bounded if there is a number M such that the absolute value of every element in the sequence is less than or equal to M .

Theorem(Constant Limit). *If $x_n = k$ for some number k , then*

$$\lim_{n \rightarrow \infty} x_n = k$$

Theorem(Generalized Decreasing Sequence). *If $x_n = b_n/n$, where $\{b_n\}_{n=1}^{\infty}$ is a bounded sequence, then*

$$\lim_{n \rightarrow \infty} x_n = 0$$

Theorem(Power Limit). *If $x_n = k^n$ with $|k| < 1$, then*

$$\lim_{n \rightarrow \infty} x_n = 0$$

As an example of the limit of the generalized decreasing sequence pattern consider the sequence $\{x_n\}_{n=1}^{\infty}$, where $x_n = \sin(n)/n$. In this case $b_n = \sin(n)$. The only thing we need to do is check that $|b_n|$ is bounded. This is easy as the absolute value of the sin of anything is less than or equal to 1, for instance. Therefore,

$$\lim_{n \rightarrow \infty} \sin(n)/n = 0$$

As another example, consider the sequence $z_n = 1/n + \sin(n)/n$. We can view z_n as a sum of two sequences $1/n$ and $\sin(n)/n$. Since each of these sequences have limits their sum does also and the value of the limit is the sum of the limits of $1/n$ and $\sin(n)/n$. Therefore,

$$\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} 1/n + \lim_{n \rightarrow \infty} \sin(n)/n = 0 + 0 = 0$$

Finally, if a given sequence, $\{x_n\}_{n=1}^{\infty}$, converges to a , is it the case that $\{f(x_n)\}_{n=1}^{\infty}$ converges to $f(a)$? This is a nice computational property to have as it says: $\lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n) = f(a)$. In other words, if a function has this property, limits “pass through” it. This property holds for “nice” functions. In fact, we will call a function *continuous* at a if this property holds. Functions that are continuous at every input value in their domain are called continuous functions.

So, what do continuous functions look like and how can one spot them. They have the property that while drawing their graph, one would not have to lift the pen off the page. All polynomials are continuous for all inputs; the logarithm functions are continuous for positive inputs; the exponential functions, a^x , for $a > 0$ are all continuous for all inputs. Sums, products, and differences of continuous functions are continuous. Ratios of continuous functions are continuous provided the denominator stays away from 0. For an example of a discontinuous function see example 3 in appendix A.

One more fact ties continuity to the derivative: a function which is *differentiable* at a is automatically *continuous* at a . Why? Let $\{x_n\}_{n=1}^{\infty}$ be a sequence converging to a . For the terms with $x_n \neq a$ we may write

$$f(x_n) - f(a) = \left(\frac{f(x_n) - f(a)}{x_n - a} \right) (x_n - a)$$

The first factor is an approximating slope, which tends to $f'(a)$ since f is differentiable at a ; the second factor tends to 0. By the Product of Limits Theorem their product tends to $f'(a) * 0 = 0$. (For any terms with $x_n = a$ the difference $f(x_n) - f(a)$ is already exactly 0.) That is, $\lim_{n \rightarrow \infty} f(x_n) = f(a)$ – which is precisely the definition of continuity at a .

Convergence Based on Properties of a Sequence

We have seen how to show a sequence converges to a given value. What if we don’t have a candidate for the limit? Can we tell if a sequence will converge just by examining properties of the sequence itself? The answer is yes, in certain situations we know that a limit of the sequence exists.

Theorem (Monotone Sequence). Suppose $\{x_n\}_{n=0}^{\infty}$ is a sequence which is bounded above by M ; meaning, $x_n \leq M$ for all n . If the sequence is also non-decreasing ($x_{n+1} \geq x_n$ for all n) then the limit of the sequence exists. Similarly, if a sequence is bounded below and is non-increasing it also has a limit.

There are other criteria, but they are outside the scope of this book.

An Application: The Number e

The chapter on Differential Calculus used, on faith, the fact that the sequence $x_n = (1 + 1/n)^n$ has a limit – the number called e . The Monotone Sequence Theorem is exactly the right tool to pay this debt: we need only check that the sequence is bounded above and non-decreasing. We sketch both checks.

Bounded above. Expanding by the binomial theorem and regrouping each term,

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^n \frac{1}{k!} \overbrace{\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)}^{k-1 \text{ factors, each less than } 1}$$

Each product of factors is at most 1 (for $k = 0$ and $k = 1$ the product is empty and, by convention, equals 1), so the k^{th} term is at most $1/k!$; and since $k! \geq 2^{k-1}$ for $k \geq 1$,

$$x_n \leq \sum_{k=0}^n \frac{1}{k!} \leq 1 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots \leq 3$$

Non-decreasing. In the displayed sum, passing from n to $n + 1$ does two things: every factor $(1 - j/n)$ grows to $(1 - j/(n + 1))$, so no existing term shrinks (the $k = 0$ and $k = 1$ terms stay fixed at 1; every later term grows); and one more (positive) term appears, at $k = n + 1$. Neither effect can push the sum down, so $x_{n+1} \geq x_n$ – and non-decreasing is all the theorem asks.

Bounded above and non-decreasing: by the Monotone Sequence Theorem the limit exists. The value of the limit, $2.718281828\dots$, is the number e . (The chapter on The Derivative As Approximation offers the fast route to its digits: $e = \sum_{n=0}^{\infty} 1/n!$, from Taylor's Theorem.)

Two Trigonometric Limits

In the chapter on Differential Calculus, computing the derivative of the sine function came down to two limits: for any sequence $\{h_n\}_{n=1}^{\infty}$ which converges to 0 with the property that $h_n \neq 0$ for all n ,

$$\lim_{n \rightarrow \infty} \frac{\sin(h_n)}{h_n} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{\cos(h_n) - 1}{h_n} = 0$$

In the chapter these were taken on faith; here we pay the debt. The first limit is a fine application of the Squeeze Limit Theorem of this appendix; the second follows from the first.

Examining the figure below and comparing the areas of the triangular region ABC to the sector ADC to the triangle ADE we find that

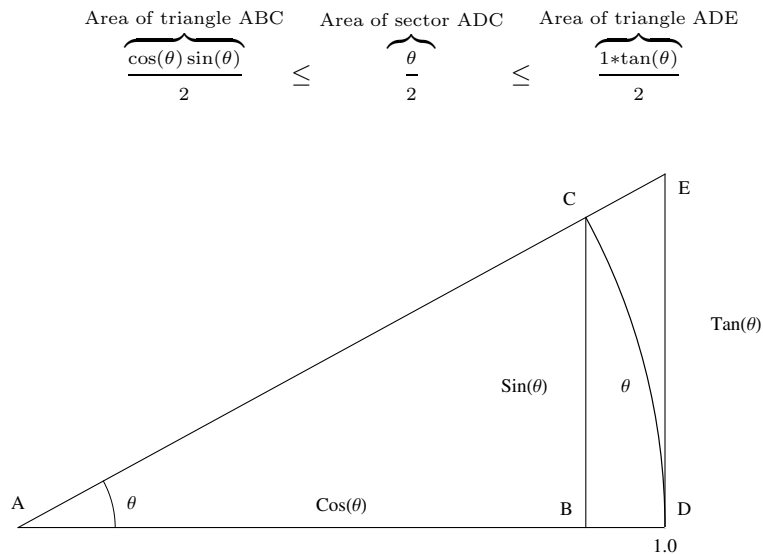


Figure 1: Comparison of triangle ABC, sector ADC, and triangle ADE.

Here, we are assuming θ is positive, measured in radians, and is less than $\pi/2$. The area of each triangle is found using the formula, base * height / 2, while the area of the sector is determined by the formula $\theta r^2/2$. In our case, $r = 1$, giving $\theta/2$; again, provided θ is measured in radians. We briefly describe radian measure for those who have not seen it before.

The radian measure of an angle is the arc length of the angular sector formed by this angle on the unit circle. That is, if a sector has angle θ in radians, then the corresponding arc (on the unit circle) has length θ (see the figure

above). The arc length of a complete circle (360 degrees) is 2π ; therefore, there are 2π radians in 360 degrees. This means that there are $\frac{360}{2\pi} = \frac{180}{\pi}$ (which is approximately 57) degrees per radian. Also, the area of the unit circle is π ; so, the area of a sector of θ radians (on the unit circle) is a fraction of the area of the full circle π . What fraction? Well, the sector of θ radians has an arc length of θ while the arc length of a complete circle is 2π . Since the area of a complete circle is π , the area of the sector must be $\frac{\theta}{2\pi}$ of π . So, the area of the sector is $\theta/2$. One can use other angular measures (such as degrees or grads) but the formulas for the area of a sector have messy constants; and, in turn, the derivatives of the trig functions using these angular measures also have messy constants. Note that this is not unlike the derivative of the logarithmic functions; any choice of base other than e leads to a messier formula.

Continuing with our calculations, the first inequality gives the relation:

$$\frac{\sin(\theta)}{\theta} \leq \frac{1}{\cos(\theta)}$$

while the second inequality yields the relation:

$$\cos(\theta) \leq \frac{\sin(\theta)}{\theta}$$

Combining these gives:

$$\cos(\theta) \leq \frac{\sin(\theta)}{\theta} \leq \frac{1}{\cos(\theta)}$$

A similar argument shows that the relation holds when θ is negative and larger than $-\pi/2$. We may replace θ with the n^{th} element of our sequence h giving:

$$\cos(h_n) \leq \frac{\sin(h_n)}{h_n} \leq \frac{1}{\cos(h_n)}$$

Now, since the limit of the first and third terms exists and is 1, it is the case that the middle term has the same limit.[†] Therefore,

$$\lim_{n \rightarrow \infty} \frac{\sin(h_n)}{h_n} = 1$$

We can use this result to compute the other limit of interest, $(\cos(h_n) - 1)/h_n$. First note that

$$\frac{\cos(h_n) - 1}{h_n} = \frac{(\cos(h_n) - 1)(\cos(h_n) + 1)}{h_n(\cos(h_n) + 1)} = \frac{\cos^2(h_n) - 1}{h_n(\cos(h_n) + 1)}$$

This may be written

$$\frac{-\sin^2(h_n)}{h_n(\cos(h_n) + 1)} = -\frac{\sin^2(h_n)}{h_n^2} \frac{h_n}{\cos(h_n) + 1}$$

Since the last expression is a product of expressions which have limits, we may write:[‡]

$$\lim_{n \rightarrow \infty} \frac{\cos(h_n) - 1}{h_n} = - \lim_{n \rightarrow \infty} \frac{\sin^2(h_n)}{h_n^2} \lim_{n \rightarrow \infty} \frac{h_n}{\cos(h_n) + 1} = -(1^2) * (0/2) = -1 * 0 = 0$$

These are the two limits used in the chapter on Differential Calculus to show that $\sin'(x) = \cos(x)$ and $\cos'(x) = -\sin(x)$.

Sequences without Limits

Not all sequences have limits, in fact most do not. In this section we give examples of sequences which don't have limits.

Example 7: Let $x_n = (-1)^n$. This sequence bounces between 1 and -1 and so doesn't converge to anything.

Example 8: Let $x_n = n$. This sequence gets larger and larger and so it cannot converge to anything.

Example 9: Let $x_n = \sin(n)$. This sequence bounces around the interval $[-1, 1]$ but doesn't converge to anything.

[†] This follows from the Squeeze Limit Theorem of this appendix.

[‡] This follows from the Product of Limits Theorem of this appendix.

Appendix D: Mathematical Induction

Domino Theory

Several times in the main text we claimed a formula for *every* n – for example $(x^n)' = n x^{n-1}$, or $\sum_{i=1}^n i = n(n+1)/2$ – after checking only the first few cases. How can a finite amount of work prove infinitely many statements? That is exactly what Mathematical Induction buys us. Mathematical Induction is a technique used to determine the truth of a sequence of statements. The sequence is usually large or infinite. Normally, determining the truth of such a collection would be time consuming – if not impossible. However, if there is a strong connection between successive elements of the sequence, then it can be much easier. By connection I mean that when a given statement in the sequence is true it is relatively easy to show that the next statement is true.

In this sense, Induction is a generalization of what might be called “The Domino Theory”. If you have ever played with dominoes then, in a sense, you already understand Mathematical Induction. To see why, consider placing a number of (upright) dominoes in a room. Knocking them down may be easy or hard depending on how they are laid out. If they are put next to each other in the “standard domino way”, then knocking one domino down will knock all the rest down. How would you describe this to a person who has never seen dominoes fall in this way? You might start by labeling the dominoes; that is, after setting up the dominoes they have a natural ordering – they form a sequence. Let’s label them $D_1, D_2, \dots, D_n$. You then tell the domino newcomer the dominoes are arranged so that whenever any one domino falls the domino after it falls. By this we mean domino D_2 is placed after domino D_1 so that when D_1 falls D_2 falls. In general, if domino D_k falls then D_{k+1} falls as well, if $1 \leq k < n$. Now you ask the newcomer: “Without looking into the room with the dominoes, if I told you that the first domino, D_1 , has fallen, what can you say about the rest of dominoes?” So, you’ve given the newcomer two facts:

1. Domino D_1 has fallen.
2. Whenever domino D_k falls, domino D_{k+1} falls provided $1 \leq k < n$. That is, when one domino falls the next in line falls.

This is all he needs to conclude that all of the dominoes in the room have fallen.

This is the essence of mathematical Induction. Mathematical Induction concerns the *truth* of propositions labeled $P_1, P_2, \dots, P_n$. Like Dominoes, the propositions are either easy or hard to prove depending on how they are “arranged”. If the propositions are arranged like dominoes; that is, if

1. Proposition P_1 is true;
2. Whenever proposition P_k is true, proposition P_{k+1} is true provided $1 \leq k < n$;

then the conclusion is that all of the propositions are true.

In moving from dominoes to propositions all we’ve done is to replace statements like “domino D_k falls” with “proposition P_k is true”.

This result extends to the case where there are an infinite number of propositions (dominoes). In this case the induction conditions (domino conditions) are:

1. Proposition P_1 is true.
2. Whenever proposition P_k is true, proposition P_{k+1} is true provided $1 \leq k$.

The conclusion under these conditions is that *all* of the P statements are true.

One may ask how easy is it to show that any given proposition implies the next? Just as one may ask how one verifies that a given domino falling knocks the next one down? For many collections of statements, P_n , it is the case that the proof of one proposition implying the next is generic, and essentially one only has to do one proof!

Applications

We apply Mathematical Induction to a few problems below. It is worth keeping in mind what must be shown in the second step of induction. For each $k \geq 1$ you must show: *if P_k is true, then P_{k+1} is true.* You are not assuming that P_k actually is true – this step alone does not tell us that P_k is true or that P_{k+1} is true. If you get confused think of dominoes. To show domino D_{k+1} falls when D_k falls you would need to *assume* domino D_k has fallen and show that D_{k+1} falls as a consequence.

A Formula for Adding the First n Integers

By adding up the numbers from 1 to 2, then 1 to 3, and so forth, someone guesses that the formula for the sum of the first n integers is:

$$\sum_{i=1}^n i = n(n+1)/2$$

How can one prove this? We do it by Mathematical Induction. Let P_n be the statement* “ $\sum_{i=1}^n i = n(n+1)/2$ ”. We want to show that every statement, $P_1, P_2, \dots$, is true. Mathematical Induction tells us that if P_1 is true and it is the case that whenever P_k is true it follows that P_{k+1} is true as well, then all of the statements in the sequence $\{P_n\}_{n=1}^\infty$ are true.

First we show P_1 is true. But P_1 is the statement “ $\sum_{i=1}^1 i = 1(1+1)/2$ ”. The left hand side is 1 and the right hand side is 1; therefore, P_1 is true, since “ $1 = 1$ ” is a true statement.

The next thing to do is to *assume* for any given k ($k \geq 1$) P_k is true. This means we assume “ $\sum_{i=1}^k i = k(k+1)/2$ ” is a true statement; that is we assume: $\sum_{i=1}^k i = k(k+1)/2$. We then use this to show “ $\sum_{i=1}^{k+1} i = (k+1)(k+2)/2$ ” is a true statement; that is, $\sum_{i=1}^{k+1} i = (k+1)(k+2)/2$. We proceed as follows: Take the sum on the left and relate it to P_k .

$$\sum_{i=1}^{k+1} i = (k+1) + \sum_{i=1}^k i$$

We use the assumed truth of P_k to rewrite the sum on the right hand side so that

$$\sum_{i=1}^{k+1} i = (k+1) + \sum_{i=1}^k i = (k+1) + k(k+1)/2 = (2(k+1) + k(k+1))/2 = (k+1)(k+2)/2$$

That is,

$$\sum_{i=1}^{k+1} i = (k+1)(k+2)/2$$

We have just shown the truth of the statement P_{k+1} , “ $\sum_{i=1}^{k+1} i = (k+1)(k+2)/2$ ”, from the assumed truth of P_k . We see that we have done so for every $k \geq 1$; the same argument works for *every* k .† Mathematical Induction

* We surround statements in double quotes to indicate that they are statements with no assigned truth value. You can think of our usual mathematical statements such as, $3 = 2 - 1$, as consisting of two things: The first is the statement “3 is equal to 2 minus 1”. The second thing that is conveyed is that the statement is true. In this section we wish to make the separation of a statement from its truth value. So we will refer to statements, P_n , in quotes to indicate that they are statements without (yet) an assigned truth value.

† Notice that this is the same *magic* we have when finding formulas for the derivative of certain functions; namely, we find a way to compute the formula for the derivative at a given “x” and that same formula works for all other “x”s. If this wasn’t the case for a lot of elementary functions, calculus wouldn’t be useful. This is the same for induction: If the way to show the truth of statement P_k implies that P_{k+1} is true *changes* for each k , then induction isn’t useful. This is analogous to writing a function in a programming language that is parameterized. Sometimes rather than writing many such functions, one can write *one function* that takes one or more parameters while the body of the function is *generic*. This generic logic doesn’t always happen in programming nor does it always happen in mathematics, but when it does and we can identify the generic argument, Induction gives us tremendous power.

tells us that P_n is true for all n . That is, we have shown that

$$\sum_{i=1}^n i = n(n+1)/2 \quad \text{for } 1 \leq n$$

Notice, by the way, that the inductive step amounts to checking that Δ of both sides agree – induction is the engine that makes our telescoping (discrete Fundamental Theorem) arguments rigorous.

Notice what helped here was that we had a natural ordering of the propositions (like our room of dominoes) and showing the truth of a given P_k implying P_{k+1} was the same for all k . Most successful applications of induction behave similarly. If the proofs were different for each k , then induction wouldn't be as useful; we would have a lot of work to do.

The Derivative of the Power Functions

We know the derivative of the function x ; it is 1. What is the derivative of x^n when n is an integer? We stated earlier in the book that the answer is $(x^n)' = n x^{n-1}$. We can use Mathematical Induction to show this for $n \geq 1$. Let P_n be the statement: “ $(x^n)' = n x^{n-1}$ ” First, $(x^1)' = x' = 1$; while $1x^{1-1} = 1 * x^0 = 1 * 1 = 1$. So, P_1 is true. We *assume* now for any k ($k \geq 1$), P_k is true and try to show P_{k+1} is true. Statement P_k true means $(x^k)' = k x^{k-1}$. Now, the derivative of x^{k+1} can be related to x^k since $x^{k+1} = x x^k$. Using the product rule and the assumed truth of the formula for the derivative of x^k gives: $(xx^k)' = x'x^k + x(x^k)' = 1x^k + x(k x^{k-1}) = x^k + k x x^{k-1} = (k+1)x^k$. Or, $(x^{k+1})' = (k+1)x^{(k+1)-1}$. That is, the statement P_{k+1} is true when P_k is true (for *any* $k \geq 1$). By Mathematical Induction, P_n is true for all $n \geq 1$. That is,

$$(x^n)' = n x^{n-1} \quad \text{for } n \geq 1$$

What about when n is 0 or negative? We haven't proved the formula in those cases. One can easily check the formula for $n = 0$. For negative n we can use induction again. In this case let P_n be the statement “ $(x^{-n})' = -n x^{-n-1}$ ” for $n \geq 1$ and $x \neq 0$. Induction requires statements labeled from 1, so we index the statements by $n \geq 1$, where P_n concerns the power $-n$. To check P_1 – the claim $(x^{-1})' = -x^{-2}$ – one can use the definition of the derivative directly. This requires work, but once done, one can mimic the induction proof above. Given that P_1 has been proved true we next show that P_k implies P_{k+1} . Consider an $x \neq 0$, then

$$(x^{-(k+1)})' = (x^{-1}x^{-k})' = (x^{-1})'x^{-k} + x^{-1}(x^{-k})'$$

The last statement is true by the product rule. Using the assumed truth of P_1 and P_k we may write

$$(x^{-(k+1)})' = -1x^{-1-1}x^{-k} + x^{-1}(-kx^{-k-1}) = -x^{-k-2} - kx^{-k-2} = -(k+1)x^{-(k+1)-1}$$

We have just shown that P_{k+1} is true given that P_k is true. By mathematical induction P_n must be true for $n \geq 1$. The only thing we left out was the proof that P_1 is true. We leave this as an exercise for the reader.

Appendix E: Taylor's Theorem

In the chapter “The Derivative As Approximation” we approximated a function near a point by its tangent line – $f(x+h) \approx f(x) + f'(x)h$ – and asked two questions: exactly how big is the error, and can we shrink it by adding a quadratic correction, or a cubic one, and so on? The chapter's answer, *Taylor's Theorem*, was reached by a derivation that cheated: it assumed the various “ μ ” values were all the same. In this appendix we prove the theorem carefully. Its simplest cases return exactly the tangent line approximation – but now with a formula, not a hand wave, for the error.

Generalization of the Mean Value Theorem

We start by stating and proving a generalization of the Mean Value Theorem:

Theorem (Generalized Mean Value Theorem). *Let f and w be continuous functions on the interval $[a, b]$. If f has minimum and maximum values m and M while $w(s) \geq 0$ for $s \in [a, b]$ and $\int_a^b w(s) ds > 0$, then there exists a value $\mu \in [a, b]$ such that*

$$\int_a^b f(s)w(s) ds = f(\mu) \int_a^b w(s) ds$$

proof: Since f is greater or equal to m and less than or equal M we must have

$$\int_a^b m w(s) ds \leq \int_a^b f(s)w(s) ds \leq \int_a^b M w(s) ds$$

Since m and M are constants they may be taken out of the integrals (linearity property of integrals), so that

$$m \int_a^b w(s) ds \leq \int_a^b f(s)w(s) ds \leq M \int_a^b w(s) ds$$

Dividing by the integral in w gives

$$m \leq \frac{\int_a^b f(s)w(s) ds}{\int_a^b w(s) ds} \leq M$$

Note that the middle term is an intermediate value of the function f . Since f is a continuous function on $[a, b]$, the intermediate value theorem tells us that there is a $\mu \in [a, b]$ such that $f(\mu) = \frac{\int_a^b f(s)w(s) ds}{\int_a^b w(s) ds}$. This can be rewritten as

$$\int_a^b f(s)w(s) ds = f(\mu) \int_a^b w(s) ds$$

Corollary (Mean Value Theorem). *If f is a function such that f' exists and is a continuous function on the interval $[a, b]$, then there exists a $\mu \in [a, b]$ [†] such that*

$$f(b) = f(a) + f'(\mu)(b - a)$$

This follows from the generalized Mean Value result substituting f' for f and setting $w(x) \equiv 1$ for all $x \in [a, b]$. That is, we have for some $\mu \in [a, b]$:

$$\int_a^b f'(s)1 ds = f'(\mu) \int_a^b 1 ds$$

[†] The Mean Value Theorem is usually stated with weaker hypotheses – f need only be differentiable on the *open* interval (a, b) – and with a sharper conclusion – μ lies strictly between a and b . Proving that version requires other tools; this integral version is all we need here.

Or,

$$f(b) - f(a) = f'(\mu)(b - a)$$

Proof of Taylor's Theorem

In the chapter “The Derivative As Approximation” we derived Taylor's Theorem by pretending that the various “ μ ” values handed to us by the Mean Value Theorem were all the same – and we confessed there that this was a cheat. Here we take an honest route. The only tools we need are the FTC and Integration by Parts – both from the chapter on integral calculus – together with a *single* application of the Generalized Mean Value Theorem at the very end.

Suppose f' is continuous on the interval $[a, b]$ and let $x \in [a, b]$. The FTC tells us that

$$f(x) = f(a) + \int_a^x f'(s) ds \quad (1)$$

Squint at this equation and you can already see the shape of Taylor's Theorem: a polynomial approximation to f (here just the constant $f(a)$) plus an integral that records the error *exactly*.

To do better we apply Integration by Parts to the error integral, using the two functions $f'(s)$ and $v(s) = -(x - s)$. Keep in mind that we are integrating with respect to s , so the value x is a constant as far as the integration is concerned; therefore, $v'(s) = 1$, and we may write $\int_a^x f'(s) ds = \int_a^x f'(s) v'(s) ds$. Why the minus sign in v ? An anti-derivative of 1 is only determined up to a constant, and this choice makes v vanish at the endpoint $s = x$ – watch what that does for us. If f'' is also continuous on $[a, b]$, then Integration by Parts gives

$$\begin{aligned} \int_a^x f'(s) v'(s) ds &= (f'(x) v(x) - f'(a) v(a)) - \int_a^x f''(s) v(s) ds \\ &= f'(a)(x - a) + \int_a^x f''(s)(x - s) ds \end{aligned}$$

The endpoint terms collapsed to $f'(a)(x - a)$ because $v(x) = -(x - x) = 0$ while $-v(a) = (x - a)$; and the minus sign in front of the last integral canceled the minus sign inside v . Plugging this back into (1) gives

$$f(x) = f(a) + f'(a)(x - a) + \int_a^x f''(s)(x - s) ds \quad (2)$$

This is the tangent line approximation from the chapter “The Derivative As Approximation” – but now with an exact formula for the error.

Let's do it once more, applying Integration by Parts to the error integral in (2), this time with the two functions $f''(s)$ and $v(s) = -\frac{(x-s)^2}{2}$. By the chain rule $v'(s) = (x - s)$ – differentiating $(x - s)$ with respect to s produces a factor of -1 that cancels the minus sign out front – and once again $v(x) = 0$. If f''' is also continuous on $[a, b]$, then

$$\begin{aligned} \int_a^x f''(s)(x - s) ds &= (f''(x) v(x) - f''(a) v(a)) - \int_a^x f'''(s) v(s) ds \\ &= f''(a) \frac{(x - a)^2}{2} + \int_a^x f'''(s) \frac{(x - s)^2}{2} ds \end{aligned}$$

so that (2) becomes

$$f(x) = f(a) + f'(a)(x - a) + f''(a) \frac{(x - a)^2}{2!} + \int_a^x f'''(s) \frac{(x - s)^2}{2!} ds \quad (3)$$

Each round of Integration by Parts peels one more polynomial term off of the error integral and leaves behind an error integral of the same shape. The pattern that emerges is this: if $f^{(k+1)}$ exists and is continuous on $[a, b]$, then

$$f(x) = f(a) + \sum_{i=1}^k f^{(i)}(a) \frac{(x - a)^i}{i!} + \int_a^x f^{(k+1)}(s) \frac{(x - s)^k}{k!} ds \quad (4)$$

Call this statement P_k . Notice that the hypothesis is honest bookkeeping: each round of Integration by Parts differentiates f once more under the integral, so reaching P_k costs us $k + 1$ continuous derivatives – no hidden assumptions about mysterious “ μ ” functions.

How do we prove P_k for every k ? We use induction (see the appendix on induction): we must show that P_0 is true and that P_k implies P_{k+1} for any $k \geq 0$.

P_0 is true since it is just equation (1) – when $k = 0$ the sum is empty while $\frac{(x-s)^0}{0!} = 1$.

For the inductive step, assume that P_k is true and that $f^{(k+2)}$ exists and is continuous on $[a, b]$. We apply Integration by Parts to the error integral in (4) exactly as before, this time with the two functions $f^{(k+1)}(s)$ and $v(s) = -\frac{(x-s)^{k+1}}{(k+1)!}$. The power rule and chain rule give

$$v'(s) = \frac{(k+1)(x-s)^k}{(k+1)!} = \frac{(x-s)^k}{k!}$$

– as usual the two minus signs cancel – and, as usual, $v(x) = 0$. Therefore

$$\begin{aligned} \int_a^x f^{(k+1)}(s) \frac{(x-s)^k}{k!} ds &= (f^{(k+1)}(x) v(x) - f^{(k+1)}(a) v(a)) - \int_a^x f^{(k+2)}(s) v(s) ds \\ &= f^{(k+1)}(a) \frac{(x-a)^{k+1}}{(k+1)!} + \int_a^x f^{(k+2)}(s) \frac{(x-s)^{k+1}}{(k+1)!} ds \end{aligned}$$

Substituting this into (4) extends the sum to $i = k + 1$ and produces exactly the statement P_{k+1} . The two requirements for induction have been fulfilled; therefore (4) is true for every $k \geq 0$.

One task remains. The error integral in (4) is exact, but it is not easy to look at. We convert it into a friendlier form by applying the Generalized Mean Value Theorem – just once – on the interval $[a, x]$, with $f^{(k+1)}$ playing the role of f and with the weight function $w(s) = \frac{(x-s)^k}{k!}$. (For $x = a$ both sides of (4) are $f(a)$, so there is nothing to prove; since $x \in [a, b]$, we may assume $x > a$.) Let us check the hypotheses: $f^{(k+1)}$ and w are continuous, and $w(s) \geq 0$ for $s \in [a, x]$ since $s \leq x$ there. Also, an anti-derivative of $\frac{(x-s)^k}{k!}$ with respect to s is $-\frac{(x-s)^{k+1}}{(k+1)!}$ – our friend v again – so by the FTC

$$\int_a^x \frac{(x-s)^k}{k!} ds = -\frac{(x-x)^{k+1}}{(k+1)!} + \frac{(x-a)^{k+1}}{(k+1)!} = \frac{(x-a)^{k+1}}{(k+1)!}$$

which is positive since $x > a$. The Generalized Mean Value Theorem then hands us a value $\mu \in [a, x]$ – a value that depends on x , so it is really a function $\mu(x)$ – such that

$$\int_a^x f^{(k+1)}(s) \frac{(x-s)^k}{k!} ds = f^{(k+1)}(\mu(x)) \int_a^x \frac{(x-s)^k}{k!} ds = f^{(k+1)}(\mu(x)) \frac{(x-a)^{k+1}}{(k+1)!}$$

Substituting this into (4), and then writing n for $k + 1$, turns (4) into the celebrated result:

Theorem(Taylor’s Theorem). *If f is a function such that $f^{(n)}$ exists and is a continuous function on the interval $[a, b]$, then for any $x \in [a, b]$ †*

$$f(x) = f(a) + \sum_{i=1}^{n-1} f^{(i)}(a) \frac{(x-a)^i}{i!} + f^{(n)}(\mu(x)) \frac{(x-a)^n}{n!}$$

where μ is a function such that $\mu(x) \in [a, x]$.

Notice that when $n = 1$ the theorem says $f(x) = f(a) + f'(\mu(x))(x-a)$ – the Mean Value Theorem with which we began, now seen as the first rung of a ladder.

Let us make the error term earn its keep.

† Note, when $n = 1$, $\sum_{i=1}^{n-1} g_i = \sum_{i=1}^0 g_i = 0$, regardless of the values of g_i .

Example 1: How good is the approximation $\sin(x) \approx x$ when x is small?

With $f(x) = \sin(x)$ and $a = 0$ we have $f(0) = 0$, $f'(0) = \cos(0) = 1$, and $f''(0) = -\sin(0) = 0$ – so the tangent line at 0 is the line $y = x$, and the quadratic correction vanishes for free. Taylor's Theorem with $n = 3$ then says

$$\sin(x) = x + f'''(\mu(x)) \frac{x^3}{3!}$$

for some $\mu(x)$ between 0 and x . Now $f''' = -\cos$, which is never bigger than 1 in size – no matter what $\mu(x)$ turns out to be. So for $|x| \leq 0.1$,

$$|\sin(x) - x| \leq \frac{|x|^3}{3!} \leq \frac{(0.1)^3}{6} \approx 1.7 \times 10^{-4}$$

Not bad for a straight line – and we never had to compute $\mu(x)$.

Appendix F: How to Study from This Book

This book was written to give you a map of calculus: the four strategic moves, the discrete and continuous boards, and the dictionary that ties them together. But a map you never unfold fades. This appendix is about keeping the map alive after you close the book. It rests on a small number of findings about human memory – findings which are as well established as anything in the study of learning[†] – and on a piece of good luck: you have something no student in history has had, a tireless examiner. Used correctly, an AI will keep this material alive in you for years. Used incorrectly, it will do your thinking for you and you will retain nothing. The difference between the two is the subject of this appendix.

Four Findings About Memory

Just as the calculus in this book was organized around four strategic moves, the practice of *keeping* it is organized around four findings about memory.

1. *Retrieval beats rereading.* Rereading a chapter feels like learning – the material seems familiar, and familiarity masquerades as knowledge. But the act that strengthens memory is pulling the idea *out* of your head, not putting it in again. Testing yourself is not merely a way to measure knowledge; it is the way to create it.
2. *Spaced beats massed.* Ten problems tonight do less for you than three tonight, three next week, and three next month. Memory is strengthened most when you retrieve something just as you are on the verge of losing it. Forgetting a little, then recovering, is not a failure of the method – it *is* the method.
3. *Mixed beats blocked.* Twenty integration-by-parts problems in a row teach you to execute integration by parts. They do not teach you to *recognize* when it applies – because in a block of twenty, recognition is never in question. Real problems arrive unlabeled. Your practice should too.
4. *Struggle first, then check.* An answer you fought for – even if you lost – sticks far better than an answer you were shown. Commit to an answer, in writing, *before* you see the solution. The discomfort of not knowing is not an obstacle to learning; it is the sensation of learning.

Notice what these findings say about the traditional arrangement – read a large book slowly, do a block of like problems after each section, review everything the week of the exam. It is almost perfectly backwards. It produces what it optimizes for: a few weeks of buffered material, released on exam day and gone by summer.

The Method

Here is the whole method. For each chapter of this book:

1. *Read the chapter*, working the examples with pen and paper as you go. This is the only step that resembles ordinary studying.
2. *Close the book and rebuild the summary.* Every chapter ends with a summary. Before you reread it, write your own from memory: the definitions, the theorems, the moves that were played. Then compare. What you missed is not a mark against you – it is the most valuable information the exercise produces, because it tells you exactly what to revisit.
3. *Work the exercises cold.* Commit to an answer before checking anything. If you are stuck for ten minutes, that is fine; finding 4 above says the struggle is doing the work.

[†] They come from a century of experiments, starting with Ebbinghaus's measurements of his own forgetting in the 1880s, through the modern work on "retrieval practice" by Roediger and Karpicke (2006), the "spacing" and "desirable difficulties" work of the Bjorks, and Skemp's 1976 comparison of relational and instrumental understanding. The findings have held up; education fashion comes and goes around them. Full citations are collected in the References section at the end of the book.

4. *Then bring in the examiner* – the next day, the next week, and the next month, as laid out below.

The whole apparatus costs perhaps two hours a week. It is not the grind; it is the thing the grind was always pretending to be.

Rules for the Examiner

An AI can play two roles: oracle or examiner. As an oracle it answers your questions, solves your problems, and explains what you ask it to explain. This feels wonderful and teaches you almost nothing – it is rereading, with better production values. As an examiner it asks *you* the questions, waits for your committed answer, and only then tells you the truth. Everything good in this appendix comes from holding the AI in the examiner’s chair. Three rules accomplish this:

1. *Never let it solve before you commit.* Tell it so, in the prompt, explicitly: “Do not show me the answer or any hint until I have given my answer.” A good examiner is told the rules of the exam.
2. *Make it grade the setup separately from the algebra.* In this book the hard step was never the arithmetic; it was choosing the move – recognizing the telescoping sum, seeing that an integrating factor is called for. Instruct the examiner to grade “did you choose the right move” and “did you execute it” as two separate marks. An AI will happily absorb the arithmetic burden anyway; what must remain yours is the choice of move.
3. *Make it probe for understanding, not recall.* The test of understanding is not whether you remember a formula – it is whether you can *rebuild* it from the move that produced it, and whether you know what it is *for*. Years ago I watched a room of bright engineering students fail an exam problem with the formula for integration by parts written on the board in front of them. They had memorized everything and understood what none of it was for. The examiner’s best questions are the ones that would have exposed this a year earlier: “What does this formula trade for what?” “When would you reach for it?” “Derive it from the product rule.”

Prompts That Work

What follows are prompts you can give an AI, nearly verbatim. They are grouped by chapter, and each one embodies the rules above. You will naturally drift from these scripts as you and your examiner develop a working relationship; that is as it should be. What should not drift is the structure: examiner not oracle, commit before checking, setup graded apart from algebra.

After the chapters on Difference and Summation Calculus

You are my examiner on difference calculus. Give me sequences, one at a time, and I will compute their difference sequences. Then reverse the game: give me difference sequences and I will find antidifferences. Mix in sums I should evaluate by recognizing a telescoping difference and applying the Fundamental Theorem of Difference Calculus. Do not show answers or hints until I commit to an answer. Keep track of the kinds of errors I make and return to those kinds later without telling me you are doing so.

The last sentence quietly gives you interleaving and spacing inside a single session: the examiner reintroduces your weak spots when you are no longer expecting them.

After the chapters on Differential and Integral Calculus

Quiz me on differentiation and integration, mixed together and unlabeled, one problem at a time. After each of my answers, grade two things separately: whether I chose the right technique (linearity, product rule, chain rule, substitution, integration by parts) and whether I executed it cor-

rectly. Do not tell me which technique a problem needs -- recognizing it is the point.

And a second game, which no textbook can play with you:

Show me a worked evaluation of an integral that contains exactly one subtle error. I will find the error and explain why it is wrong. Start easy and get harder as I succeed.

Finding a planted error exercises precisely the verification skill you will need in the age of AI: the machine will do the computation, and *you* must be the one who can tell whether it is right.

After the chapter on the Derivative as Approximation

Give me quantities to estimate with the tangent line approximation -- no calculator -- and then have me bound the error using the next term of Taylor's Theorem. Then ask me conceptual questions: why is the tangent line the best line? What does the error term buy us? Make me derive the first two terms of Taylor's Theorem from the Mean Value Theorem.

After the chapters on Difference and Differential Equations

Give me word problems -- loans, populations, cooling coffee, radioactive decay -- without telling me whether they are discrete or continuous. I will set up the difference or differential equation, name the strategic move (isolate and invert) and the tool that executes it (summing factor or integrating factor), and solve it. Grade the setup, the move-and-tool choice, and the algebra as three separate marks. Occasionally give me a problem where the model breaks down and ask me where and why.

The Monthly Examination

Once a month, hold a real examination -- oral, in the old style:

You are my oral examiner for twenty minutes on the whole of discrete and continuous calculus. Ask me to: derive the Fundamental Theorem of Calculus from the telescoping sum picture; explain what integration by parts trades for what, and derive it from the product rule; state the discrete/continuous dictionary from memory; solve one difference equation and one differential equation and point out that they are the same game. Push back on vague answers. At the end, tell me which parts of my understanding are load-bearing and which have gone soft.

The one-month spacing is chosen on purpose: by finding 2, you want to retrieve this material roughly when you are starting to lose it. If the viva (as the British call an oral exam) goes badly in places, that is the system working -- you have found the soft spots while they are still cheap to repair.

The Drill Companion

This book deliberately contains few exercises; it is a map, not a gymnasium. For raw volume of practice, pair it with any inexpensive problem collection -- the Schaum's Outline of Calculus is the traditional choice and costs less than lunch.[†] Two rules keep the drill useful:

- Do not work the problems in blocks. Open two chapters at once and alternate, or have your examiner select page numbers for you at random. Finding 3 again: real problems arrive unlabeled.

[†] Any edition will do. The problems have not changed in fifty years because the calculus has not changed in three hundred.

- Before each problem, say – out loud or on paper – which strategic move it will need. *Then* work it. This one habit converts a thousand grind-problems into a thousand recognition drills, which is the skill the grind never taught.

A Schedule

Collecting all of the above into one place. After finishing a chapter:

When	What
same day	rebuild the chapter summary from memory; exercises, cold
next day	examiner session on the new chapter
next week	mixed session: new chapter plus everything before it
each month	the twenty-minute oral examination, whole book

Miss a session, pick up where you left off; the schedule is a rhythm, not a debt. And notice what is absent from it: there is no “review week,” because there is nothing left to cram – the material never had a chance to leave.

The Test

A year from now, sit down with a blank sheet of paper and try to write out the dictionary that closed the last chapter – Δ and d/dt , summation and integration, the two Fundamental Theorems, the summing factor and the integrating factor, $a = 1 + r$ tying geometric to exponential growth. Then pick one row and rebuild it: derive the continuous entry from the discrete one by the *approximate, then take a limit* move.

If you can do that, you have what this book was written to give you. Not a buffer of formulas – those were never the point, and the machine holds them better than you ever will – but the map: which game is being played, which move applies, and the ability to rebuild any tool you have lost. That is what “knowing calculus” can still mean, and it is the part no examiner, human or artificial, can hold for you.

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